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Paleolakes and socioecological implications of last glacial "greening" of the South African interior

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Supplementary information

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Study approach

We investigated four palaeolake basins in detail, spanning from the western interior Nama-Karoo near Brandvlei, to the southern margins of the Savanna Biome near Kimberley (Fig. S1). In the Karoo, we dated pan (playa) fill deposits from the Sak River catchment (Grootvloer -29.984°S, 20.522°E) and pan fill and shoreline deposits from Swatkolkvloer (30.793°S, 20.069°E) and the adjacent pan, Hoek van Spruit se Vloer (-30.880°S, 20.252°E) (1, 2). At Kimberley we revisited the Alexandersfontein pan site (-28.838°S 24.782°E; 3) and provide a first suite of robust age constraints for high stands identified in this original study. Today, these basins are arid to semi-arid (aridity indexes of 0.09 (Swartkolkvloer) to 0.3 (Alexandersfontein)), and experience either mixed or predominantly summer rainfall. Swartkolkvloer was studied in the most detail and here several shoreline deposits up to 18 m above the modern pan floor were identified on the eastern margins of the pan. We applied several geochronological methods, employing – where possible – multiple methods, resulting in a total of 18 new radiocarbon dates and 26 new luminescence ages, the latter using both quartz and K-feldspar methods.

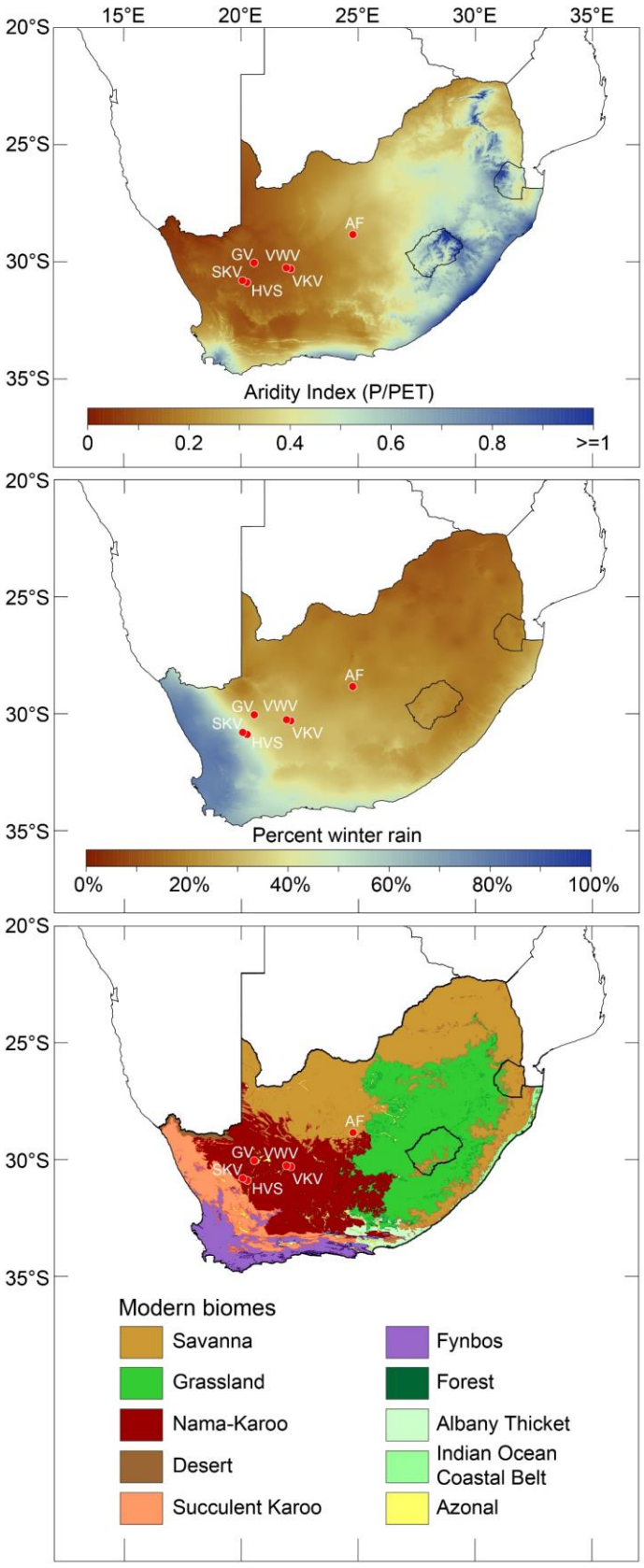


Fig. S1: Maps of South Africa showing major biomes, aridity index (precipitation (P)/potential evapotranspiration (PET)), and rainfall seasonality. Biome data from Mucina and Rutherford (4) and aridity index calculated according to, and with the data from, Trabucco and Zomer (5) and WorldClim 2.1 (6). Study site codes are: SKV – Swartkolkvloer; GV – Grootvloer; HVS – Hoek van Spruit se Vloer; VWV – Vanwyksvlei; VKV – Verkeerdevlei; AF – Alexandersfontein.

Archaeological surveys, preliminary geomorphological observations, and several new age constraints on one shoreline deposit are also reported for two pans in the Vanwyksvlei area (-30.317° S 21.849°E).

Methods

Field sampling and surveying

Sediment samples were obtained from freshly cleaned exposures where available. Samples from pan floor sedimentary fills were obtained with a Dormer sand augering system. Sediments were logged and described using standard methods. Additional localized 3D mapping of the sample sites was achieved using photogrammetry from vertical drone images, analysed within the Agisoft Metashape software. Surficial archaeological scatters were mapped and photographed in place and were not field collected. Emphasis was placed on assessing the depositional relationships between observed material culture and landforms or sediments diagnostic of lacustrine activity. All artifacts photographed in hand were replaced precisely where encountered. No excavation was conducted.

Radiocarbon dating

Results summarised in Appendix A.

Luminescence dating

Samples were opened under subdued light in the University of Leicester and were prepared for coarse grain (90-250 µm) quartz and/or K-feldspar analysis (7, 8). Sample tube ends were retained to assess the water contents and radionuclide concentrations. The remaining material was treated with 10% HCl to remove carbonates and 30% H_2O_2 to remove organics. The samples were then wet sieved and density separated using LST Fastfloat to isolate the fractions: 1) >2.58 g cm^{-3} and <2.72 g cm^{-3} (for quartz) and 2) <2.58 g cm^{-3} (for K-feldspar) The quartz fraction was treated with 45 minutes of 48% hydrofluoric acid, while the K-feldspar fraction (<2.58 g cm^{-3}), for most samples (Appendix A, Table S12), was etched for 10 minutes in 10% HF. The samples were rinsed in HCl, dried and re-sieved.

All luminescence measurements were performed on a Risoe DA20 TL/OSL reader. All samples were analysed using small (2mm) aliquots, which in most cases contained 100-150 grains. For quartz, the integration windows were the first 0.2 seconds and the subsequent 0.9 seconds (9). For K-feldspar analyses, integration windows (300 second stimulation for post-IR) comprised the first 2.6 seconds and the last 7 seconds, with the samples run “one at a time” in the Risoe reader. Dose-response curves were fitted with saturating exponential fits, with D_e uncertainties incorporating counting statistics, curve fitting uncertainties and a 1.5% systematic uncertainty (all calculated within the Risoe *Analyst* software (10). The uncertainty in the final D_e estimates incorporates a beta source calibration uncertainty (3%). For both minerals, aliquots with recycling ratios (repeat dose points in the dose response curve) outside 10% of unity, or recuperation (zero dose signal) >5% of the natural signal were removed prior to averaging of the D_e distributions.

Single grain quartz analyses were undertaken for several samples from the Alexandersfontein site. These measurements employed a focused 532 nm Nd:YVO_4 solid state diode-pumped laser emitting at 532 nm focused to a ~20 µm spot. Dose-response curves were fitted with saturating exponential fits in

the *Analyst* software and their uncertainties included the addition of a 3% systematic instrumental uncertainty (determined following 11). In the subsequent analysis, grains were rejected for the following reasons (e.g. 12): 1) if the test dose signal was less than 3 sigma above background; 2) if the relative test dose uncertainty was larger than 20%; 3) if the recycling ratio or IR depletion ratios were inconsistent with unity at two sigma; 4) if recuperation (zero dose L_x/T_x) was greater than 5% of L_n/T_n; 5) if the dose-response curve did not grow with dose; 6) if a finite D_e or D_e uncertainty could not be obtained due to saturation of the dose response curve, or a failure of the L_n/T_n measurement to intercept the dose response curve (so called "oversaturating grains; 13).

Environmental Dose Rates

Dose rates were determined using elemental concentrations obtained via inductively coupled plasma mass spectrometry (ICP-MS; for U and Th) and ICP-OES (for K) analyses of the dried sample tube ends. External beta dose rates were further evaluated using GM beta counting on the same tube-end material (following 14). Where *in-situ* gamma spectrometry measurements were made (AF17/1/1, AF17/1/2, AF17/5/3, AF17/5/2, VKV17/1/1 and SKV17/1/2) the resulting U, Th and K contents (obtained using the "windows" method; (15)) were used to derive the sample dose rate.

Table S1: Beta dose rate comparisons. The all-sample average ratio is 0.97 ± 0.05 (n=14). **in situ* gamma spectrometry windows method.

Sample	Beta dose rate: ICP-MS (Gy ka^{-1})	Beta dose rate: GM beta counting (Gy ka^{-1})
SKV18/4/2	1.30 ± 0.08	1.21 ± 0.07
SKV19/5/1	1.11 ± 0.08	1.06 ± 0.07
SKV18/3/1	1.21 ± 0.09	1.08 ± 0.07
SKV18/2/9	0.86 ± 0.06	0.79 ± 0.05
SKV19/2/10	1.38 ± 0.09	1.40 ± 0.07
SKV19/2/11	1.82 ± 0.12	1.91 ± 0.10
HSV19/3/4	2.07± 0.14	2.12 ± 0.11
GV18/5/1	1.41 ± 0.10	1.29 ± 0.08
GV18/5/2	0.68 ± 0.05	0.68 ± 0.04
GV18/5/3	1.92 ± 0.13	1.87 ± 0.11
GV18/4/2	1.55 ± 0.11	1.44 ± 0.09
AF17/1/1	0.92± 0.07*	0.91± 0.05
AF17/1/2	0.88 ± 0.05*	0.87 ± 0.05

Final age uncertainties incorporate 3% relative uncertainties for the dose rate conversion factors, grain size attenuation factor, water attenuation and HF etching. ICP-MS-derived external beta dose rates were compared to those obtained from GM Beta counting and show good correspondence (Table S1). For sites already exposed by artificial excavation (Breek Been Kolk and SKV Quarry) the measured sample water contents were rather low (~0.6%) and potentially unrepresentative of the burial water content. For these samples, the water content was set to 3 ± 3%. For all other samples (including those excavated while sampling or rapidly eroding (e.g. Alexandersfontein)) the measured water content was utilized, also with

a 3% (absolute) uncertainty. The ages (particularly for the K-feldspar measurements) are relatively insensitive to sample water content; a 1% change in water content typically results in a 200-300 year difference in the final age.

Cosmic dose rates were calculated following (16) using the *Luminescence* R package (17), and are based on measured sediment overburdens. While potentially relevant to some of the shoreline sites (e.g. the Quarry at Swartkolkvloer), we did not apply any correction for additional water attenuation of the cosmic dose rate (i.e. submergence of the sites). The OSL ages from these sites are relatively insensitive to this issue. For example, assuming an additional water absorber was present for the entire burial duration, the difference in cosmic dose rate to a sample buried at 1 m sediment depth is ~6% per meter of water depth. Given that the cosmic dose contribution is typically 6% of the total dose rate, the resulting impact on the final ages is not significant (1-2% differences in age). Given the uncertainties in both water depth and the duration of any submergence (it is very unlikely the higher marginal shorelines were submerged for the duration of sample burial) we therefore elected not to include any such corrections.

Sample luminescence properties - Quartz

Quartz from Swartkolkvloer and Grootvloer was relatively insensitive and, particularly when using a late background signal subtraction, large numbers of aliquots were rejected for excessive recuperation. Rejection rates were reduced and equivalent doses increased with the use of early background subtraction (9). Fast ratios, indicating the degree of dominance of the quartz fast component (18) were variable and for some aliquots, rather low (e.g. GV18/4/2 had a median fast ratio of 12, but exhibited values as low as 1.5). The fast ratio as a screening criterion resulted in large numbers of aliquot rejections (Appendix A; Table S13). At Grootvloer, a quartz OSL age for GV18/4/2, obtained using early background subtraction and a fast ratio rejection threshold of 10, did however produce an age close to the overlying radiocarbon date and accompanying K-feldspar ages. At Swartkolkvloer, quartz ages were consistently lower than both accompanying radiocarbon dates and the K-feldspar ages, with up to 45% of aliquots rejected using standard criteria (recycling ratio, recuperation < 5% of the natural sensitivity corrected OSL). Thus, for both Grootvloer and Swartkolkvloer the luminescence chronology is based only on IRSL analysis of the K-feldspar fractions. At Alexandersfontein the quartz OSL signal was brighter (Fig. S2) and consistently passed standard aliquot rejection criteria. Dose recovery experiments further implied the SAR protocol was suitable for quartz at this site (Table S2). The luminescence ages at Alexandersfontein are therefore based on single aliquot or single grain quartz OSL measurements.

Table S2: Summarized single aliquot (2 mm) dose recovery results for selected quartz samples from Alexandersfontein.

Sample	Dose recovery ratio across all preheating temperature (160-260°C) (n)	Overdispersion for all measurements (%)	Dose recovery ratio for chosen preheat combination (n)
AF17/1/1	0.97 ± 0.04 (23)	0	0.99 ± 0.04 (3)
AF17/5/1	0.96 ± 0.05 (24)	0	0.96 ± 0.01 (3)
AF17/5/2	0.96 ± 0.07 (27)	4 ± 1	0.96 ± 0.04 (3)

Sample luminescence properties - K-feldspar

Samples were analysed using both IR-IRSL methods combining an initial 50°C stimulation with either a 225°C ($pIRIR_{225}$) or 290°C ($pIRIR_{290}$) post-IR measurement. The K-feldspar IRSL signals for both Grootvloer and Swartkolkvloer were bright, exhibited low signal recuperation and generated reproducible sensitivity-corrected dose-response curves, which typically grew well beyond 350 Gy. The results of dose recovery experiments were also acceptable (Table S3).

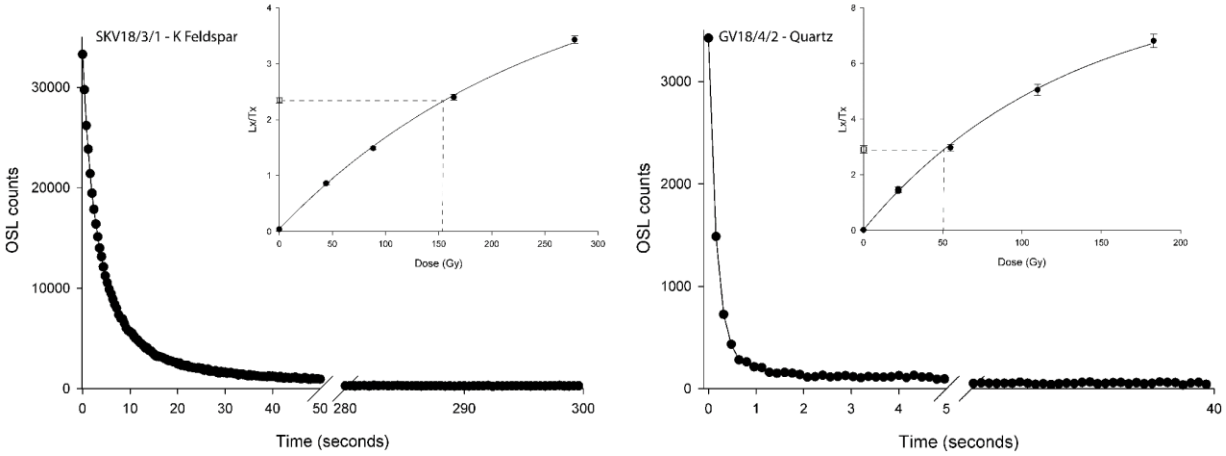


Fig. S2: IRSL ($pIRIR_{225}$) and OSL shine downs curves for 2 mm aliquots of K feldspar and quartz (respectively) from SKV18/3/1 and GV18/4/2. These are regenerated signals in response to 38 Gy (SKV18/3/1) and 55 Gy (GV18/4/2) laboratory beta doses. The insets show the dose response curves for the same samples. The example from GV18/4/2 is a relatively sensitive quartz aliquot for this sample, lying on approximately the 66th percentile of the SAR protocol test dose signals.

K-feldspar, particularly the pIRIR signal, bleaches more slowly than quartz (e.g. 19, 20, 21). Measured residuals following two days of natural (UK) light exposure were typically < 1 Gy for IR50 measurements, 6-8 Gy for the $pIRIR_{225}$ measurements and 11-12 Gy for $pIRIR_{290}$ measurements (Table S3). For the lower temperature measurements this represents a small proportion of the typical equivalent doses. There is a more substantial residual for the $pIRIR_{290}$ measurement, although in most instances there is good agreement between the $pIRIR_{290}$ ages and the fading-corrected $pIRIR_{225}$ ages and /or the radiocarbon dates (see below), without residual subtraction. We therefore did not apply any residual subtractions. Fading rates (g-$value_{2days}$) ranged between 3 and 8% per decade for the IR50 signal, but were substantially reduced for the post-IR IRSL measurements (1.2-2.7% per decade for the $pIRIR_{225}$ signal and 1.3-2.0% for the $pIRIR_{290}$; Appendix A; Table S13). Where performed fading correction followed Huntley and Lamothe (2001) and was performed in the *Luminescence* R package (17).

Equivalent dose distributions and luminescence ages – Swartkolkvloer and Grootvloer

The equivalent dose distrubutions were visualised using radial plots, as created using the Radial Plotter software (22). The majority of the Swartkolkvloer and Grootvloer samples were analyzed using multi-grain aliquots of K-feldspar, which regardless of measurement protocol, showed low to moderate amounts of inter-aliquot scatter (as measured by the overdispersion parameter in Appendix A; Table S13). In most

instances there was little evidence for strongly positively skewed equivalent dose distributions that might suggest significant issues with incomplete bleaching, although in some cases individual high or low outlier D_es were observed. These were formally identified by applying the nMAD (Median Absolute Deviation) approach (e.g. 23) and are highlighted in the relevant radial plots. Critically, in most cases there was good age agreement between the radiocarbon dating and IRSL ages obtained using the Central Age Model (CAM) weighted mean equivalent dose (24).

Table S3: Post-IR IRSL dose recovery experiment results and residual dose measurements for a selection of samples. Each result is the average of 3 aliquots unless stated. Samples were bleached in natural (UK) daylight prior to analysis.

Sample	IR 50 DRR	$pIRIR_{225}$ DRR	$pIRIR_{290}$ DRR	IR 50 Residual (Gy)	$pIRIR_{225}$ Residual (Gy)	$pIRIR_{290}$ Residual (Gy)
SKV Quarry						
SKV18/2/9	1.08 ± 0.02	1.05 ± 0.03		0.8 ± 0.2	5.5 ± 2.3	11.6 ± 2.1
SKV19/2/10	0.96 ± 0.03	0.96 ± 0.03				
SKV18/3/1	1.08 ± 0.09		1.15 ± 0.1			
SKV Dam (Breek Been Kolk)						
SKV18/4/1	1.03 ± 0.02	1.00 ± 0.03				
SKV18/4/2	0.97 ± 0.01	0.99 ± 0.02				
SKV19/5/1 (n=4)	1.04 ± 0.02	1.01 ± 0.01	1.04 ± 0.04	1.1 ± 0.1	6.2 ± 1.4	11.5 ± 0.6
Hoek van Spruit se Vloer						
HSV19/3/4	1.11 ± 0.01	1.04 ± 0.01		1.3 ± 0.4	6.4 ± 1.8	
Grootvloer						
GV18/3/1	0.97 ± 0.01	1.01 ± 0.04				
GV18/5/1	1.00 ± 0.02	1.00 ± 0.01				
GV18/5/2 (n=4)	0.97 ± 0.1	0.96 ± 0.11	1.06 ±0.04			

As anticipated from the measured fading rates, the IR50 ages are substantially younger than the $pIRIR_{225}$ and $pIRIR_{290}$ ages. The question remains as to whether fading correction should be applied to the $pIRIR_{225}$ ages. At Swartkolkvloer and Grootvloer the weighted sample mean $pIRIR_{225}$ fading rate (g-$value_{2days}$) was 1.9 ± 0.2% per decade (n = 14, statistically consistent with a single value (25, 26)). This is markedly lower than the IR50 fading rate (5.1 ± 0.4% per decade n = 14). Some studies have shown that samples analysed with $pIRIR_{225}$ protocol with measured fading rates of 1.5-2.5% may require fading correction (e.g. 27, 28). In this instance it is noteworthy that the uncorrected $pIRIR_{225}$ ages were lower than the accompanying ^{14}C dates in several cases (i.e. SKV18/4/2, SKV18/5/1, SKV17/1/2 and GV18/4/2), but were brought into good agreement with the ^{14}C dates following fading correction (using a mean g-$value_{2days}$ for each locale, where SKV n=8 and GV n=4). The uncorrected $pIRIR_{290}$ ages (mean g-$value_{2days}$ 1.9 ± 0.4% per decade; n=3) were in most instances in good agreement with the radiocarbon dates, implying the $pIRIR_{290}$ signal does not require fading correction (see also 29, 30, 31). Given the additional uncertainty imparted by fading correction, where possible, the $pIRIR_{290}$ ages were utilized rather than fading-corrected $pIRIR_{225}$ ages. An exception was the $pIRIR_{290}$ age for SKV18/2/9, which produced a clear age inversion, which we assume reflects incomplete bleaching. Despite this, and the typical ~11 Gy measured residuals for the $pIRIR_{290}$ signals (Table S3), in most instances the $pIRIR_{290}$ ages are otherwise in

good agreement with the fading-corrected $pIRIR_{225}$ ages and the radiocarbon dates. Thus, no residual subtraction was applied to the $pIRIR_{290}$ data.

Luminescence ages – Alexandersfontein

Equivalent dose distributions for multi-grain (2mm) aliquot quartz analyses exhibited moderate (AF17/1/1/ and AF17/1/2; 14 ± 2%) to significant (AF17/5/1 and AF17/5/3; 30 ± 4 and 24 ± 4% respectively; Table S7) amounts of overdispersion (inter-aliquot scatter unaccounted for by individual measurement uncertainties). As such, additional single grain analyses were undertaken on samples AF17/1/2 (Donga site), AF17/5/1 and AF17/5/3 (Mauritzfontein). The resulting equivalent dose distributions were assessed visually using radial plots and selections of the most appropriate estimate of the burial age D_e are discussed in the context of each sample below. In the case of one sample (AF17/5/1), which showed the most single grain D_e overdispersion (66 ± 4%; Table S7), we utilised the finite mixture model (FMM; 32) to identify distinct equivalent dose populations. The model was run with a range of "intrinsic" overdispersion values (varying sigma from 5-40%) and a range of component numbers, with the solution that maximised the log likelihood and minimised the Bayes Information Criterion (BIC) utilised for final component identifications and associated age estimations (see e.g. 33).

Study sites and associated chronologies

In this study, we focus our efforts on the reconstruction of palaeohydrological change using evidence from unambiguous lake shorelines and deposits at Swartkolkvloer and dated lacustrine sedimentary fills at Swartkolkvloer and Grootvloer. In addition, we present a revised chronology for lacustrine deposits at Alexandersfontein pan. The latter site attests to the regional significance of the identified hydrological changes.

Swartkolkvloer and Grootvloer

Swartkolkvloer and Grootvloer are located ~50 km south and north (respectively) of Brandvlei (Fig. S1). The area is characterized by subdued terrain, underlain by Lower Ecca Group shales, with Jurassic dolerite dykes providing local relief elements (de Witt and Bamford, 1993; de Witt, 1999). The regional drainage system comprises the northwestward then northwards flowing Sak River and the northwards flowing Fish River, with the confluence of the two located at Twee Rivier, ~60 km south of Brandvlei. This is immediately north (downstream) of the Swartkolkvloer system. Approximately 50 km north of Brandvlei the Sak River flows into the wide expanse of the Grootvloer depression – a major basin more than 35 km wide with an area ~275 km^2 – before the river turns to the northeast and flows to the Orange River via the Hartbees River. The headwaters of the Fish River are located near to Sutherland, while those of the northwest flowing Sak River are located near to Loxton (34).

Today, both the Sak and Fish Rivers are ephemeral. Flow data (35) (1972-2019) from the Sak River at the Sak Hol Pads Leegte gauging station (-30.47471°S 20.51917°E) reveal a highly variable regime, with a mean annual flow volume of 56 ± 86 million m^3. Flow events are largely confined to March and April (late summer). Upstream in the Fish River catchment, the Hardeheuwel gauging station (-31.80686°S 20.35805°E) record (1927-2019) is similarly variable, with a mean annual flow volume of 16 ± 30 million

m^3. Here a greater proportion of the flood events occur during the winter (June-August) with a secondary peak in March, implying flow events today are generated by both winter and summer rainfall (Fig. S3).

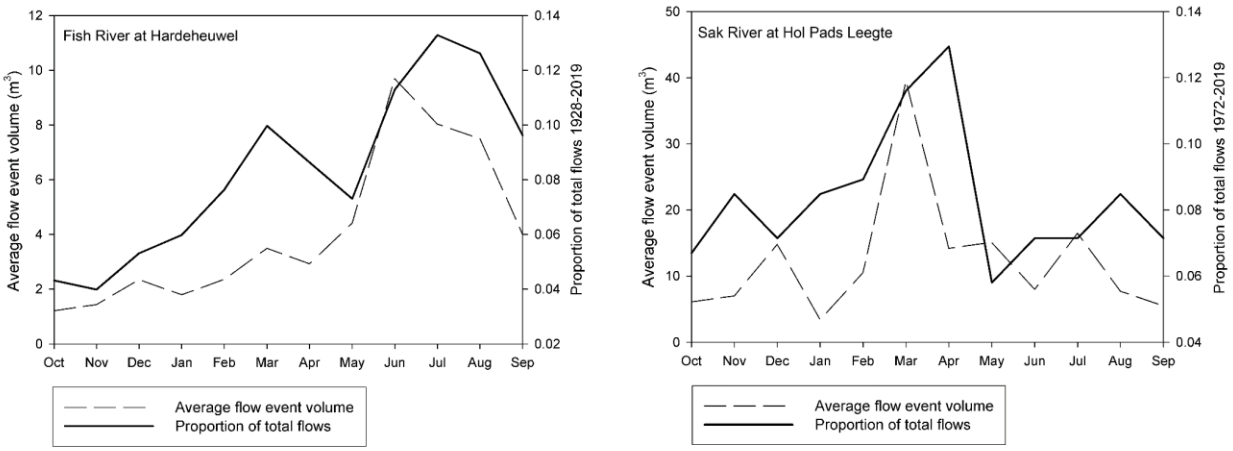


Fig. S3: Average monthly flow events within the Sak (1972-2019) and Fish Rivers (1927-2019) as obtained from gauging data.

Swartkolkvloer

At an altitude of 921 m above sea level (asl), Swartkolkvloer forms the end point of an endoreic drainage system comprising three pans at successively lower altitudes (the others being Hoek van Spruit se Vloer (945 m) and Voëlvlei se Vloer (~925 m) (Fig. S4). They are connected to the Fish River immediately east of Hoek van Spruit se Vloer, via a narrow (~300 m) gap between two bedrock slopes, at an altitude of 949 m asl. Here the floodplain of Fish River also widens significantly, and Kent and Gribnitz (1) proposed this area could be treated as a fourth pan (Fig. S4). Hoek van Spruit se Vloer (945 m) and Voëlvlei se Vloer (~925 m) are separated by a sill (947 m) at -30.896°S 20.175°E. Voëlvlei se Vloer slopes gradually to the northwest from 936 m to 921 m, where it connects to Swartkolkvloer via a narrow canyon. All the pans are susceptible to wind erosion under modern conditions, with significant dust lifting at both Swartkolkvloer and Hoek van Spruit se Vloer observed 31.07.2017.

Swartkolkvloer is a large oval pan (10 x 8 km), approximately 71 km^2 in area, with a pan surface at 921 m asl. Swartkolkvloer is the only pan in the chain of three to experience seasonal inundation. This is intermittent and typically occurs for ~30% of years, within the lowest points of the pan (Figure S4; 36). Inundation is strongly seasonal and, as with flow in the Sak River (Fig. S5), most likely to occur in March - May.

The pan sediments (at the pan centre) - excavated to a depth of 1.1 m - comprise clayey sands (to ~0.5 m) overlying silty clays (Appendix A; Table S11) mixed with evaporates (to > 1.1 m). A layer rich in *Tomichia ventricosa* shells was consistently identified within the 0.45-0.50 m depth range. Two whole shell specimens from this layer were submitted for radiocarbon dating and these date to >43,500 ^{14}C yr. BP and 43,140-44,610 cal yr BP (Appendix A; Table S10). This species was identified at all Swartkolkvloer study locales. It is an aquatic gastropod with a wide salinity tolerance (freshwater to hypersaline; 37). Its

presence is thus not diagnostic of freshwater conditions, although Kent and Gribnitz (1) interpreted the relatively low Mg content of their analysed specimens to be indicative of fresh rather than saline water.

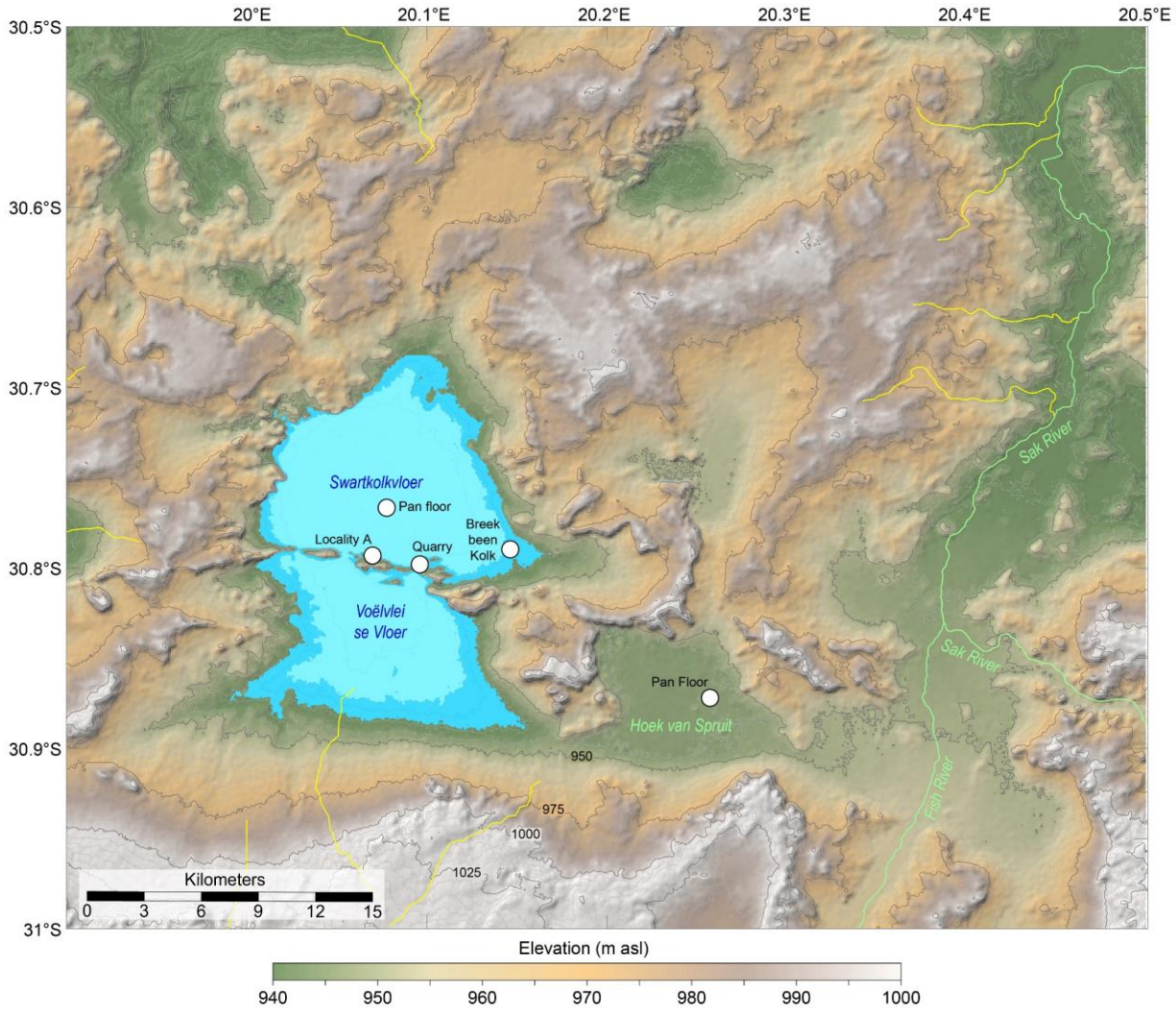


Fig. S4: Map of Swartkolkvloer region. Sampling locations and 939 m and 932 m lake levels are shown. Primary dry (yellow) and non-perennial (green) stream systems are shown. No perennial streams currently exist in the region. Topography from the Multi-Error-Removed Improved-Terrain DEM (38).

A sample for luminescence dating (SKV19/6/1) was obtained from these same pan floor sediments at a depth of 0.52 m. The post IR-IRSL measurements produced a rather scattered (Fig. S6), but un-skewed D_e distributions, with the $pIRIR_{225}$ CAM D_e producing an age markedly younger than other Swartkolkvloer sites (Table S5) and much younger than the radiocarbon dates from the *T. ventricosa* shells from a nearly identical depth (0.48 m). Given its pan floor context, the intrusion of younger grains via desiccation cracks may account for some of this variability (introducing younger grains) and the incorporation of the (older) shells into younger sediments as a reworked or lag deposit also cannot be ruled out. In this respect it is interesting that the fading corrected $pIRIR_{225}$ CAM age (17 ± 2 ka) agrees well with the two young radiocarbon ages reported by Kent and Gribnitz (1; 16,090-16,940 cal. yr BP (13,670 ± 130 14C year BP)

and Beaumont (2; 16,370-17,130 (13,900 ± 130 14C year BP), potentially providing supporting evidence (as will be shown, in contrast to other shoreline data) for a latest Pleistocene inundation phase.

Several localities on the eastern and southeastern margins of the Swartkolkvloer preserve evidence of palaeo-shorelines and lacustrine deposits.

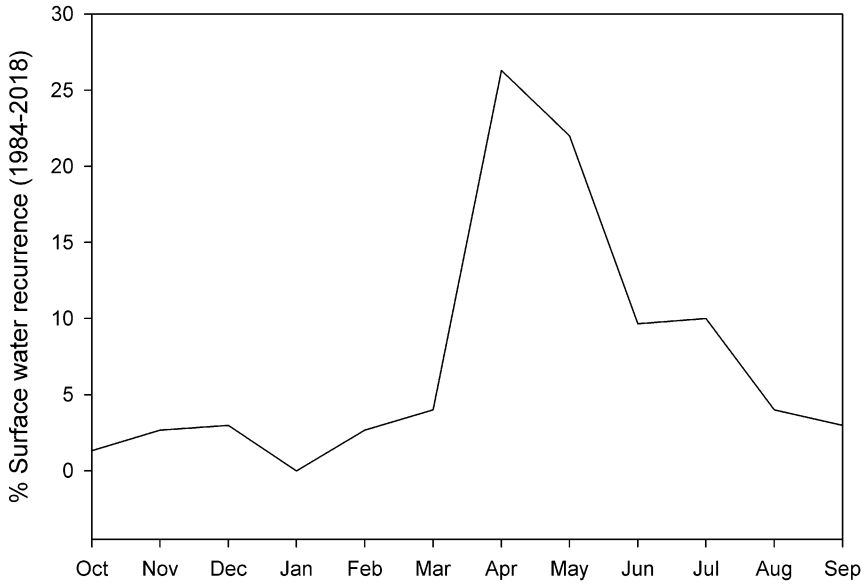


Fig. S5: Monthly % surface water recurrence for Swartkolkvloer, as extracted from the Global Surface Water Explorer dataset(36).

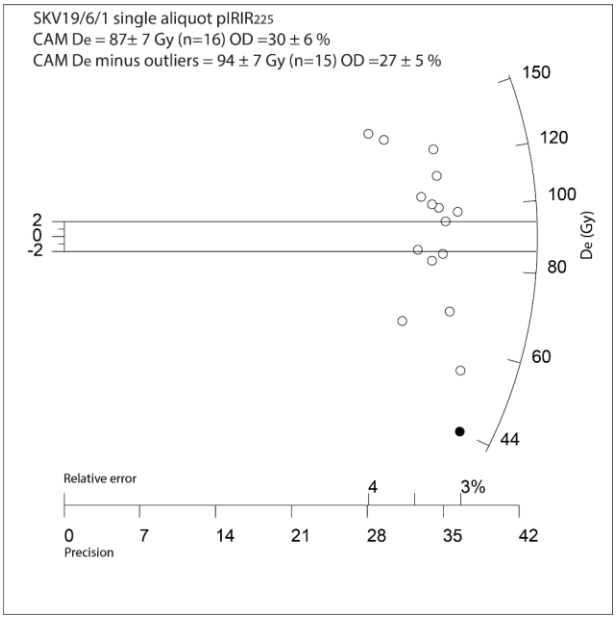


Fig. S6: Radial plot illustrating the equivalent dose distributions for the $pIRIR_{225}$ measurements undertaken for the Swatkolkvloer panfloor sediments (SKV19/6/1). Outlier aliquots are marked by filled circles and where present and the CAM D_es and overdispersion (OD) values are reported without and without the nMAD outlier(s) included.

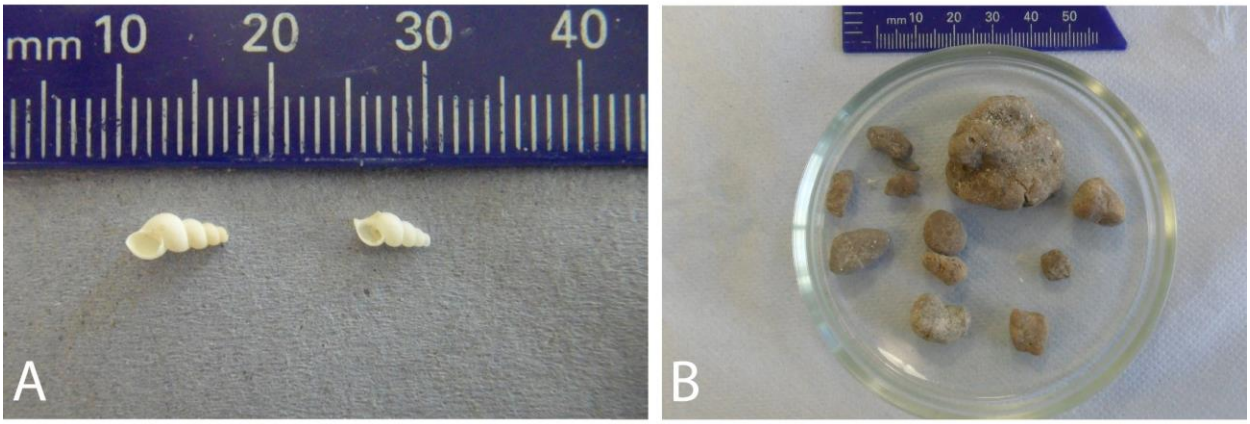


Fig. S7: Intact and cleaned *Tomichia ventricosa* specimens from SKV19/5 (depth 1.15 m) B: rounded gravel clasts from SKV19/5 (depth 1.1 m).

These were described in detail by Kent and Gribnitz (1) and later discussed by Beaumont (2). Beaumont reported four radiocarbon dates from deposits at the site Breek Been Kolk (BBK - the farm dam on the eastern shoreline; Fig. S4 and Table S4), while Kent and Gribnitz (1) provided detailed stratigraphic and sedimentological descriptions of several sites, reporting a single radiocarbon date of unclear affinity (most likely the Breek Been Kolk area; Table S4). The locations these authors described were located and resampled for this study. In addition, sediment fill in the upstream pan Hoek van Spruit se Vloer was cored and sampled to a depth of ~2 m for luminescence dating.

Table S4: Reported radiocarbon dates for Breek Been Kolk obtained by Beaumont (2). These have been calibrated using SHCal20 (39, 40). Depths are estimated from figure 3 of Beaumont (2). Although we cannot be certain to have sampled the same face of his large (~4 x 2 m) excavation, as they were analysed on the same materials as our radiocarbon dates we have included these ages in our site synthesis. The sample on the bottom row of this table, which lacks a laboratory code, is the radiocarbon date reported in Kent and Gribnitz (1). The specific provenance of this sample was not explained in this publication, but the calibrated range is reported here for comparison with Beaumont's ages.

Sample	Material	Depth (m)	^{14}C age	2 sigma lower range (cal yr. BP)	2 sigma upper range (cal yr. BP)	Cal median (cal yr. BP)
Breek Been Kolk						
Pta1760	*T. ventricosa*	0.1	13,900 ± 130	16,370	17,130	16,820
Pta4006	*T. ventricosa*	1.0	24,800 ± 290	28,340	29,840	29,000
Pta4005	*T. ventricosa*	2.0	29,100 ± 590	31,860	34,530	33,380
Pta3958	*T. ventricosa*	2.8	38,300 ± 1700	39,980	44,510	42,320
N/A	*T. ventricosa "consolidated whole shells" **	?	13,670 ± 130	16,100	16,940	16,486

Swartkolkvloer Farm Dam (Breek Been Kolk)

The farm dam (referred to as "Breek Been Kolk" by Beaumont (2) and henceforth BBK) is largely a natural shoreline that forms an impoundment for a channel following into the pan from the east. The crest of this feature lies at 939 m asl, that is, 18 m above the modern pan floor surface. Based on multiple exposures

and boreholes, Kent and Gribnitz (1) described beds of consolidated and loose shells (consisting largely of the aforementioned *Tomichia ventricosa*), which appeared as pod-like lenses interbedded with clays, "shingle" (gravels) and nodular calcretes. The shell beds tended to dip to the east. The stratigraphy across this site is complex and varied, as illustrated by the two sites sampled in the present study. These deposits were described as extending to depths of at least 3 m below the modern crest of the dam. Later, Beaumont (2) excavated at this locale to more than 5 m, recording shelly sand and gravels overlying a basal clay (his figure 3) and reporting four radiocarbon dates (Table S4).

The remains of Beaumont's excavation (henceforth SKV18/4) were identified for us by the landowner. We were able to re-sample the upper 1.4 m of this otherwise largely infilled excavation. The exposed section comprises 0.8 m of calcareous, negatively skewed silty sands (Appendix A; Table S11) with abundant roots and root casts, and occasional granule-sized shale fragments. These grade over the depth range ~0.8 m to 1.2 to increasingly shell-rich sands, with a sharp contact to well-cemented shelly, finer grained (negatively skewed) silty sands at 1.3 m (base of the exposure). *Tomichia ventricosa* shells were scattered but present across the depth range sampled. We obtained radiocarbon dates of 42,030-42,540 cal yr BP and 42,920-49,000 cal yr BP for specimens sampled at 0.23 and 0.9 m (Fig. S8). Two samples were obtained for luminescence dating. The upper sample (SKV18/4/2) exhibited considerable inter-aliquot scatter for both the IR50 and $pIRIR_{225}$ measurements (Fig. S9), particularly compared to the subjacent SKV18/4/1. Given that the latter was obtained from bedded, clearly in situ and partially cemented material, whereas SKV18/4/2 was obtained from unconsolidated sands, within which infilled roots and root casts were observed, we assume that the inter-aliquot scatter for SKV18/4/2 most likely reflects post-depositional disturbance, or perhaps mixing/disturbance during the Beaumont excavations in the 1980s. This sample was excluded from further analyses. The lower sample (SKV18/4/1), obtained

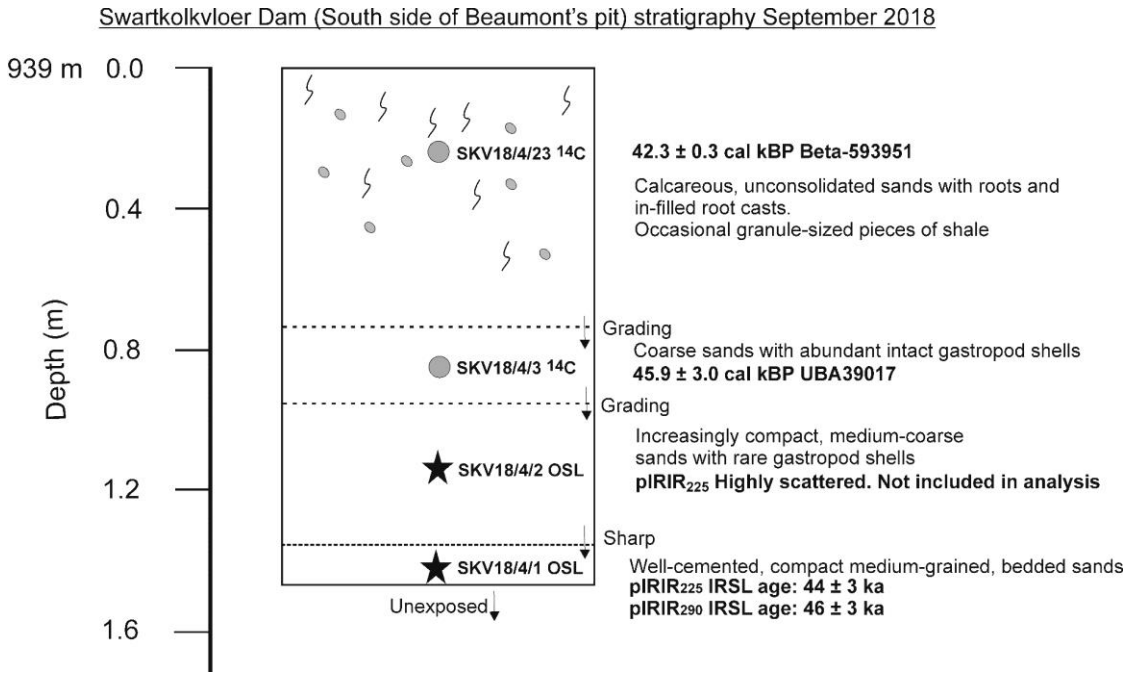


Fig. S8: Stratigraphy and radiocarbon / luminescence dating results for the upper section of Beaumont's (2) (deeper by now partly infilled) excavations at Breek Been Kolk.

from the well-cemented basal sands exhibited far less inter-aliquot scatter and produced (fading-corrected) $pIRIR_{225}$ (44 ± 3 ka) and $pIRIR_{290}$ (46 ± 3 ka) ages in good agreement with each other and the radiocarbon dates (Fig. S8). It is worth noting that the exposed section (1.5 m depth) available in this study is approximately one third of the ~ 5 m excavated by Beaumont (1985; his figure 3). The ages presented here therefore probably represent the final stages of shoreline formation at this location.

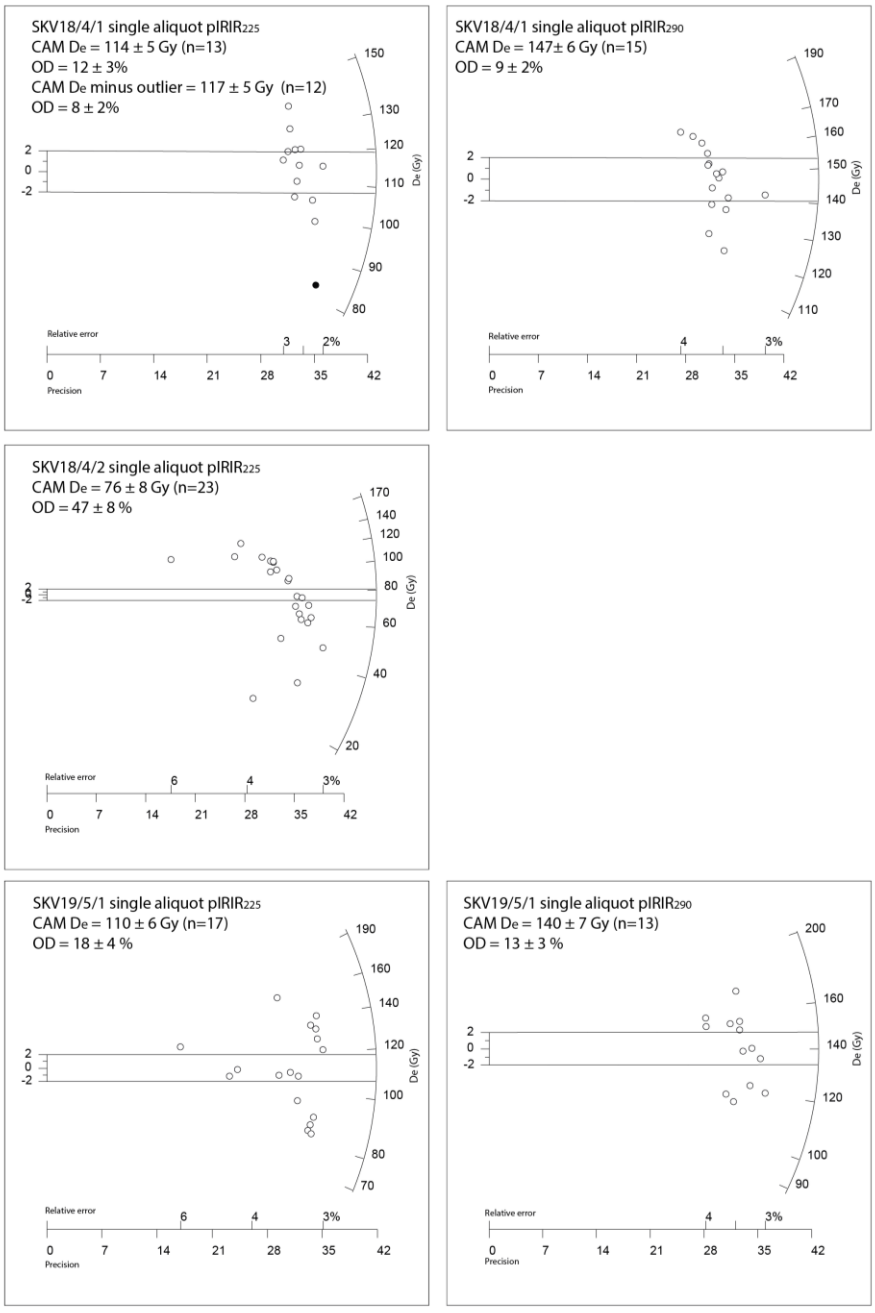


Fig. S9: Radial plots illustrating the equivalent dose distributions for the $pIRIR_{225}$ and $pIRIR_{290}$ measurements undertaken from SKV18/4 and SKV19/5. Statistical outlier aliquots (defined using nMAD) are marked by filled circles where present and the CAM D_es and associated overdispersion (OD) values are reported both with and without these outlier(s) included.

Approximately 100 m west southwest of Beaumont's site, additional excavations have exposed the upper ~1.25 m of the Breek Been Kolk sequence. Sedimentologically, this site (SKV19/5) reveals a substantially different sequence to SKV18/4, exemplifying the stratigraphic complexity of this locale (1). The sequence exposed here is, however, closely akin to Kent and Gribnitz's (1) description of "Pit 3" and it also provides convincing evidence for the water-lain origin (and indeed the prolonged action of water) of the sediment in the upper parts of the Breek Been Kolk sequences. The upper ~0.75 m variously comprises well-bedded and partially cemented shelly sands and silty sands. However, below 0.75 cm, very coarse, rounded granules grade with depth to well-rounded pebbles (both including gastropod shells), the latter showing weakly-defined bedding that nonetheless clearly dips pan-wards. While the coarse sediment fraction may have in part been originally derived from the catchment to the east (Kent and Gribnitz, 1985) (which may also acount for some rounding of the gravel fraction clasts), given the observed sedimentary structures and the inter-bedded nature of the gravels, sands and shells we infer, like Kent and Gribnitz (1985, p366), that these deposits represent high energy depositional events within the lake itself (i.e. wave-deposited). In this respect it is telling that the Breek been Kolk locale, for which a well-developed shoreline morphology can be observed, has formed at the point on the eastern pan margin with the longest possible westerly or northwesterly fetch (> 11 km).

The two radiocarbon dates from *T. ventricosa* and the associated luminescence age are in good agreement with one another and with the nearby SKV18/4 (Beaumont excavation) ages. The radiocarbon dates increase with depth, while the fading-corrected $pIRIR_{225}$ (44 ± 3 ka) and $pIRIR_{290}$ 48 ± 3 (ka) ages

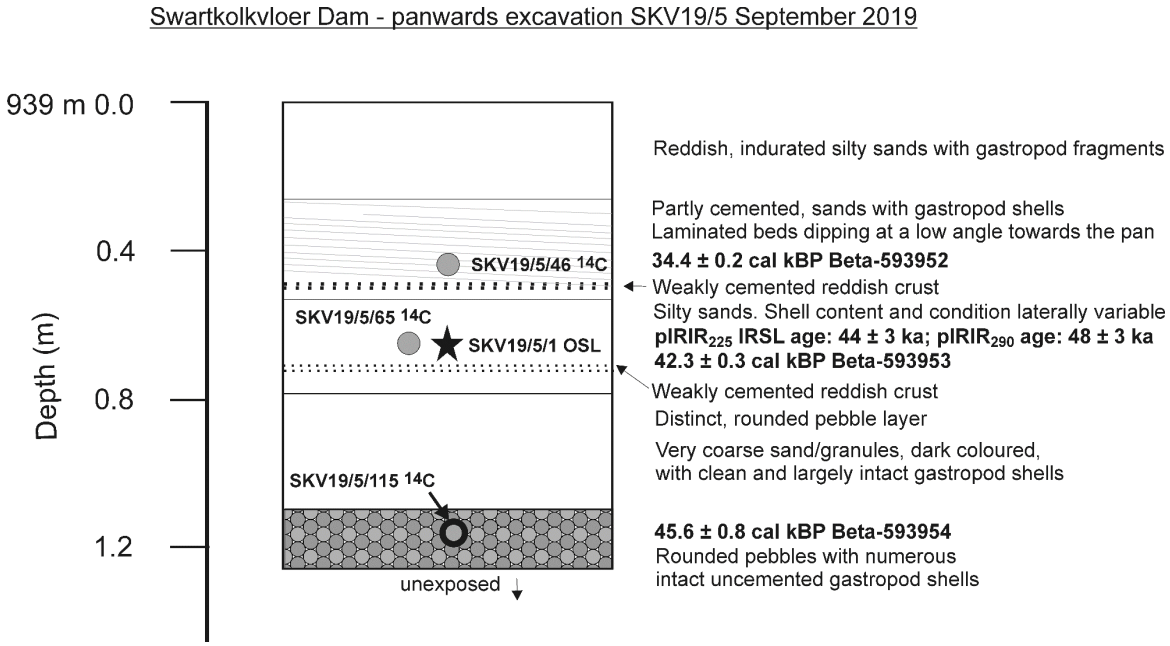


Fig. S10: Stratigraphy and radiocarbon / luminescence dating results for the second Breek Been Kolk excavation west of Beaumont's (2) excavation. The two horizontal dotted lines indicate the depths of the two reddish, pedogenically-modified crusts referred to in the text.

agree, given their uncertainties. The $pIRIR_{290}$ age is somewhat older than the adjacent radiocarbon data (Beta-593953; 42,051-42,575 cal. yr. BP), although they agree within the 2 sigma luminescence age uncertainties. Of note are two reddish crusts at 0.48 and 0.71 m (dotted horizontal lines in Fig. S10), which suggest phases of surface stability and pedogenesis. In the case of the lower-most crust, the timing of its

formation is not resolved by the radiocarbon and luminescence ages. However, for the uppermost crust at 0.48 m there is a marked age difference between the radiocarbon/luminescence ages either side, suggesting a break in deposition between approximately 42 and 34 ka.

Swartkolkvloer Quarry

Substantial and laterally continuous exposures of lacustrine sediments are exposed at the Quarry site, 4.3 km west southwest of Breek Been Kolk (SKV18/2 and SKV18/3; Fig. S4). The Quarry was excavated for agricultural lime in the 1940s, and Kent and Gribnitz (1) noted that in 1945 the site foreman had reported fish and crab remains within these deposits. Since then, large sections of the Quarry, particularly at the southern end, have been in-filled (Fig. S11b). Fresh exposure presents at the northern end where the deposits are cut by a small east-west orientated channel (Fig. S11a). The surface immediately above the Quarry lies at ~934 m, although the wider deposit, distinguished by light coloured outcroppings (Fig. S11a) extends to the south and east, and reaches 937 m. The ~100 m long northeast-southwest orientated quarry cutting lies between two bedrock hillslopes north and south of the site (Fig. S11a). The former was potentially an island during lake high stands (Fig. 2 main text), with the Quarry site forming a (potentially sheltered from the westerly fetch) shoreline between the two high points.

The Quarry deposit is exposed to a depth of ~2 m today. The stratigraphy exposed at the northern end of the site (Figs. S11c and S12) broadly accords with that described by Kent and Gribnitz (1) . The upper 0.1-0.2 m are covered by a veneer of shale clasts, which cover moderately-cemented medium-grained (D_{50} = 2.6 phi) negatively skewed sands (Table S11), with sporadic, but largely fragmentary, *T. ventricosa* shells. The sands are bedded and dip down slope towards the west-northwest. Beneath, and separated by a clear lag or stone line, is a largely marl unit. The stone line above this shows some lateral continuity but cannot be traced across the whole of the quarry exposure. The marls are weakly bedded, with a consistent but rarer (compared to the units above and below) occurrence of *T. ventricosa*. From 0.9 m, thin, more laterally continuous (over scales of metres) lenses of *T. ventricosa* shells are observed. Below 1.26 m a distinct package of finely alternating (centimetre scale) layers of marl and shell-rich sand is observed. The shells at this depth are well preserved and largely intact. From 1.58 m, the sediment grades from a shelly marl, to a pure greyish marl, which continues to 1.98 m. We did not observe the basal contact of this section. Kent and Gribnitz (1) estimated this to lie ~2.5 m below the upper surface. Across the Quarry section as a whole, there is an increased contribution of variably bedded and rounded shale fragments and sands in the upper sediments, with the top of the section covered in angular talus deposits. In detail, the site stratigraphy shows lateral variability, but the section depicted from the northern end of the site is broadly representative. The northern section was sampled for radiocarbon dating (four samples) and OSL dating (three samples). Two additional samples (one radiocarbon, one OSL) were obtained in the upper sediments of the southern part of the quarry (eastern face; SKV18/3) at depths of 1.0 and 1.5 mm respectively (Fig. S11b). We anticipate these ages represent a significant proportion of the site's depositional history, given its likely thickness.

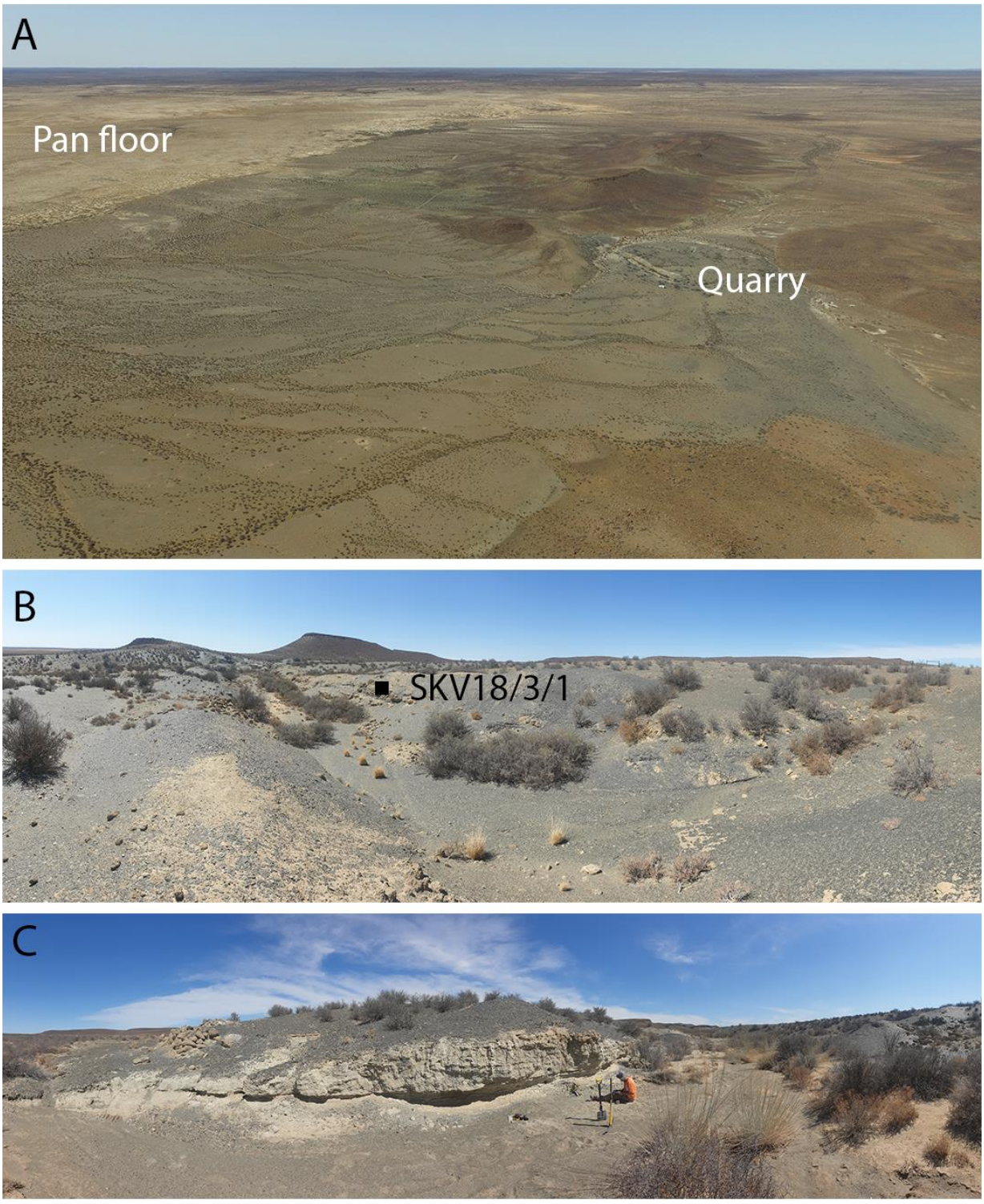


Fig. S11: SKV Quarry site. **A**) Oblique aerial view from a drone, looking approximately northeast. Lacustrine marls appear as greyer-hued sediments in the landscape from this perspective. For scale, a vehicle is visible immediately southeast of the quarry. **B**) view looking north from the southern end of the quarry exposure. This section is less well exposed. The location of OSL sample SKV18/3/1 and radiocarbon sample UBA-39018 is shown with the black square **C**) View south from the northern end of the quarry showing the section (SKV18/2) sampled and described in **Fig S8**.

The four radiocarbon dates range from 42,740-48,450 cal yr BP to > 47,900 14C yr, with their calibrated ranges generally wide and overlapping. All of the fading-corrected $pIRIR_{225}$ and the $pIRIR_{290}$ ages for the lower samples (SKV19/2/10 and SKV19/2/11) lend support to an antiquity of ~50 ka (Tables S5, S13, Fig. S12). The uppermost $pIRIR_{290}$ age (67 ± 5 ka) is however markedly in excess of the other age estimates for this site. The fading corrected $pIRIR_{225}$ age of 51 ± 4 ka from the same sample (and indeed the fading corrected IR50 age of 51 ± 5 ka) is however consistent with the underlying luminescence and radiocarbon ages, suggesting the $pIRIR_{290}$ age is affected by incomplete bleaching. The $pIRIR_{290}$ equivalent dose distribution is more overdispersed than the $pIRIR_{225}$ (Fig. S13) and it is noteworthy that the 4 lowest D_e aliquots (mean 136 Gy) produce an age (54 ka) in agreement with the other $pIRIR_{290}$ ages from this section. Both the $pIRIR_{225}$ and the $pIRIR_{290}$ ages for SKV19/2/11 are somewhat younger than the other luminescence ages for this section and the inter-aliquot scatter is relatively high for the $pIRIR_{225}$ measurements, although in each case they are within 2 sigma uncertainties of the ages of the adjacent luminescence samples, and broadly consistent with the radiocarbon dates from this section.

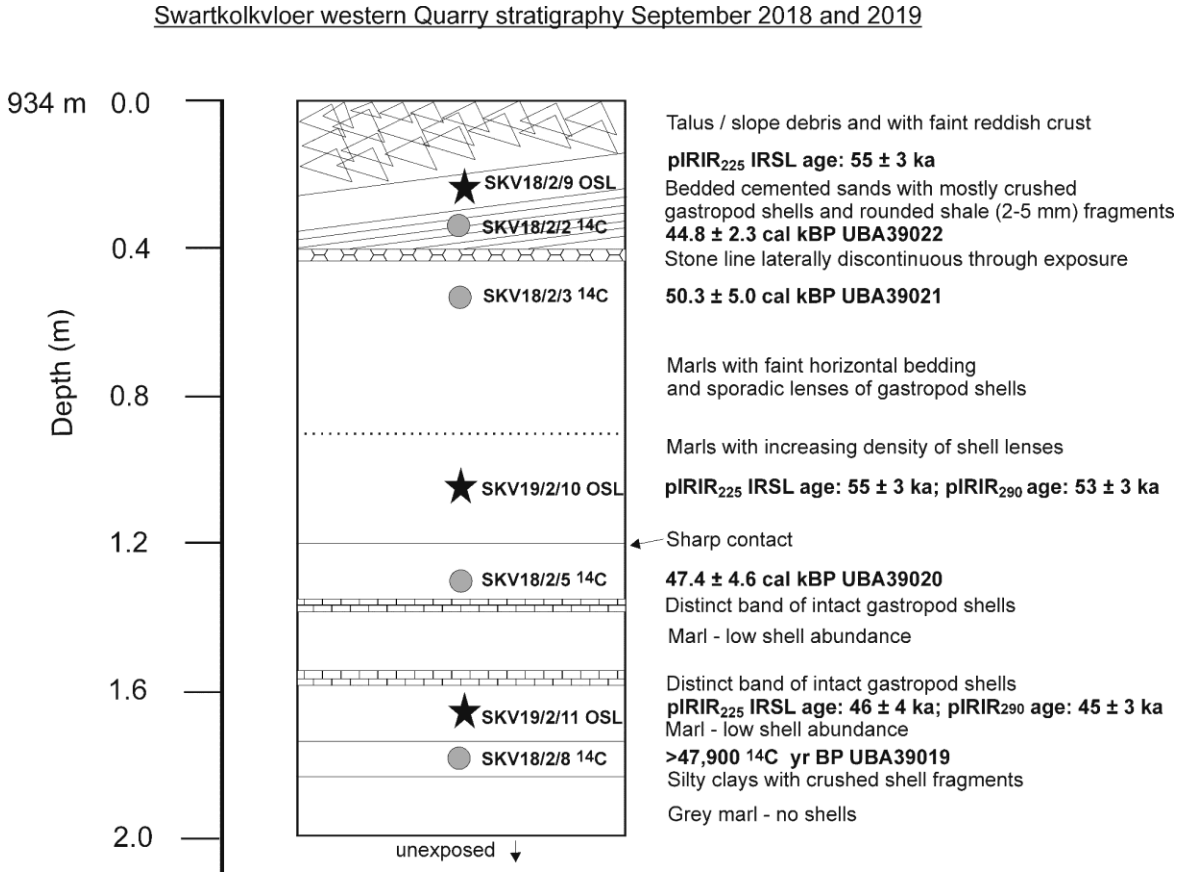


Fig. S12: Stratigraphy and radiocarbon / luminescence dating results for northernmost exposure of at the quarry site.

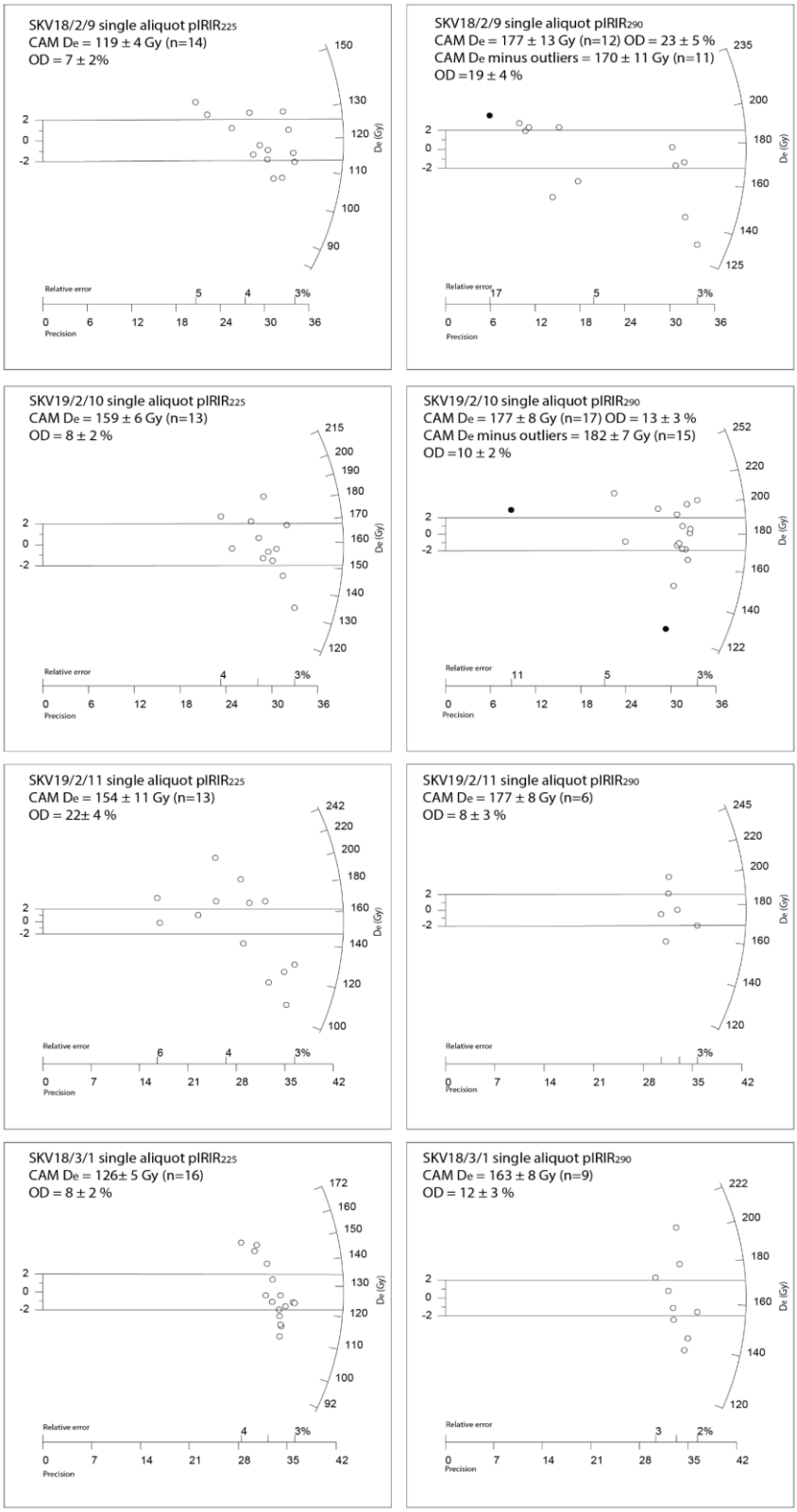


Fig.S13: Radial plots illustrating the equivalent dose distributions for the $pIRIR_{225}$ and $pIRIR_{290}$ measurements undertaken from SKV18/2 and SKV18/3. Statistical outlier aliquots (defined using nMAD) are marked by filled circles where present and the CAM D_es and associated overdispersion (OD) values are reported both with and without these outlier(s) included.

Locality A

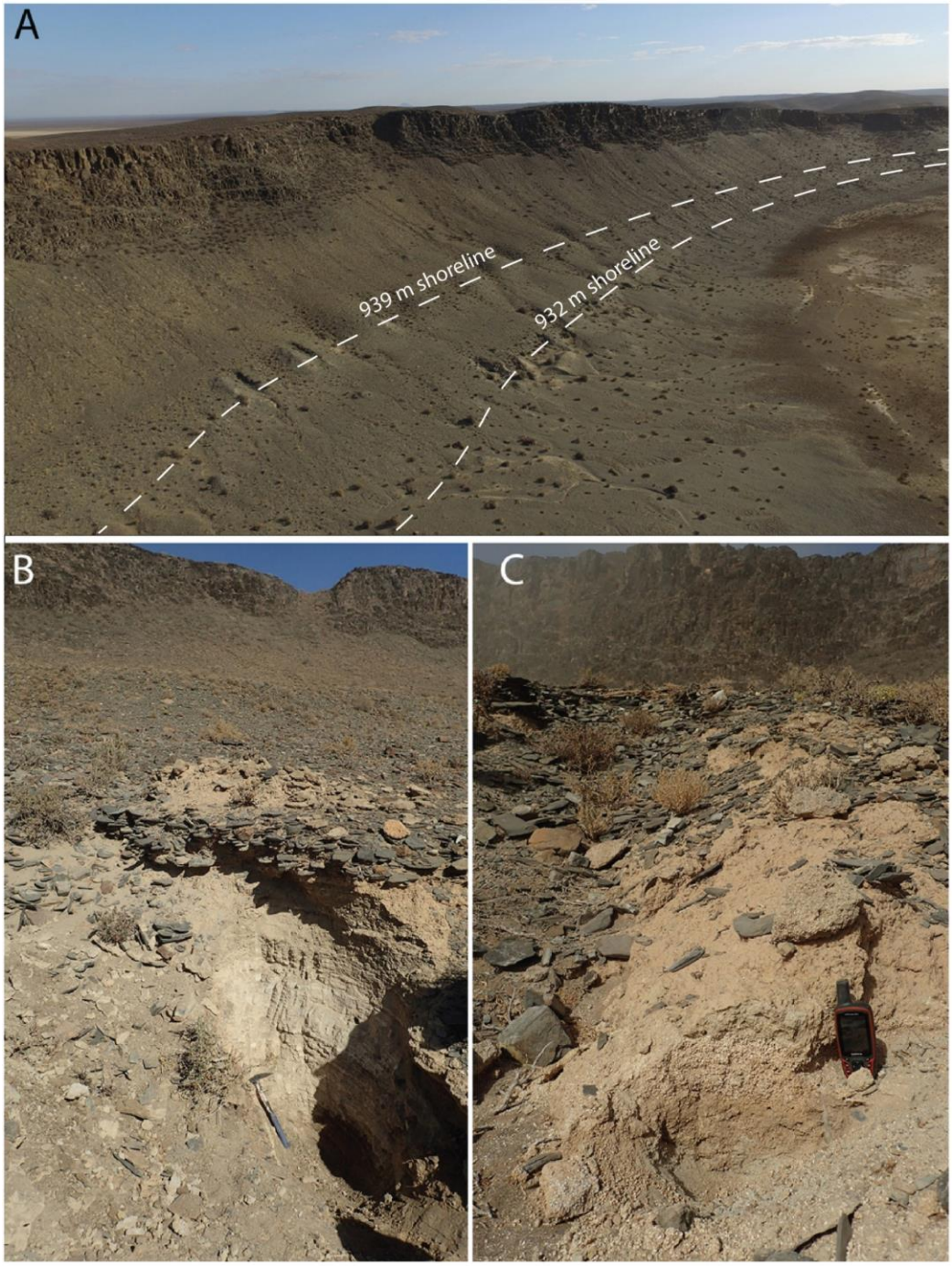


Fig. S14: Locality A imagery. **A)** Oblique aerial view showing the upper and lower deposits, looking approximately west. Note the break of slope at the 939 m shoreline accompanied by significant talus accumulation (see also panel C); **B)** Close up of the cleaned section in the lower shoreline, with the marl component readily apparent (geological hammer for scale); **C)** Close up of the thin veneer of sediment that comprises the upper deposit. The OSL sample from this locale was taken below and to the left of the handheld GPS in this photograph.

The western most site (SKV17/1) corresponds to "Locality A" in Kent and Gribnitz (1). Here two distinct shorelines are deposited upon the steeply sloping bedrock topography that frames the southeastern shore of Swartkolkvloer. These lie at 939 m asl and 932 m asl (Figs. S14 and S15 and main text figure 1). The upper shoreline (939 m) (Fig. S14) is a thin (< 0.5 m) deposit, but is laterally continuous over more than 500 m. It is partially buried by shale clasts worked downslope from the surrounding slopes. It comprises sandy silts, with some gypsum crystals and laterally discontinuous bands of *T. ventricosa* shells, as well as layers of rounded, water-rolled shale gravels. The age of this upper shoreline was constrained with one radiocarbon and one K-feldspar luminescence age (SKV17/1/2). Compared with other samples from Swartkolkvloer the latter showed relatively high inter-aliquot scatter (overdispersion 24 ± 4%) and it was noteworthy that this sample was also markedly less sensitive (average 2 mm aliquot test dose sensitivity of ~35 cts s^{-1} Gy^{-1} compared to ~1000 cts s^{-1} Gy^{-1} for a typical Quarry sample). However, the fading-corrected $pIRIR_{225}$ CAM age estimate (39 ± 3 ka) is in agreement with the accompanying radiocarbon date (40,099-41,741 cal yr BP).

The lower shoreline deposit (upper surface 932 m) is also laterally continuous over ~500 m and presents a thicker sequence than the upper (939 m) shoreline (Fig. S14). Overall, the exposed sequence is ~1.5 m thick and comprises (from the top) weakly bedded silty fine sands (with gypsum crystals), above a unit (at 0.65-1.10 m) of massive, compact marls interbedded with distinct shell layers (coquina). The lowermost unit, below a coquina layer, is rich in sub-rounded shale fragments and dips towards the pan floor (Fig. S16). This closely matches the descriptions provided by Kent and Gribnitz (1). The two radiocarbon ages from this site suggest formation of this lower shoreline between 31,113-31,740 and 33,741-35,219 cal yr BP. This age is consistent with the finding of a single Levallois core (13 x 12 x 5.5 cm) in hornfels directly sitting atop this lower shoreline (Fig. S17). Encountered approximately 70 m west of the lower shoreline sampling section, this Middle Stone Age (MSA) core has a light reddish-brown patina forming on both the platform and flaking surfaces. The technology is non-centripetal lineal Levallois, with the last removal a large irregularly shaped flake that flares distally.

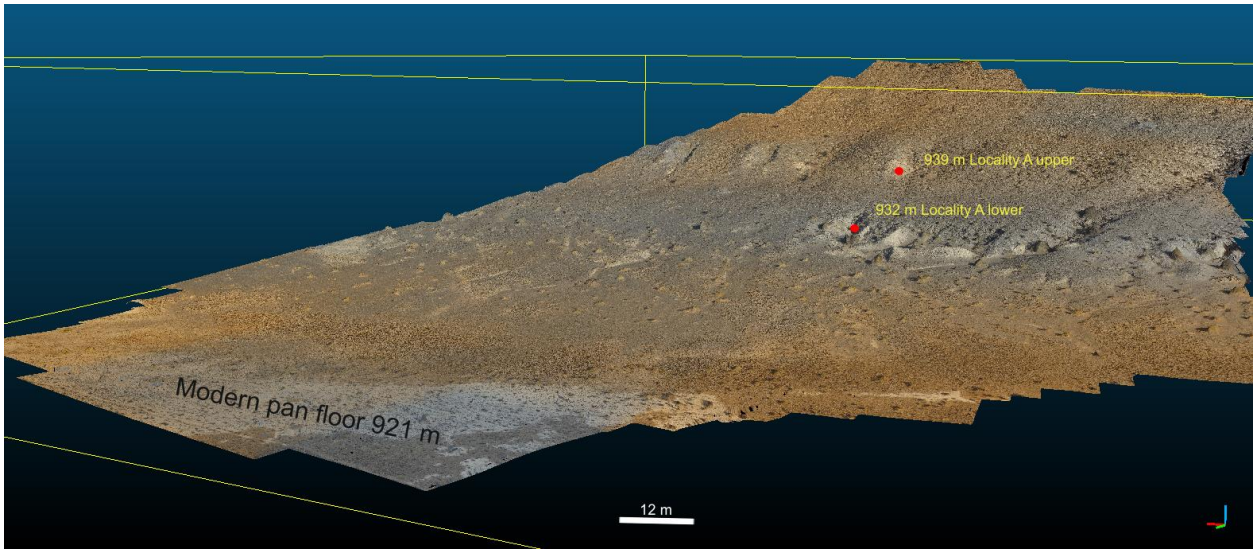


Fig. S15: 3D visualization of Locality A upper and lower shorelines, with sampling locations, as obtained using structure-from-motion photogrammetry.

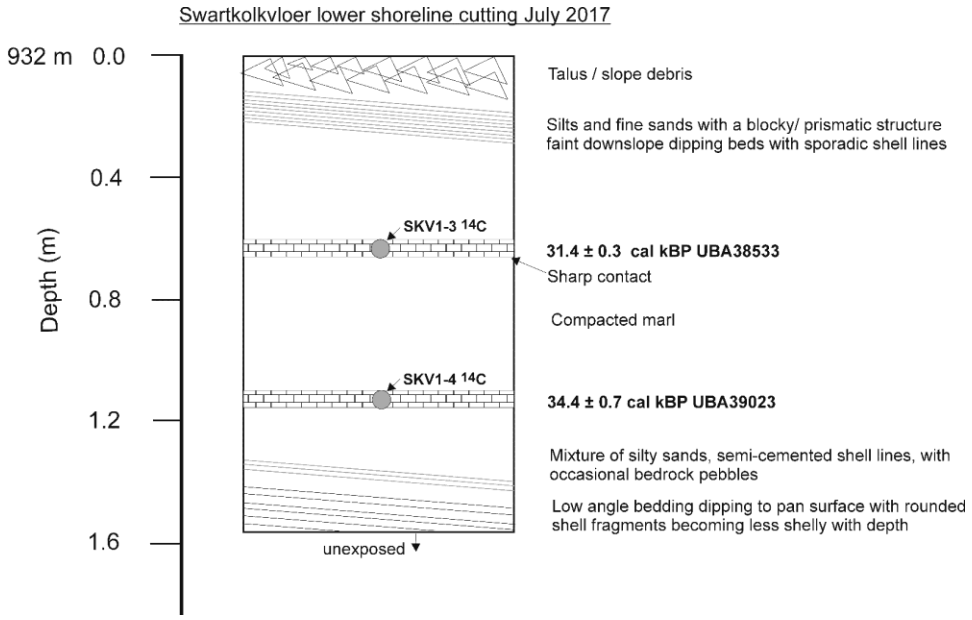


Fig. S16: Stratigraphic diagram with calibrated radiocarbon age ranges for the lower shoreline at Locality A.

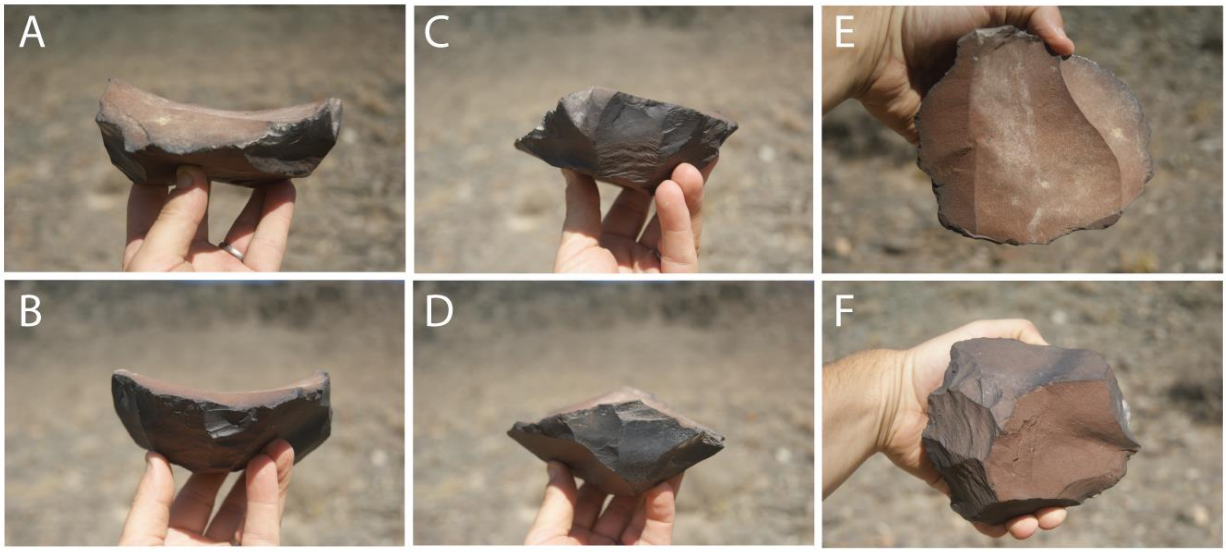


Fig. S17: Side (**A-D**), flaking surface (**E**) and platform surface (**F**) views of the hornfels Levallois core encountered on Swartkolkvloer's lower (934 m) shoreline at Locality A.

Hoek van Spruit se Vloer

To consider the wider evolution of pans in this area we also investigated the highest altitude pan, Hoek van Spruit se Vloer (also marked as "Twee Kop" on some maps) (Fig. S4). We were unable locate shorelines at this site, but sampled the pan sediment fill. Kent and Gribnitz (1) proposed that the higher altitude pan surface at Hoek van Spruit se Vloer (relative to Swartkolkvloer) was the result of recent infilling by sediments from the Fish River. In fact, they argued that when Swartkolkvloer was inundated, water would have flown *from* Swartkolkvloer *to* Hoek van Spruit se Vloer (i.e. that the palaeo-gradient in pan surface

altitudes was the *opposite* of that seen today). This implies the pan fill at Hoek van Spruit se Vloer should postdate the shorelines at Swartkolkvloer. To test this and to more broadly ascertain the timing of pan filling across the region, three luminescence dating samples were obtained via augering. The clastic pan floor sediment (Fig. S18) comprises in excess of 1.9 m of sands and silty sands, with no gastropod specimens or evaporites observed. The absence of the latter is consistent with the lower water tables at this locale (Fig. S52). The upper 0.8 m consists of a poorly sorted clay-silt-fine sand mixture, with the fill becoming increasingly sandy below (peaking at median = 1.9 ϕ at 1.48 m; Appendix A; Table S11). At 1.55 m a layer of rounded to sub-angular pebbles of mixed composition was observed. Three luminescence samples were taken from depths of 0.4, 1.32 and 1.85 m and produced indistinguishable (fading-corrected) $pIRIR_{225}$ ages of 59 ± 2 ka (weighted mean n=3; Table S5), with a $pIRIR_{290}$ age from the basal sample HSV19/3/4 dating to 68 ± 4 ka. These imply 1) a phase of pan filling that may predate shoreline formation in Swartkolkvloer; 2) the modern difference in pan floor altitudes is not a recent phenomenon; 3) a (palaeo) connection to the Fish River. The gravel layer suggests relatively high flow energy events within this phase of pan filling (the ages below and above this layer have overlapping uncertainties).

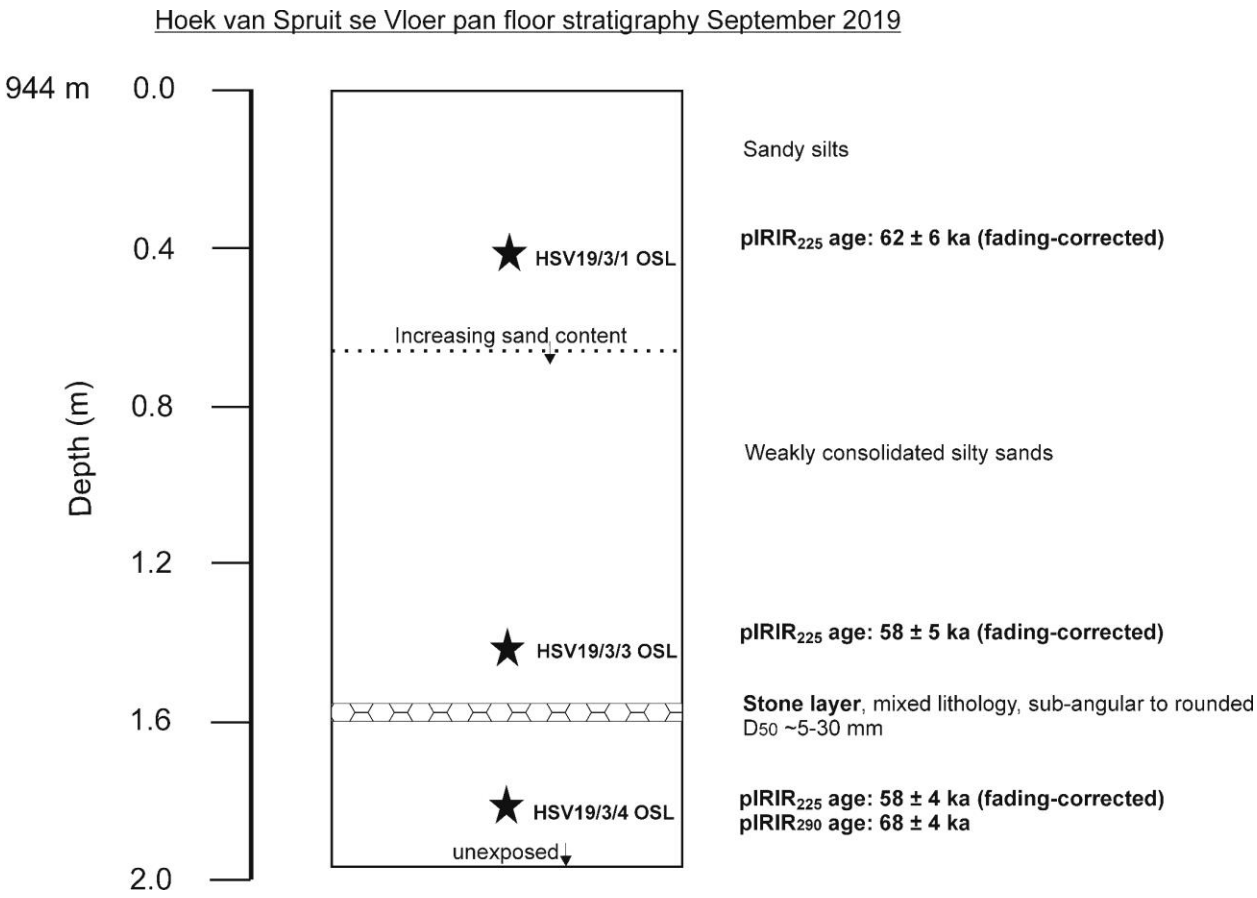


Fig. S18: Stratigraphy and dating for the Hoek van Spruit se Vloer clastic pan fill



Fig. S19: The two inselberg 'islands' punctuating the central pan floor of Hoek van Spruit se Vloer. The nearer (northern) inselberg was investigated for archaeological materials.

Table S5: Details of the quartz and K-feldspar ages for the Swartkolkvloer sites. Details of the dose rate calculations and the equivalent dose measurements and fading rate measurements are presented in Appendix A; Table S12. Ages in bold were used for the final analyses in this paper (e.g. Main text Figure 2). The quartz age for SKV18/4/1 is not considered reliable.

Sample	Dose rate K Feldspar (Gy ka^{-1})	IR50 age (ka)	$IRIR_{225}$ age (ka)	$pIRIR_{225}$ fading corrected (ka)	$pIRIR_{290}$ age (ka)	Quartz dose rate (Gy ka^{-1})	Quartz Age (ka)
Swartkolkvloer Quarry							
SKV18/2/9	2.54 ± 0.09	37 ± 2	47 ± 2	**55 ± 3**	67 ± 5	nd	nd
SKV19/2/10	3.41 ± 0.13	33 ± 2	47 ± 3	55 ± 3	**53 ± 3**	nd	nd
SKV19/2/11	3.99 ± 0.15	27 ± 2	39 ± 3	46 ± 4	**45 ± 3**	nd	nd
SKV18/3/1	3.09 ± 0.12	24 ± 2	41 ± 2	48 ± 3	**53 ± 3**	nd	nd
Swartkolkvloer dam (Breek Been Kolk)							
SKV18/4/1	3.09 ± 0.11	26 ± 2	38 ± 2	44 ± 3	**46 ± 3**	2.24 ± 0.10	33 ± 3
SKV19/5/1	2.95 ± 0.11	27 ± 2	38 ± 2	44 ± 3	**48 ± 3**	nd	nd
Locality A							
SKV17/1/2	4.25 ± 0.15	29 ± 2	32 ± 2	**39 ± 3**	nd.	nd	nd
Swartkolkvloer pan floor							
SKV19/6/1	6.33 ± 0.38	10 ± 1	15 ± 1	**17 ± 2**	nd.	nd	nd
Hoek van Spruit se Vloer pan floor							
HSV19/3/1	3.49 ± 0.13	39 ± 2	53 ± 4	**62 ± 6**	nd.	nd	nd
HSV19/3/3	4.02 ± 0.15	36 ± 2	50 ± 3	**58 ± 5**	nd.	nd	nd
HSV19/3/4	4.56 ± 0.17	37 ± 2	49 ± 3	**58 ± 4**	68 ± 4	nd	nd

Across the wider pan floor at Hoek van Spruit se Vloer we observed extremely high densities of MSA and Later Stone Age (LSA) lithic scatters of hornfels, intermingled with a pan-floor clast/lithic 'pavement' of small (<10 cm) hornfels sub-angular rocks that rings one of two inselberg 'islands' in the central parts of the pan (Figs. S19 and S20a,b). Here approximately 10% of the observed clasts were culturally modified and it is likely that groups were exploiting this source of hornfels ringing the inselberg(s). The hornfels

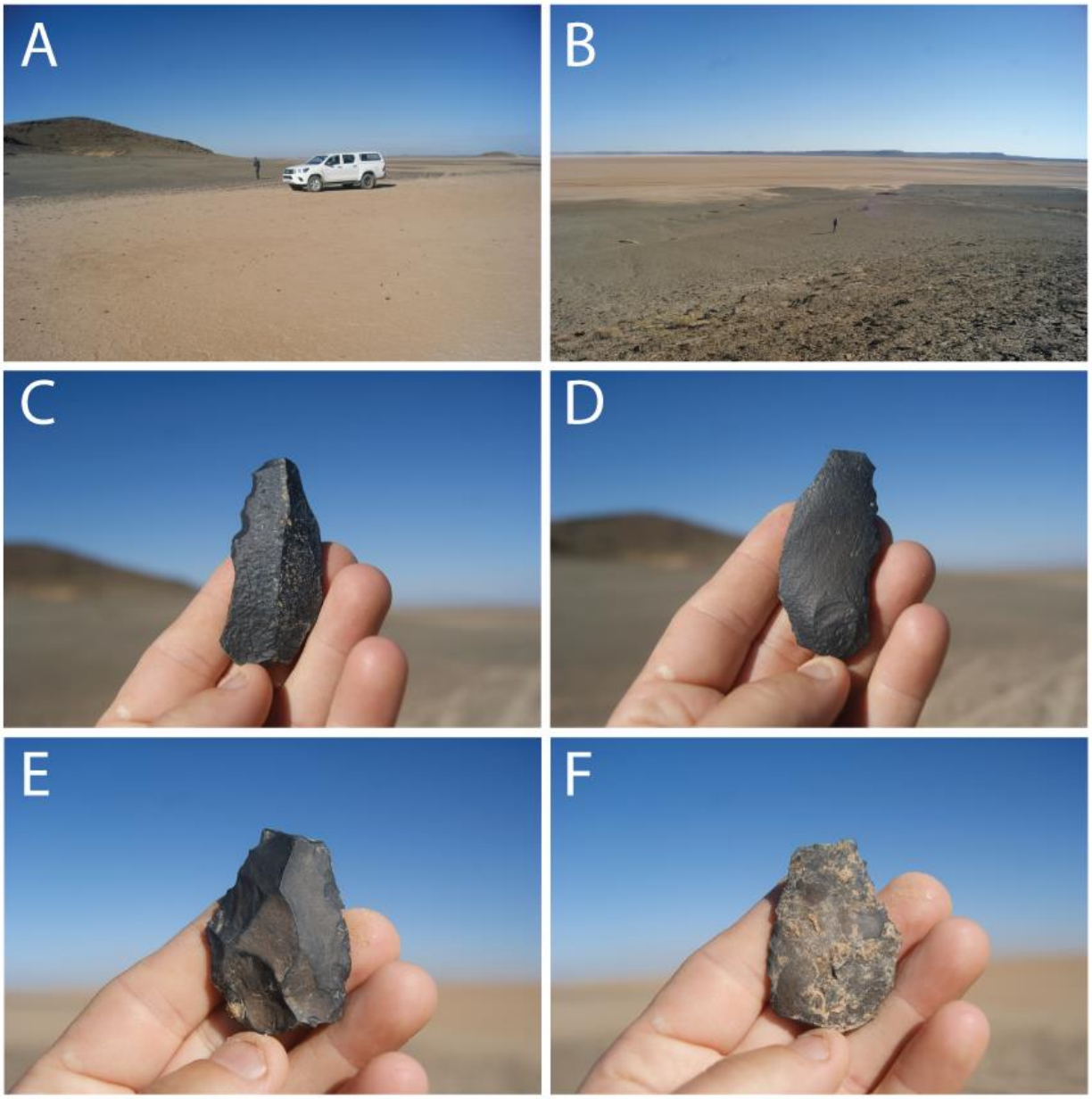


Fig. S20: Views looking south towards (**A**) and north from (**B)** Hoek van Spruit se Vloer's northern inselberg 'island', each showing the charcoal grey hornfels 'pavement' ringing the landform. The ventifacted and edge-rounded component of the high-density lithic scatter there included retouched triangular flakes produced using Levallois reduction (**C-D, E-F**).

grade from black to lighter charcoal grey. One large dark-caramel coloured quartzite core with a cortical back was noted, which is very different from anything else seen. Artifact forms are extremely diverse and mostly comprise large unretouched flakes, but numerous tools were noted. Despite the exposed location and regular lifting of dust from this pan surface, most artifacts are minimally ventifacted and patinated, and possess sharp edges and ridges, suggesting relatively recent deposition or exposure. The formal tools of this fresh component are dominated by large, "D-shaped" side-scrapers that typify southernmost Africa's early Holocene Oakhurst technocomplex, known in the interior Karoo as the Lockshoek (41). However, a second component exhibits more lustrous surfaces, including those of retouched flake scars,

and slightly more rounded edges and ridges, suggesting a greater antiquity. This inference is supported by the observation that the ventifacted component includes small side-scrapers and elongated convergent flakes with Levallois dorsal scarring and sporadic, marginal retouch (e.g. Fig. 20 c, d). Such pieces are typo-technologically similar to those that typify the "later" (smaller) variant of Sampson's (1968, 1972, 1985) Orangian Industry, the middle of three techno-cultural phases he encountered and defined in his Orange River Scheme and Zeekoe Valley research areas (between the preceding Florisian and subsequent Zeekeogat). Sampson hypothesized that the entire Orangian was deposited during warmer sub-stages of the Last Interglacials (MIS 5e, c and a), but thermoluminescence dates from Driekoppen, the Zeekoe valley's only stratified MSA site, suggest the later Orangian falls within MIS 3 (Wallsmith 1994).

Swartkolkvloer stable isotope analysis

T. *ventricosa* shells from across the Swartkolkvloer sites exhibit considerable variability in $\delta^{13}C$ (range 6.6 ‰) and $\delta^{18}O$ (range 8.6 ‰) (Fig. S21). In light of this, and the wide environmental tolerances of this species, as well as the relatively limited number of specimens (c.f. 42) and lack of modern comparative data, we interpret these data cautiously. Notwithstanding, for the total dataset there is a positive correlation between the two isotopes (Pearson's r = 0.74 p = < 0.001). This covariance, and variability, is consistent with a closed lake basin environment (e.g. 43).

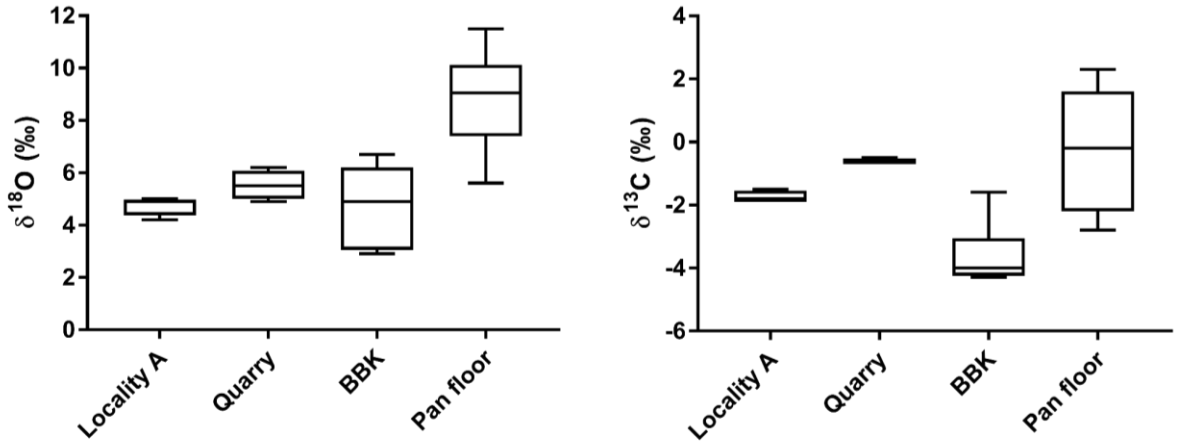


Fig. S21: Box and whisker plots of the $\delta^{13}C/\delta^{18}O$ analysis of the *T. ventricosa* specimens sampled at the various Swartkolkvloer sampling sites. For $\delta^{18}O$ we observe statistically significant differences between sampling locales (Kruskall-Wallis test statistic = 13.9, p = 0.003). Dunn's Multiple Comparisons indicate that this reflects differences between the Pan Floor samples and the Locality A/BBK. For $\delta^{13}C$ we observe statistically significant differences between sampling locales (Kruskall-Wallis test statistic = 16.4, p = 0.0009). Dunn's multiple comparisons indicate this reflects differences between the Quarry samples and the Pan floor /BBK.

Considered by individual sampling locality, some patterns are apparent. The greatest variability and the highest delta values for both isotopes are seen in specimens obtained from the Swartkolkvloer pan floor sediments (the aforementioned layer at 0.45-0.50 m depth). Given the variability in the radiocarbon and luminescence dating results for these sediments (Tables S5 and S10), this variability may reflect real differences in sample age (and environmental conditions), although the strong covariation between the two isotopes (Pearson's r = 0.84 p = < 0.01) and the high absolute $\delta^{18}O$ values (Fig. S22) in

particular are consistent with an evaporative basin, perhaps implying that at least some of these specimens were deposited during a low lake phase. Samples associated with the other three shoreline deposits (Breek Been Kolk, The Quarry and Locality A) exhibit significantly lower $\delta^{18}O$ values than the pan floor specimens, consistent with a less evaporative environment, and with their reported Mg/Ca ratios (1). The Locality A lower shoreline $\delta^{18}O$ values and those of BBK and the Quarry are not significantly different from one another (Fig. 21).

The variation in $\delta^{13}C$ is likely driven by variability in the $\delta^{13}C$ of dissolved inorganic carbon (DIC), and this will respond to exchange of CO_2 with the atmosphere and lake productivity. $\delta^{13}C$ also shows considerable variability, although as with $\delta^{18}O$, much of this variability reflects the substantial variation seen in the pan floor specimens. Individual sites such as Locality A and the Quarry show much more limited $\delta^{13}C$ variability (sample standard deviations < 0.2 ‰ are not much greater than the analytical uncertainty). In this respect it is interesting that the lowest $\delta^{13}C$ /$\delta^{18}O$ values are associated with the well-developed shoreline at Breek Been Kolk, and that the $\delta^{13}C$ values for individual sites tend to be more clustered than they are for $\delta^{18}O$. For low $\delta^{13}C$ specimens we infer reduced exchange with atmospheric CO_2, coupled with reduced evaporation/residence times or higher precipitation, as also implied by the accompanying low $\delta^{18}O$ values (e.g. 42, 43, 44). Overall, the data indicate potential variability in climatic/lake status conditions. Some of this variability may correspond to micro-habitat variation; e.g. exposed Breek Been Kolk shoreline vs. the sheltered environment implied in the northern Quarry context. The data are generally consistent with a closed lake basin, and a shallower water body during the formation of the lower shoreline at locality A (increased $\delta^{13}C$ relative to BBK).

Swartkolkvloer- Hoek van Spruit se Vloer summary:

The radiocarbon dates and luminescence ages for all of the Swartkolkvloer sites are summarized in Fig. S23.

The evidence obtained from the Swartkolkvloer system is interpreted as follows:

1. The age ranges obtained from the radiocarbon dating and post IR-IRSL dating of the Breek Been Kolk and the Quarry sites, indicate the deposition of lacustrine marls (Quarry) and rolled gravels (Breek Been Kolk) at 936 - 939 m during a lake phase broadly spanning 55±3 to 39±3 ka (Fig. S23). Excluding infinite ages, the radiocarbon ages are consistent with the luminescence ages, but the proximity of these samples to the limits of the method should be noted. Samples from the well-developed 939 m shoreline at Locality A of 39±3 ka and 41±0.8 cal kBP provide evidence for a maximum lake high stand at this time, while not precluding the possibility of other previous lake phases of this depth. The ages from Locality A are relatively young compared to the Breek Been Kolk and Quarry dates, despite an identical site altitude (939 m). This most likely implies this volumetrically less substantial deposit is associated with the final phase of deposition of the 939 m shoreline (preservation potential for older deposits would be much less on this steeply sloping surface). However, the rather insensitive K-feldspar and higher inter-aliquot scatter for this sample contributes uncertainty to the luminescence age.

2. Radiocarbon dating of one *T. ventricosa* shell obtained within the pan fill sediments is broadly consistent with the Quarry and Breek Been Kolk ages (43.1-44.6 cal kBP). The other analysed shell was radiocarbon infinite. Interestingly, the IRSL age from the directly associated sediment (despite its inter-aliquot scatter) may be markedly younger. This probably suggests a mixed or lagged shell deposit in the pan floor, buried within / by younger pan fill sediments, consistent with observed the inter-shell variability in $\delta^{13}C$ /$\delta^{18}O$.

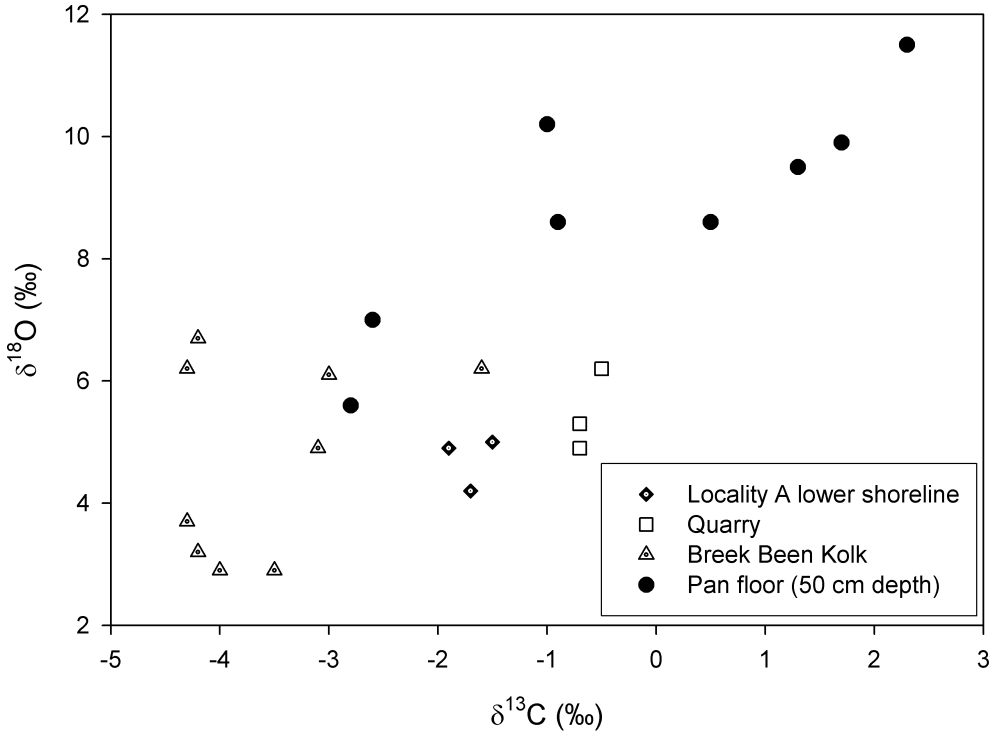


Fig. S22: Summary of whole shell $\delta^{13}C$ and $\delta 18O$ analysis of *T. ventricosa* specimens from Swartkolkvloer. The samples have been separated in sampling locales, including the lower shoreline at Locality A, Breek Been Kolk, the Quarry and the pan floor.

3. The radiocarbon dates from the Locality A lower (932 m) shoreline span 34.4±0.7 to 31.4±0.3 cal kBP, and indicate a distinct, younger, phase of inundation. The presence of gypsum suggests a more brackish lake system, although the stable isotope data do not suggest a markedly more evaporative environment than the older phase(s) of inundation (Fig. S22). These ages closely match the uppermost (0.46 m) *T. ventricosa* radiocarbon date from Breek Been Kolk (SKV19/5). Although the latter is located at 939 m, it lies stratigraphically above evidence for pedogenic alteration, and is inferred to represent a veneer of the aeolian transported sands / shells (see also 1). The implied timing for this phase of pedogenesis at SKV19/5 sequence is consistent with the age difference between the Locality A lower shoreline and the Quarry / BBK / Locality A 939 m deposits.
4. The CAM-derived luminescence age (17 ± 2 ka) from the pan floor is consistent with the 16.4-17.1 cal kBP and 16.1-16.9 cal kBP radiocarbon dates reported by Kent and Gribnitz (1) and Beaumont (2)(respectively), suggesting inundation during Heinrich stadial 1. Overdispersion is however high, however, implying a need for caution.

The stable isotope data from *T. ventricosa* specimens from across Swartkolkvloer indicate a closed-lake basin (43) with potentially more evaporative conditions associated with the pan floor specimens. The lowest $\delta^{13}C$ /$\delta^{18}O$ values are observed for specimens from the thickest shoreline deposits (including rolled gravels) at BBK and further support the inference that inundation during the 55 to 39 ka phase was the most sustained.

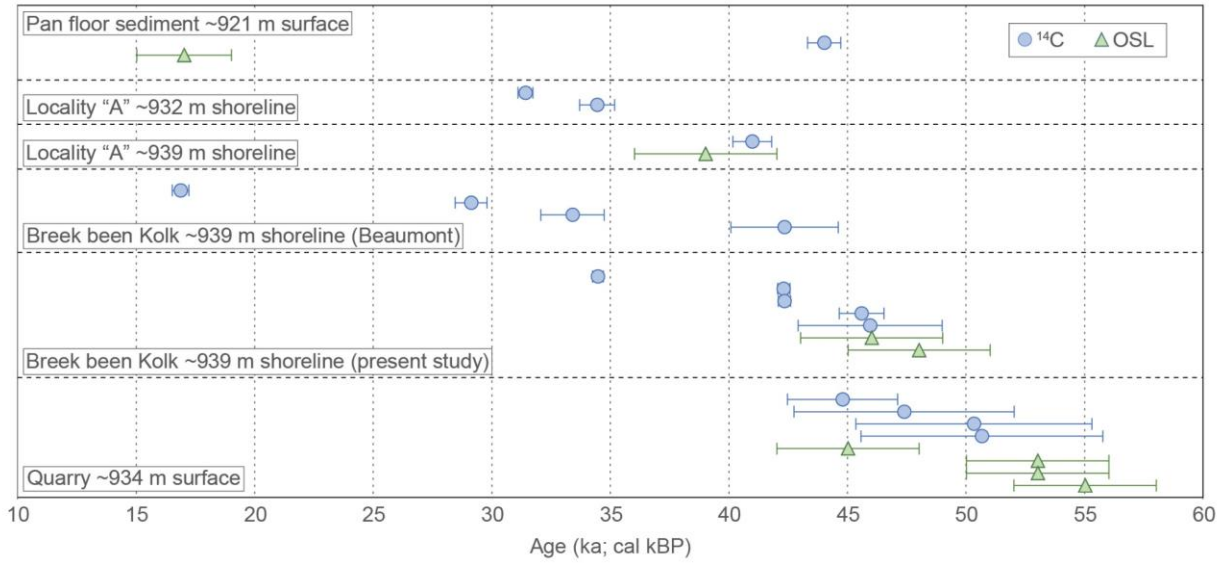


Fig. S23: Summary of the geochronological data from Swartkolkvloer, including the radiocarbon dates from Breek been Kolk reported by Beaumont (2).

Grootvloer (Hoezer Oost and Wes)

Grootvloer is an extensive (~275 km^2) pan connected to the Sak River by a broad opening on its western side (Fig.S4, S24, S25). A secondary channel (Swartbosleegte) to the southwest also connects to the pan, the catchment of which is ~220 km^2. The pan floor lies between 874 and 875 m asl. Inundation is rarer than at Swartkolkvloer; indeed, locales close to the sampling sites record no inundation for the period 1984-2019 (36). Localized ephemeral inundations occurred further north on the western margins of the pan twice during this period (2000, 2013), in April and May.

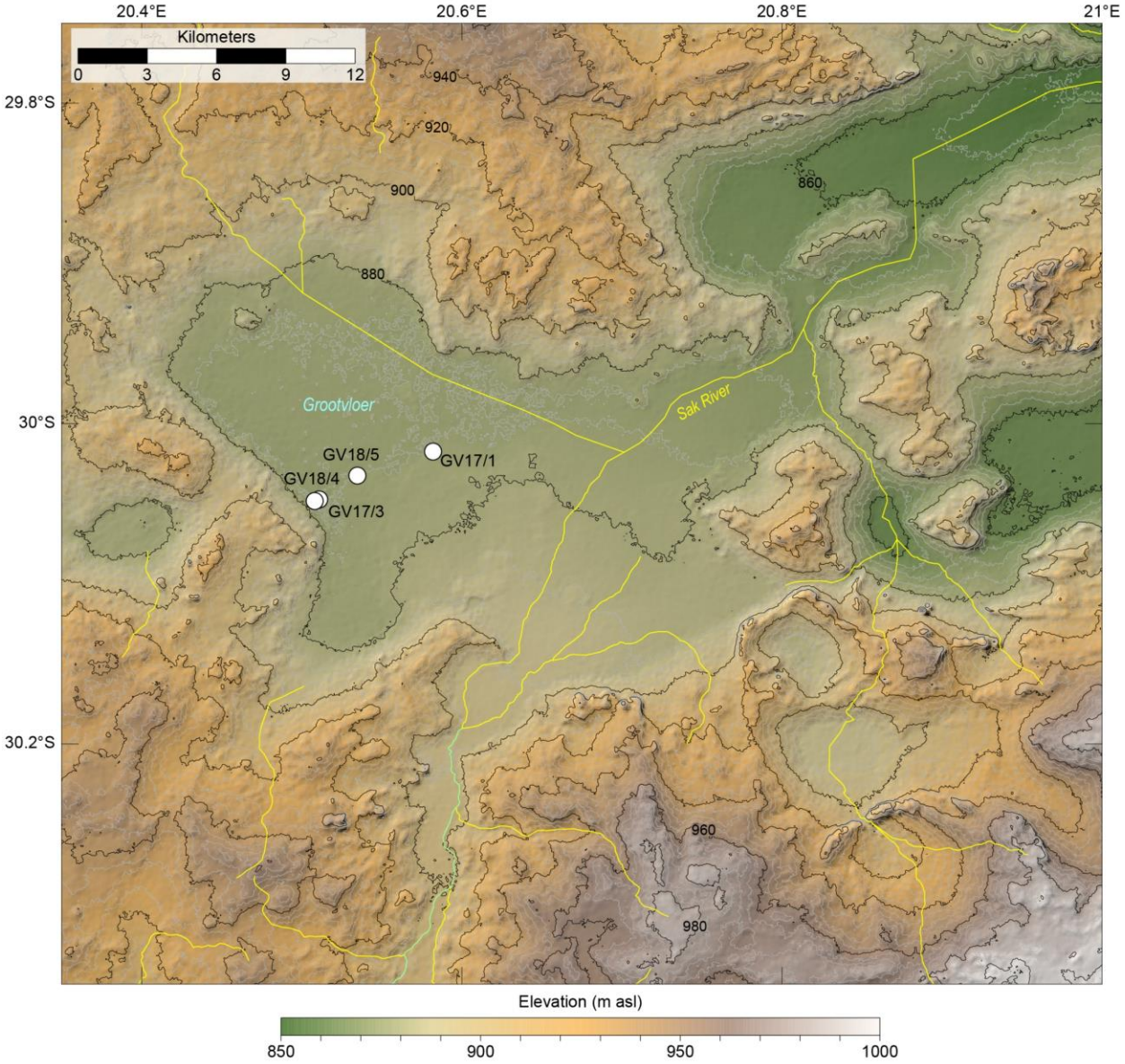


Fig. S24: Map of Grootvloer region. Sampling locations and 939 m and 932 m lake levels are shown. Primary dry (yellow) and non-perennial (green) stream systems are shown. No perennial streams currently exist in the region. Topography from the Multi-Error-Removed Improved-Terrain DEM (38).

Beaumont (2) referred to a “2 m high bar” at Hoezer Oost as a possible shoreline feature (Fig. S25). Several such features are readily identifiable on the southeastern to southern margins of the pan from satellite imagery. These track the apparent pan margin (although this is marked more by vegetation change than land surface height change) and are orientated approximately NE-SW with a curved plan form. There is some variability, including a section with a narrower and steeper cross-sectional form orientated NW-SE (~110°), which is more akin to the modern resultant wind direction. All are sparsely vegetated and subject to obvious contemporary aeolian activity, with many small nebkha dunes forming on the ridges today. Excavation into the ridge at Hoezer Oost, which we infer to be the bar that Beaumont (2) refers to, revealed at least 1.1 m of homogenous and structure-less, medium to coarse-grained sands, with occasional granule layers. A granule-sized lag also sporadically covers the surface of these features.

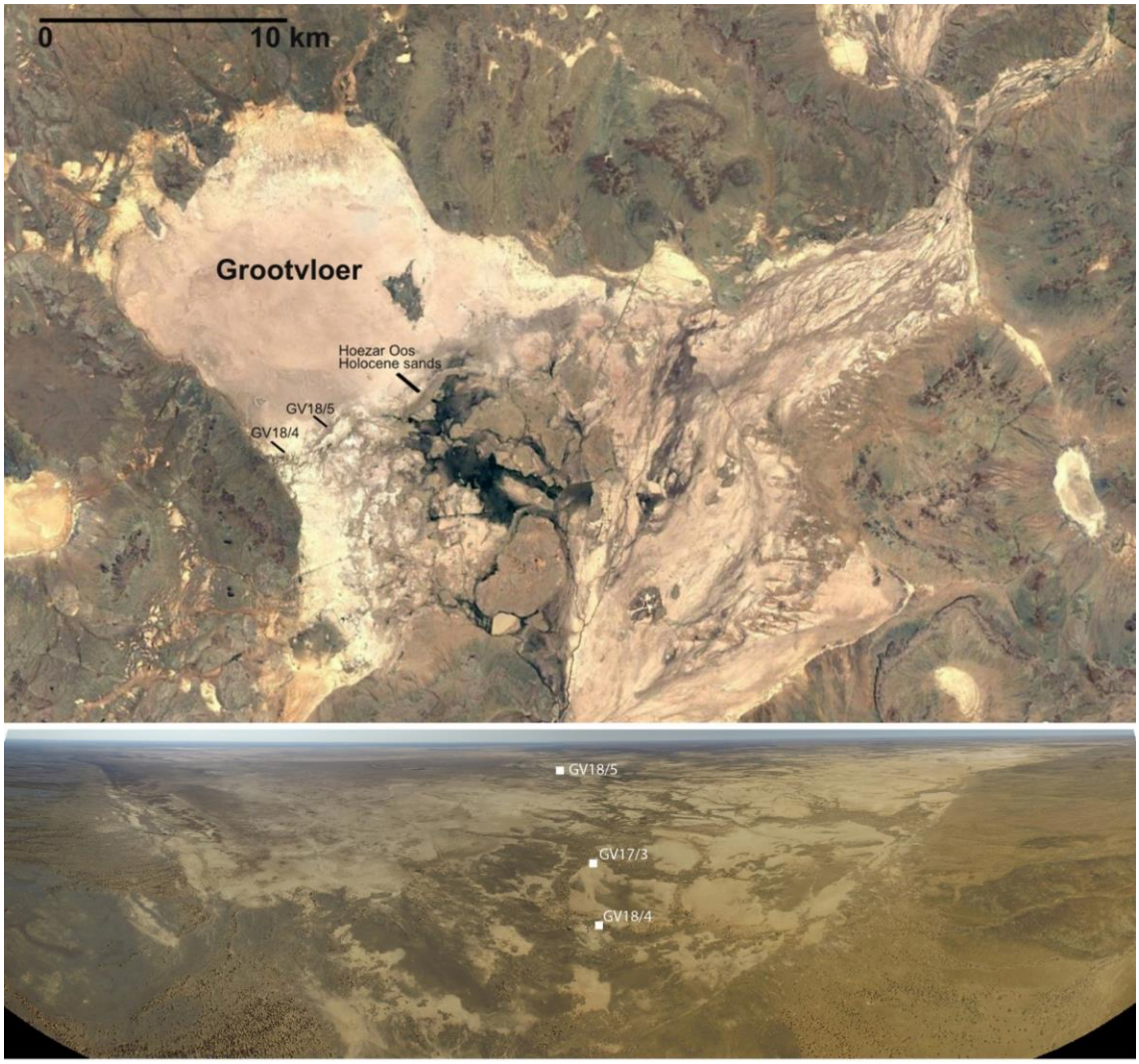


Fig. S25: **Upper**: Wider view of Grootvloer and flowing towards the northeast, the Sak River. The sandy bar referred to in Beaumont (2) at Hoezar Oos is marked (GV17/1 - for which we obtained an OSL age of 5.5 ± 0.4 ka), along with the locations of GV18/4 and GV18/5, which provided ages for LGM pan infilling. **Lower**: oblique aerial image looking to the north from the southern margin of Grootvloer. This shows sites GV18/4 (adjacent to the vehicles, which are just visible to the left of the white square) and GV17/3. The cored sand ridge GV18/5 is visible in the distance. The sands themselves, as at Hoezar Oos were found to be Holocene in age, with the underlying pan floor sediments dating to MIS 2.

A single aliquot quartz OSL age (0.9 m depth) indicated a mid-Holocene antiquity (GV17/1; 5.5 ± 0.4 ka; Table S6), and we interpret these as essentially aeolian features, or at least features significantly (re)worked by aeolian activity in their upper sections.



Figure S26: Exposures of in-situ exposure of the freshwater bivalve *Unio caffer* at GV18/4.

No shoreline deposits were identified at Grootvloer, but from a palaeohydrological perspective the presence of deposits of the freshwater bivalve *Unio caffer* Krauss, 1848 (Fig. S26) at Hoezer Wes is highly significant (see also 2). This species is associated with perennial freshwater and is reliant on fish to disperse its parasitic larvae (45). Today, it is absent from western parts of the Northern Cape Province (46). In the southernmost corner of Grootvloer several lines of this freshwater bivalve are exposed within the uppermost pan floor sediments (Fig. S26). The absence of shorelines to accompany this evidence (at least, so far identified) is inferred to reflect some combination of the low gradient landscape (limiting formation; e.g. 47) and the presence of the modern Sak River at the likely downwind margin of any lake system (limiting preservation). Aeolian reworking (see above) may also have served to limit preservation potential.

Coring at this locale (hence forth GV18/4) revealed at least 1.7 m of underlying clastic sediments, initially comprising silty fine sands, then increasingly silt-clay rich below 1.7 m (Fig. S27). A gravel lag was identified at ~1 m, which included clasts of reworked and rounded calcrete. The total depth of sediment fill in Grootvloer may be substantial, and Beaumont (2) alludes to more than 30 m of fill (evidence for this is not presented). At GV18/4 a radiocarbon date from a specimen of *Unio caffer* produced a calibrated age range of 20,510-20,890 cal yr. BP. This age is supported by coherent $pIRIR_{225}$ (fading-corrected) and $pIRIR_{290}$ ages (19 ± 1 ka and 20 ± 1 ka respectively) from the subjacent sediment fill (0.65 m). An additional

pan fill sample (GV17/3) 300 m northeast of the shell site (depth of 0.48 m), showed more inter-aliquot scatter, but produced a (fading-corrected) $pIRIR_{225}$ age of 24 ± 2 ka. The *U. caffer* radiocarbon date compares well to a date from fragments of the same species reported by Beaumont (2) from an unspecified location at Hoezer Oost (Pta3894: 17,800 ± 190 ^{14}C yr. BP; 20,970-22,050 cal yr. BP).

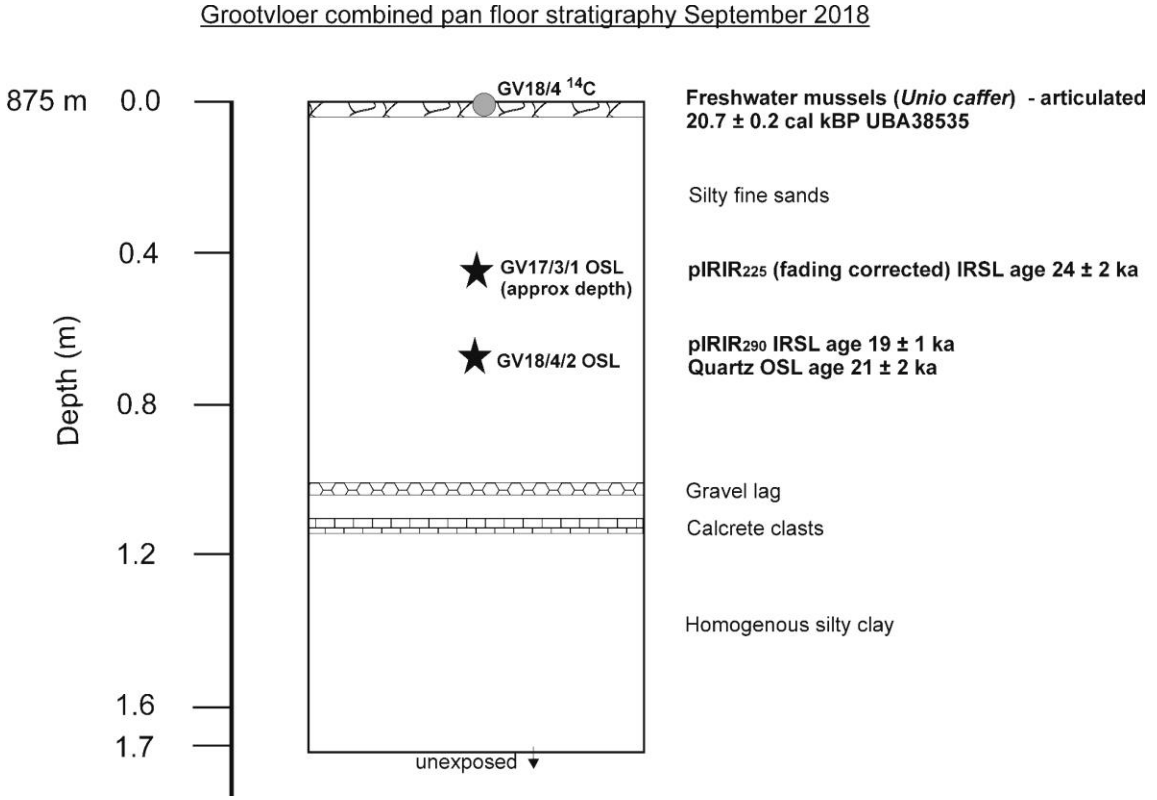


Fig. S27: Stratigraphic diagram for GV18/4. The approximate depth of the sample GV17/3/1, which was taken ~300 to the northeast is shown.

To the northeast, another example of the aforementioned (NE-SW orientated) sandy ridges (GV18/5; Fig. S25) was sampled using a dormer sand augering system. This comprises ~1.3 m of reddish coarse to very coarse sands, grading with depth to silty sands and then to silty clays (60% clay). These finer grained basal sediments, which prevented further coring, are texturally identical (~15% sand, 30% silt, 60% clay) to the pan floor sediments at GV18/4 and GV17/3, suggesting the full depth of this sandy ridge was sampled. The coarse sands are late Holocene in age (1.1 ± 0.1 ka; Table S6; Fig. S29), with these surficial sediments in many locations seemingly wind-worked under modern conditions. Consistent with this is a high-density (~30 pieces/m^2) scatter of late Holocene LSA artifacts was observed eroding from an in-situ horizon immediately beneath the apex of this ridge. This material includes fragments of mammalian bone, ostrich eggshell (one with incised decorations), flaked *Unio caffer* shell, sherds of mineral-tempered pottery, and hornfels bladelets and small end scrapers, all in extremely fresh condition. The silty and increasingly fine-grained (Table S11) sediments at the base of the GV18/5 core produced luminescence ages identical to the pan fill sediments at GV18/4, confirming the MIS 2 age of this pan fill sediment ($pIRIR_{290}$ age of 21 ± 1 ka for GV18/5/3 (Figs. S28 and S29)).

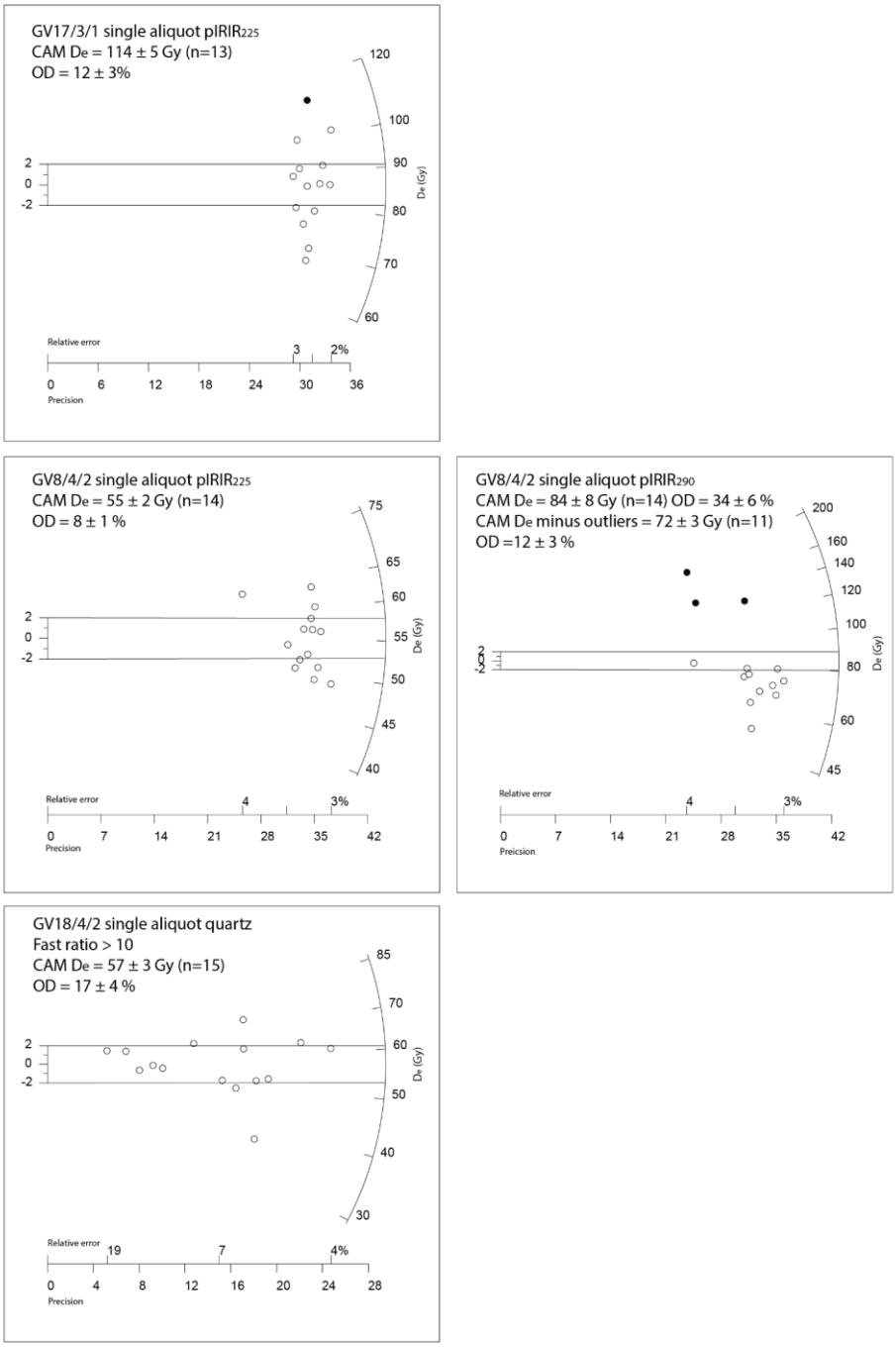


Fig. S28: Radial plots illustrating the equivalent dose distributions for the $pIRIR_{225}$ and $pIRIR_{290}$ and quartz (only aliquots with a fast ratio > 10 from a total of 32 measured) measurements undertaken from GV18/4/2 and the $pIRIR_{225}$ data obtained from GV17/3/1. Statistical outlier aliquots (defined using nMAD) are marked by filled circles where present and the CAM D_es and associated overdispersion (OD) values are reported both with and without these outlier(s) included.

Overall, the available data from Grootvloer indicate that at least on the southern margins of the pan, the upper-most pan fill corresponds to the last glacial maximum, and that deposition of this clastic fill was associated with at least one phase of sustained perennial freshwater that allowed colonization of the site by *Unio caffer*. In this light, it is significant that a bladelet-rich archaeological scatter was encountered on the surface of the pan's southeastern extremity. Located on deflated flats approximately half-way between GV18/5 and Hoezar Oos (GV17/1), this is a low, (~5 pieces/m^2) grading to medium-density (~20 pieces/m^2) mixed LSA scatter dominated by Lockshoek elements in hornfels, but with a smaller, conspicuous component of bladelets, bladelet cores and bipolar cores in diverse, high-quality raw materials (Fig. S30). These include a caramel-coloured and a banded brownish cryptocrystalline silica (CCS) and a hydrothermal chalcedony-like material with a white patina, as well as vein quartz. No formal tools in these materials were encountered. Hammerstones and upper and lower grindstones also occur, one of the latter with substantial pecking damage consistent with bipolar percussion (Fig. S30h). Ostrich eggshell fragments were also abundant, most of which were reasonably fresh, but with fewer exhibiting a light greyish ivory discolouration (Fig. S30d). These more weathered fragments were frequently encountered in and near bladelet-rich areas. While the lack of diagnostic forms demands caution, the informal character of this high-quality, bladelet-rich component, taken alongside its association with degraded ostrich eggshell, evidence for bipolar knapping, and the LGM ages for the substrate on which they sit, together point to an early Robberg archaeological occurrence.

Table S6: Details of the quartz and K-feldspar ages for the Grootvloer sites. Details of the dose rate calculations and the equivalent dose measurements and fading rate measurements are presented in Appendix A; Table S12. Ages in bold were used for the final analyses in this paper.

Sample	Dose rate K Feldspar (Gy ka^{-1})	IR50 age (ka)	$IRIR_{225}$ age (ka)	$pIRIR_{225}$ fading corrected (ka)	$pIRIR_{290}$ age (ka)	Quartz dose rate (Gy ka^{-1})	Quartz Age (ka)
Grootvloer – sand ridge (Hoezar Oos)							
GV17/1/1	nd.	nd.	nd.	nd.	nd.	2.24 ± 0.09	**5.5 ± 0.4**
Grootvloer – pan floor sediment fill (Hoezar Wes)							
GV18/4/2	4.14 ± 0.21	14 ± 1	20 ± 1	**24 ± 2**	nd.	nd.	nd.
GV17/3/1	3.62 ± 0.14	12 ± 1	15 ± 1	19 ± 1	**20 ± 1**	2.77 ± 0.12	**21 ± 2**
Grootvloer – aeolian sand ridge and subjacent pan floor sediment fill (Hoezar Wes)							
GV18/5/1	3.42 ± 0.13	0.8 ± 0.04	1.1 ± 0.1	**1.2 ± 0.1**	nd.	nd.	nd.
GV18/5/2	2.17 ± 0.08	11 ± 1	17 ± 1	**20 ± 2**	nd.	nd.	nd.
GV18/5/3	4.29 ± 0.17	13 ± 1	20 ± 1	24 ± 2	**21 ± 1**	nd.	nd.

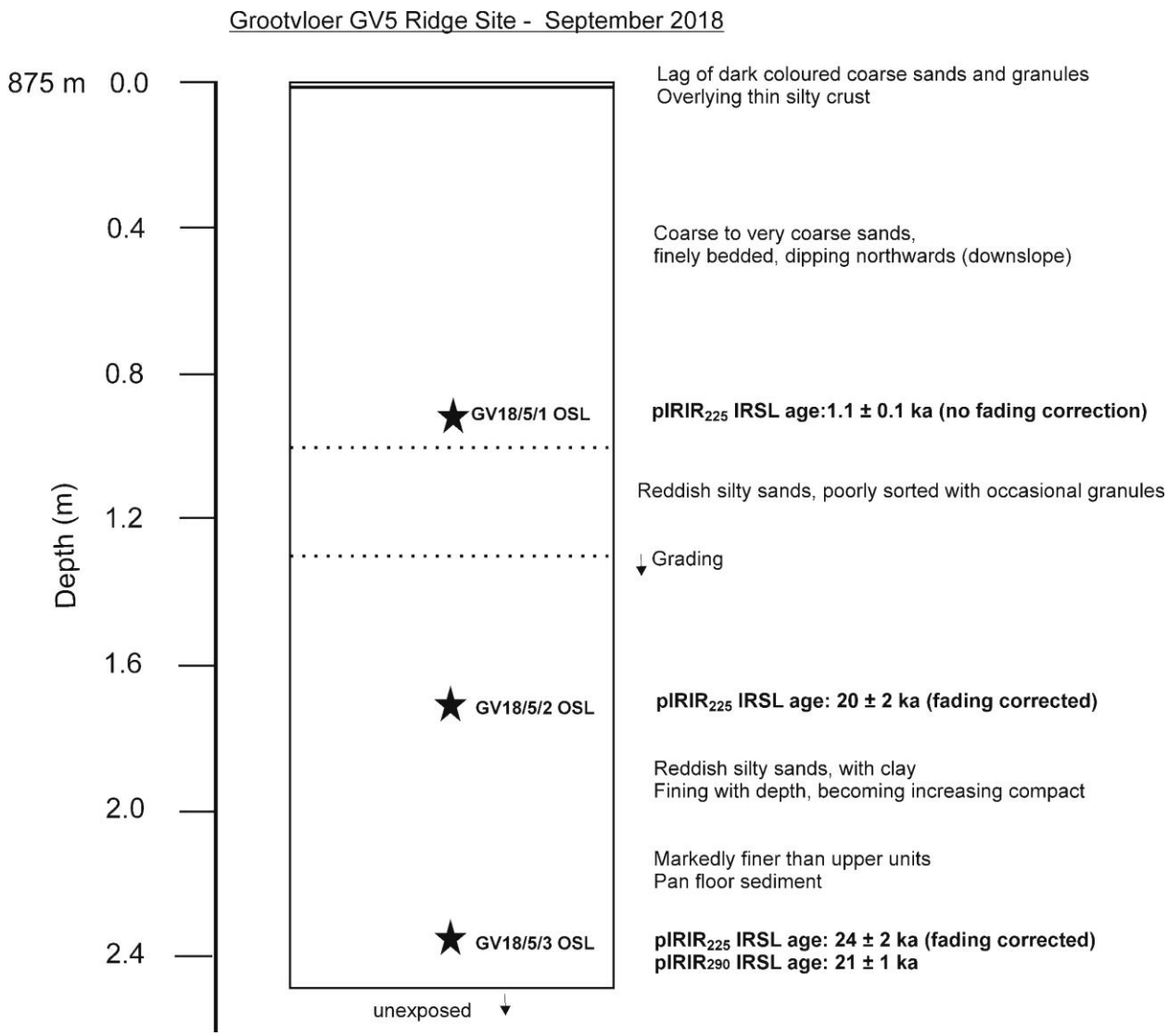


Fig. S29: Summarized stratigraphy and IRSL ages for site GV18/5. The upper sandy sediments are interpreted as largely aeolian in origin and produced a substantially younger age than the underlying, finer grained, pan fill sediments.

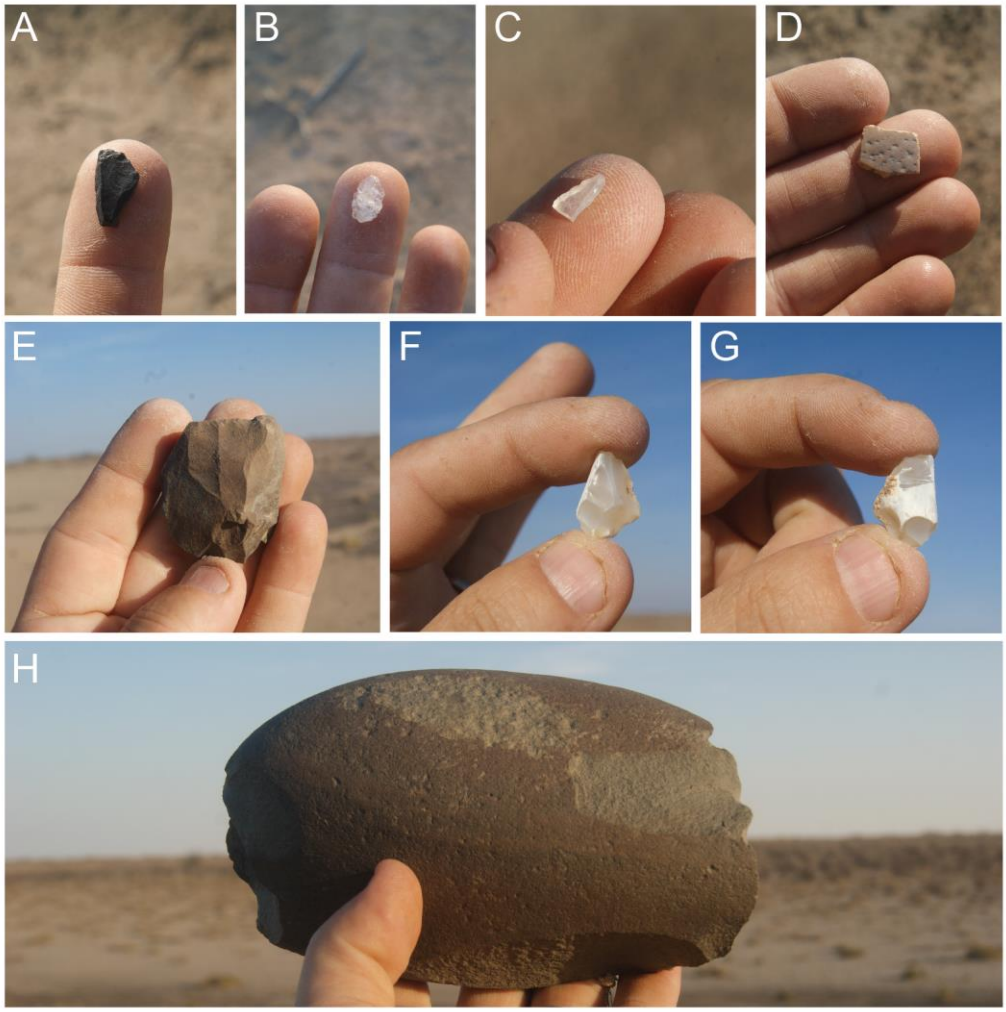


Fig. S30: Examples of artifacts from scatter situated on the deflated southern margin of the LGM-aged pan, Grootvloer, that exhibits numerous characteristics consistent with a Pleistocene microlithic LSA (Robberg) attribution. Finds include bladelet fragments (**A, C**), bladelet cores (**E**), bipolar cores (**F, G**) and products (**B**), weathered ostrich eggshell fragments (**D**), and groundstone with pecking damage likely resulting from bipolar reduction (**H**).

Alexandersfontein

Alexandersfontein pan (Fig. S31), immediately south of Kimberley, was described on several occasions by Karl Butzer and colleagues (3, 48, 49), who argued that the site presented evidence for a ~19 m lake high stand during the late Pleistocene, as well as evidence for subsequent palaeoclimatic changes. The modern pan surface (1145 m asl), which makes up a relatively small proportion of Butzer's proposed palaeo-lake area, experiences seasonal inundation during approximately 30% of years (36). The monthly inundation

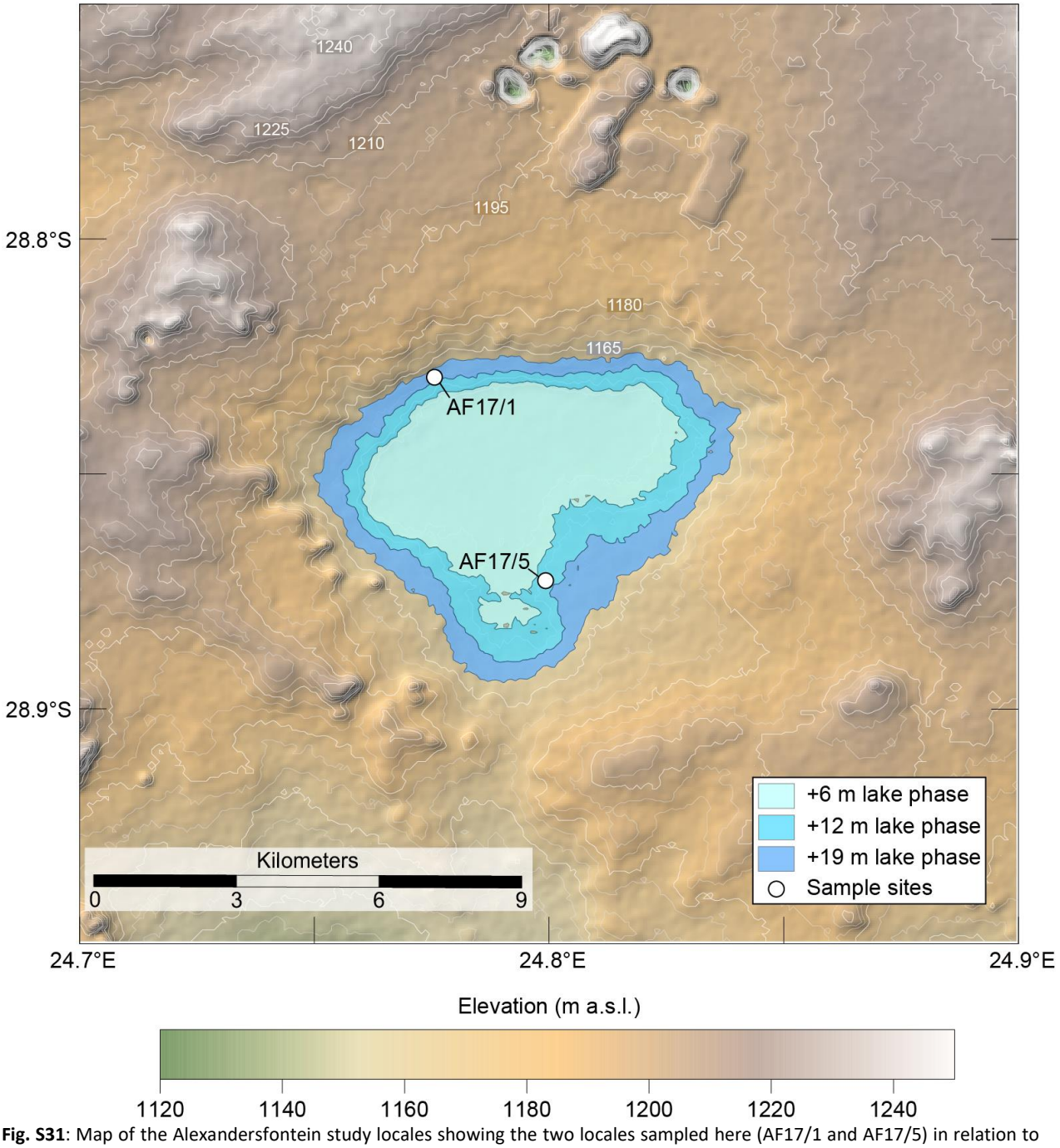


Fig. S31: Map of the Alexandersfontein study locales showing the two locales sampled here (AF17/1 and AF17/5) in relation to the extent of the proposed (3) +19 and +12 m lake phases. Topography from the Multi-Error-Removed Improved-Terrain DEM (38).

occurrence is less seasonal than observed in Swartkolkvloer and is occurs largely during the late summer (February-May).

Several locales within this area are also identified as preserving palaeo-spring deposits, and MSA and LSA archaeological materials (3, 48). Pertinent to the present study are the lacustrine beds identified by Butzer et al. (3) along the northern margin of the pan (their Transect 1 and figure 2a). Those in the northeastern pan margins at Benfontein (BT – Transect 2 and Figure 2b in 3) were later re-interpreted (48) as comprising spring tufa and colluvial deposits and were not relocated in the present study.

Alexandersfontein 'Donga' site

We located Butzer et al.'s (3) Transect 1, which was not re-described or re-interpreted in Butzer (48). The sequences here are exposed in a set of active gullies, with the un-dissected upper surface (1161 m) 16 m above the modern pan floor (1145 m). The gullies themselves are incised to bedrock (Dwyka Shale) and expose up to of 2.5 m of silt-clay lacustrine beds with an overlying poorly-sorted reddish colluvium.



Fig. S32: Panorama of the Alexandersfontein Donga site. The white arrow to the left of the image shows the 2017 sample site, with the upper reddish (early Holocene) colluvium clearly prominent. The upper surface of this sampling section lies at 1161 m or 16 m above the modern pan surface. The foreground shows the eroded remnants of the same material in the donga (gully) system, mantled by a lag of clasts that includes chemically altered late MSA lithics. The latter were later repurposed by Holocene foragers who discarded them on the overlying colluvium. This dissected surface extends to the road (from where the photograph was taken) at 1164 m, which corresponds with Butzer's (3) proposed +19 m shoreline (though we did not identify lacustrine deposits at this height).

The sequence described here (AF17/1 - "Donga site") (Figs. S32, 33) accords with that shown in figure 2a of Butzer et al. (3), though contrary to the descriptions in this previous work, none of the observed sediments were carbonate-rich. The top of the section lies at 1161 m. The uppermost 1.5 m comprises a yellowish to reddish poorly-sorted silty sand colluvium. The lower lacustrine unit comprises a greyish silty (36%) clay (58%) (Appendix A; Table S11). Infilled vertical desiccation cracks in the upper parts of lacustrine beds are visible at the contact between these units (Fig. S33). The uppermost surface of the lacustrine unit lies 14.5 m above (1159.5 m) the modern pan surface (1145 m). Thus, both the absolute altitudes and the relative height differences differ from Butzer et al. (3). In that publication lake beds at +19 m and +12 m were mapped to this locale. If a +19 m high stand is applied to modern maps, the upper lacustrine beds should extend to the 1164 m contour, where the road around the site has been

constructed (Fig. S34). It cannot be ruled out that lacustrine beds near this +19 m contour were observed by Butzer and colleagues (3; their figure 2A), but have since been destroyed or buried by construction. Notwithstanding, extending the altitude of the sampled and dated lacustrine beds in this study (+14.5 m) across the wider Alexandersfontein site produces a lake area of ~35 km^2, compared to the 44 km^2 postulated by Butzer et al. (3).

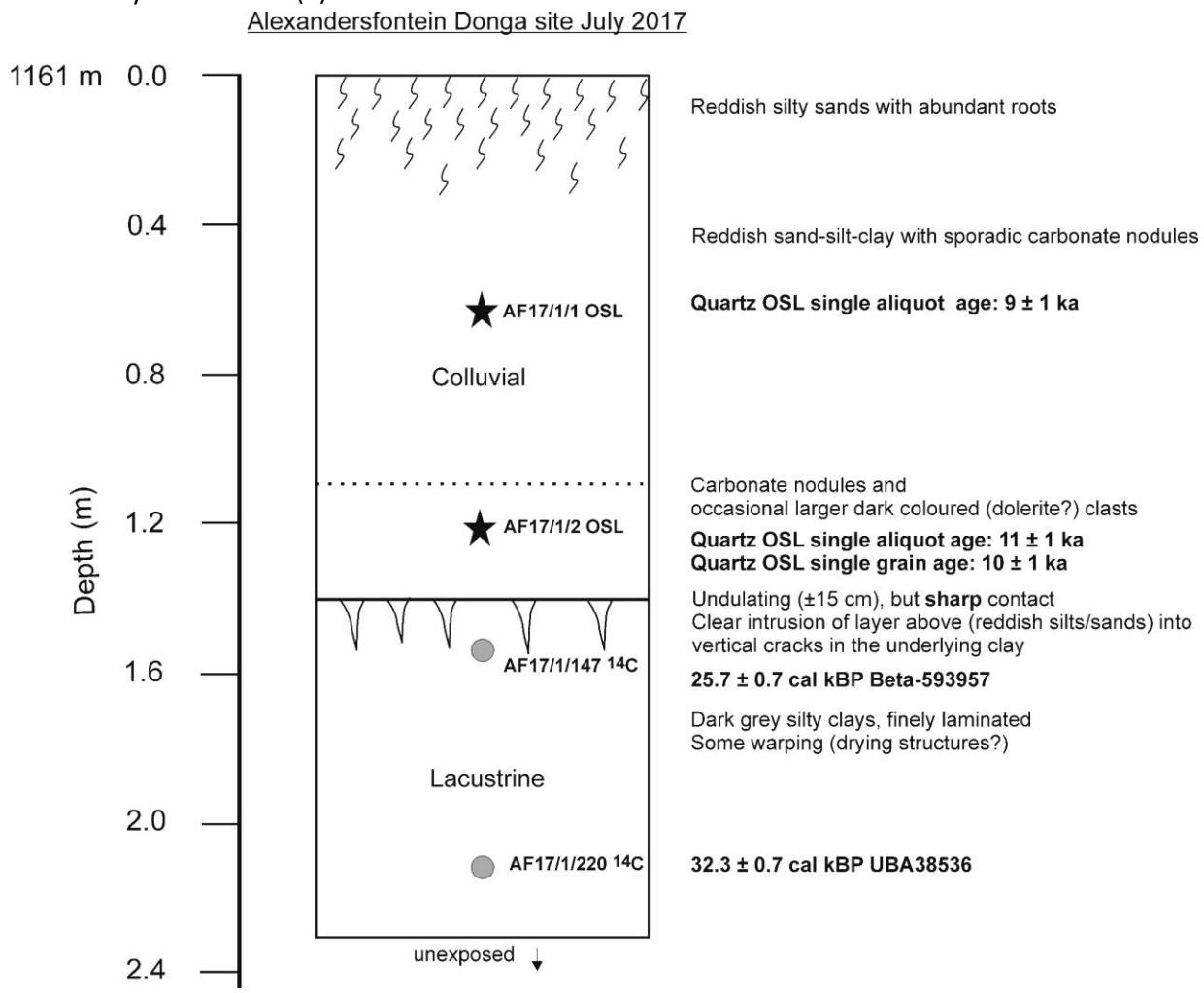


Fig. S33: Stratigraphy and dating results for of the sampled section at the donga site on the northern margins of Alexandersfontein. The lower lacustrine unit lies at ~1159 m (14 m above the modern pan surface).

A stratigraphically coherent suite of luminescence ages and radiocarbon dates provide robust chronological control for the AF17/1 Donga section. Radiocarbon dates from the lacustrine beds span 31,630-33,110 cal yr. BP (UBA-38536) to 25,530-25,870 cal yr. BP (Beta-593957). This age range is consistent with the presence of late MSA lithic artifacts around this site, observed as a discontinuous, low-density scatter on the surfaces of the exposed lacustrine beds and at the base of the dongas. These artifacts are dominated by dolerite and chemically weathered hornfels with rounded edges and flake scar ridges. They include convergent Levallois flakes, large flake blades with marginal retouch, perforators and a large single platform core in greyish brown quartzite. Similarly, Butzer and colleagues (3) encountered

"A single, convergently-patterned, large Levallois flake on lydianite [hornfels]" lying in situ along their Transect 1.

The colluvium overlying the lacustrine deposits dates to the earliest Holocene (OSL ages of 9 ± 1 and 10 ± 1 ka; Fig. S34). The single aliquot data for both samples show overdispersion (Fig. S35; Table S7). Given their interpretation as colluvial deposits, and the associated risk of incomplete bleaching, the age for AF17/1/2 (2 mm aliquot single aliquot overdispersion 18 ± 3%) was further scrutinised using single grain analysis. The single grain equivalent dose distribution is broader (overdispersion of 41 ± 4%) and somewhat positively skewed, but there is little change in CAM D_e estimate relative to the single aliquot measurements. Indeed, the resulting age following the exclusion of six high D_e outlier grains identified using nMAD (reducing overdispersion to 37 ± 3%) cannot be distinguished from the single aliquot age estimate (Fig. S35; Table S7). There are also several very low dose grains (D_e ~1 Gy). As such, we conclude that the CAM derived OSL ages are likely to be robust estimates for the burial event of interest and that the single aliquot age from the overlying sample, AF17/1/1 is also a reasonable estimate of the sample burial age. This agrees with findings of much fresher hornfels artifacts encountered on the surfaces, as well as numbers of MSA dolerite flakes with evidence of subsequent (Holocene LSA) recycling, retouch and reuse.

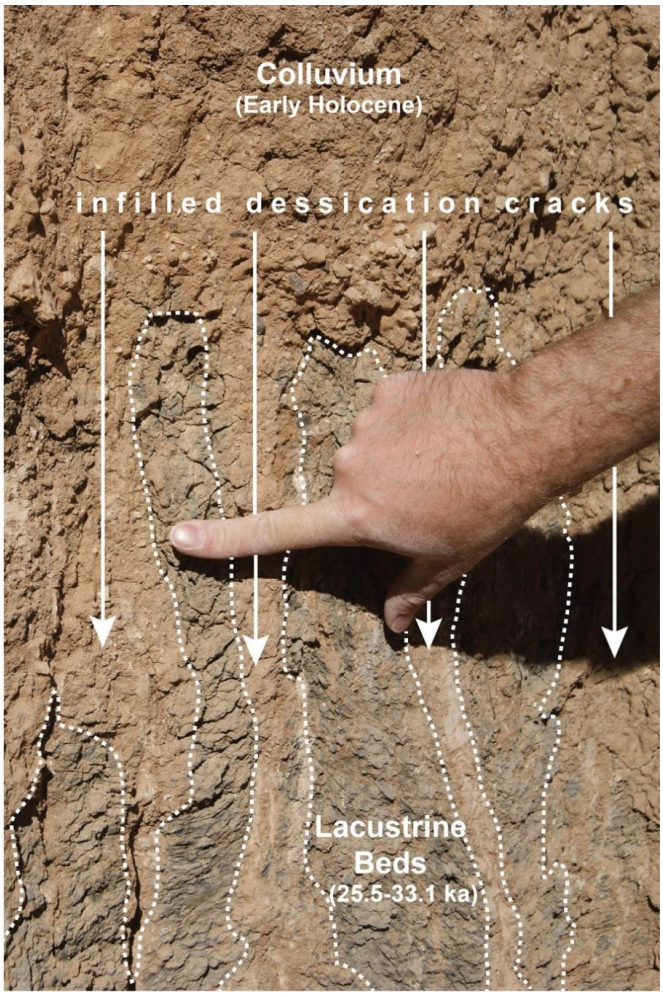


Fig. S34: Contact between the Alexandersfontein "Donga site" (AF17/1) grey clayey lacustrine beds, dated to late MIS 3 / early MIS 2, and reddish sandy colluvial overburden, dated to the early Holocene. Desiccation fissures into which the latter intruded the former are clearly visible.

The details of this wider Alexandersfontein site are subject to ongoing study, but these data provide new and more robust age control for phases of lacustrine and terrestrial deposition along this pan margin, with the former associated with a water body as large as ~35 km^2. Rather than a result of concentrated activities, the MSA artifacts here appear to represent an incidental "background" scatter resulting from human activities on or movement across, the marginal lake sediments. Alternatively, they derive from a higher density in situ occurrence that was not encountered in our investigation. Either way,

subsequent exposure through erosion concentrated artifacts in gullies, making them targets for repurposing by Holocene foragers.

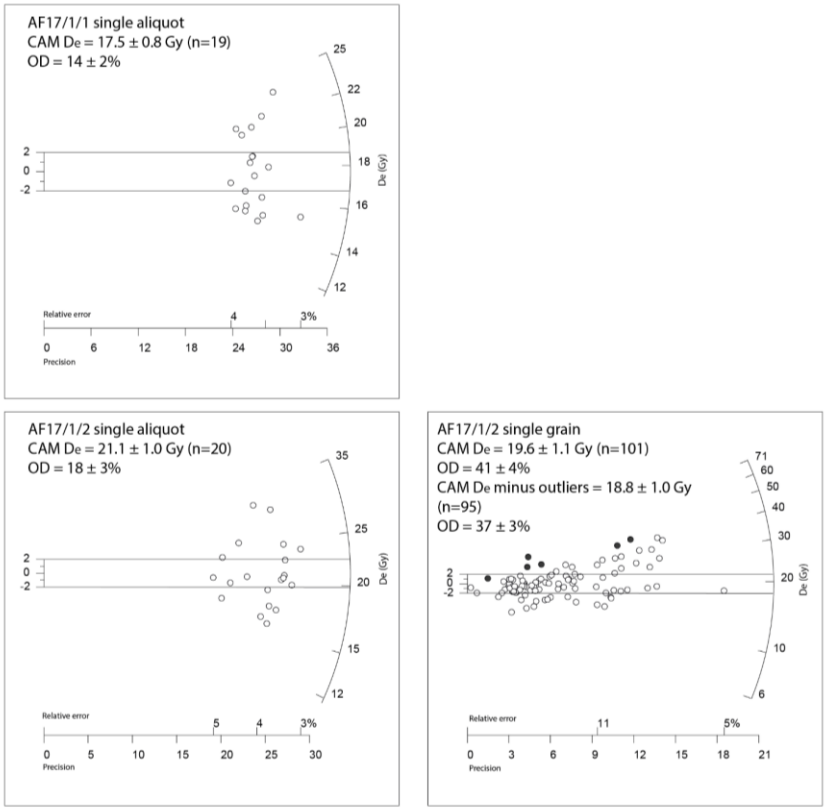


Fig. S35: Single aliquot and single grain quartz equivalent dose distributions for AF17/1/1 and AF17/1/2. Statistical outlier aliquots (defined using nMAD) are marked by filled circles where present and the CAM D_es and associated overdispersion (OD) values are reported both with and without these outlier(s) included.

Alexandersfontein 'Mauritzfontein' site

Briefly mentioned by Butzer et al. (3), and described in more detail by Butzer (48), the Mauritzfontein site lies within the wider Alexandersfontein environs, 4 km south of the modern pan surface (Fig. S36). This site is noteworthy for an array of stratified and surficial archaeological material. The site comprises a well-developed, laterally continuous laminated calcrete surface, which has been exposed by the contemporary erosion of a ~1 m thick sedimentary overburden (Fig. S36). The calcrete surface lies at ~1155 m (10 m above the modern pan floor) and is mantled by an MSA and LSA lithic lag, but also contains MSA lithics cemented within it (these were not removed or disturbed). The dissected overburden varies in thickness (typically 1-1.5 m) and comprises a sequence of silty sands, with one prominent palaeosol (B Horizon), overlain by coarser surficial sands that Butzer (48) interpreted as aeolian in origin. The site closely matches the description provided by Butzer (48; his Table 14) and the exposed sedimentary section can be mapped to his named units, *viz*: 1) a lower calcrete unit (MS2); 2) overlying B horizon (MS3C) and 3) uppermost aeolian sand unit (MS4). Butzer (48:40) argued that the lower calcrete (MS2), which he inferred to be 11.5-13.5 m above the modern pan surface (we measure it to be ~10 m), was a wave-worked feature of "highly altered shore and shore-near deposits", which includes poorly sorted "beach sand." A moderately

calcareous sandy marl unit (Butzer's MS3a) directly overlying this extensive (lithic-bearing) calcrete was sampled for OSL dating (AF17/5/1), as were units MS3c and MS4 (Figs. S36 and S37).

The quartz luminescence ages for units MS4 and MS3C confirm the occurrence of late Holocene (AF17/5/3; 2.6 ± 0.2 ka; MS4) aeolian activity with soil development prior to this, at some point after 3.2 ± 0.2 ka (AF17/5/2; MS3C; Fig. S37). The single aliquot equivalent dose distributions for both AF17/5/3 and AF17/5/2 are broad (OD 24 ± 3% and 18 ± 4% respectively; Fig. S38), most likely reflecting the occurrence of post-depositional mixing within these surficial, unconsolidated sands. Consistent with this, single grain analysis of AF17/5/2 reveals a broad spread of equivalent doses, including some zero dose grains. However, the range of radiocarbon dates provided by Butzer (48), obtained from various materials from these units (Ostrich eggshell on basal contact, 4475 ^{14}C yr BP (SI-2022); Quagga tooth, 3370 ^{14}C yr BP (SI-2586) warthog tooth 2225 ^{14}C yr BP (code not reported); with latter two "on or in top of sediment"(*sic*))



Fig. S36: A) Overhead south-facing view from a drone of Mauritzfontein (AF 17/5) with pale extensive calcrete exposures and aeolian/lacustrine deposits overlying them clearly visible. Arrows show locations and orientations of the subsequent two photos. B) View northwards with pale, MSA artifact-bearing calcretes dominating the foreground and overlying aeolian sand marl sequence from which Holocene ages were obtained in shadow (labelled "AF 17/5"). C) View (facing south) of reverse side this Holocene "island" remnant from which ages were obtained with full sedimentary sequence more clearly visible.

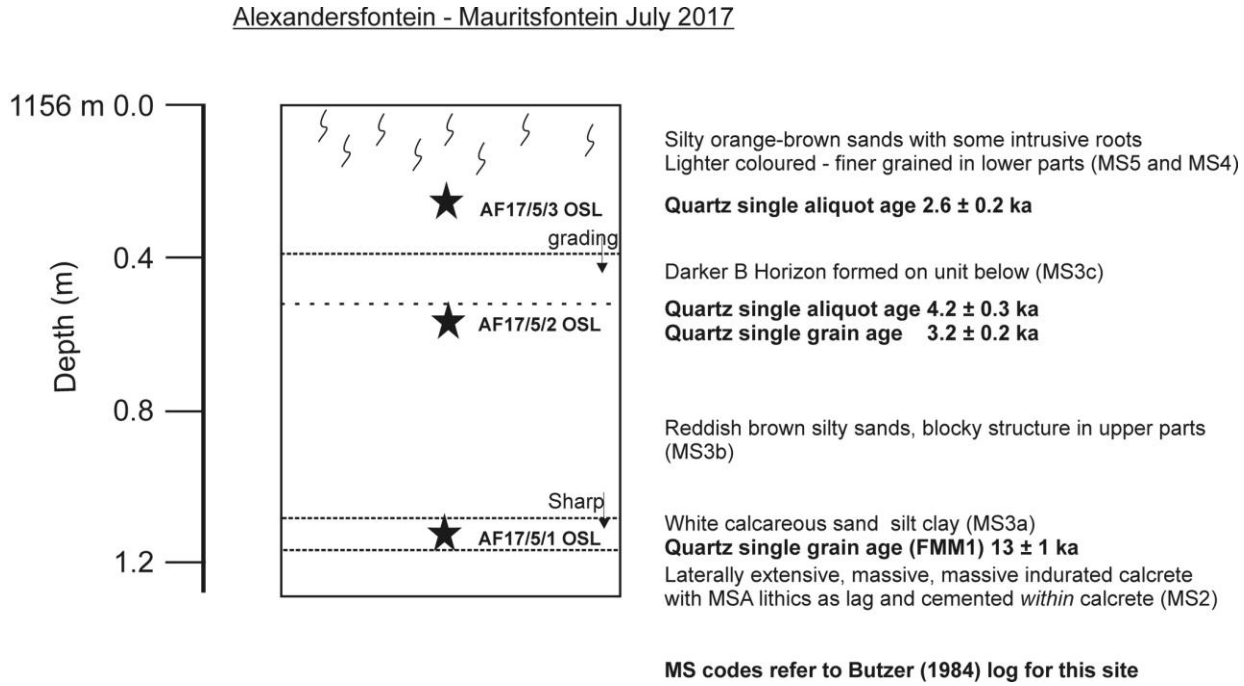


Fig. S37: Stratigraphy and dating for the Mauritzfontein section. The "MS" codes refer to Butzer's (48; table 14) units.

suggest that the CAM derived single grain age for AF17/5/2 (3.2 ± 0.2 ka using nMAD CAM; Table S7) is a reasonable age estimate for MS3C.

The sandy marl unit at the base of the exposed section, (MS3A), immediately overlying the wave-worked calcrete surface (MS2), produced a single aliquot CAM-derived luminescence age (AF17/5/1) of 29 ± 2 ka. While this may appear coeval with the ~33.1-25.5 ka lacustrine beds at AF17/1, the D_e distribution is clearly overdispersed and positively skewed (Fig. S38). A two-component finite mixture model (FMM; Galbraith and Green, 1990) applied to single grain data suggests the distribution is best characterised by two populations of grains, with ages of 13 ± 1 ka (65% of grains) and 41 ± 3 ka (35% of grains). On this basis, we interpret this marl unit (MS3A) as a later or waning lacustrine phase in the latest Pleistocene (prior to ~13 ka). This is consistent with a radiocarbon date ("on a calcareous concretion"; Butzer 1984, p40) of 10,850 ^{14}C yr BP, which can be approximately calibrated (uncertainty not quoted) to ~12,800 cal yr. BP. The higher D_e grain population may reflect the incorporation of older grains reworked from the underlying artifact-bearing MS2 calcrete during this waning lacustrine phase, tentatively placing the latter's age within MIS 3. Of interest is the overlap between this potential 41 ± 3 ka age and those we obtained at Swartkolkvloer, although at present we can only say confidently that the wave-worked and artifact-bearing MS2 calcrete predates ~13 ka.

A pre-MIS 2 age for the MS2 calcrete is however consistent with the lithics cemented within it (Figure S39, 40). These include MSA technological elements, such as radial and Levallois cores, large flakes with facetted platforms, and long, marginally retouched convergent flake blades. Burins and single platform cores were also noted. MSA artifacts here are dominated by black hornfels whose surfaces have chemically weathered to a light grey or brown patina, with another weathered, light reddish yellow hornfels also prominent (Figure S40). The overall typo-technological and raw material profile of these assemblages invokes strong parallels to Sampson's (1968, 1972, 1985) tentatively Orangian Industry, again suggestive of a MIS 3 age.

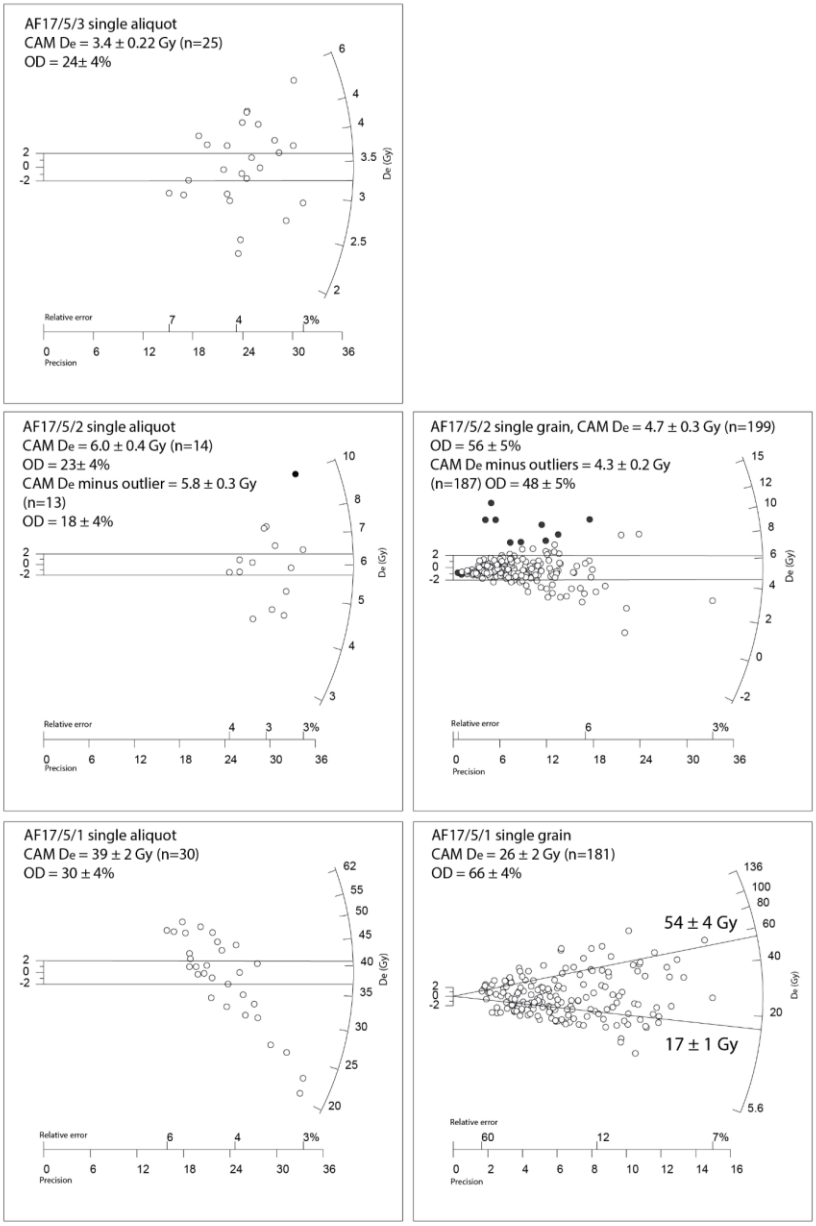


Fig. S38: Radial plots illustrating the equivalent dose distributions of the single aliquot and single grain quartz OSL dating at Mauriztfontein. Statistical outlier aliquots (defined using nMAD) are marked by filled circles where present and the CAM D_es and associated overdispersion (OD) values are reported both with and without these outlier(s) included. The single grain data for AF17/5/1 (MS3a) show the outcomes of the finite mixture modelling, which identified to populations of grains centred on 17 and 54 Gy.

As with most of the archaeological occurrences encountered at these pan-side locales, there is also abundant evidence of MSA artifacts being repurposed by LSA hunter-gatherers (who also discarded copious artifacts on primary, non-recycled materials). Here, as elsewhere, MSA pieces were targeted to produce large side-scrapers of apparent Lockshoek (Oakhurst) affiliation. In light of their likely early Holocene age, the freshness of their retouched margins (e.g. Fig. S41) further supports the calcrete-encased artifacts having been deposited considerably earlier than the *terminus ante quem* of 13 ka. Numerous lower and upper grindstones (n = 9) were also encountered, with a clear concentration

around a spring seep and high reeds on the site's southwestern margin. Most grindstones sit either atop the aeolian (later Holocene; MS4) overburden or on the surface of the MS2 calcrete, although at least one was embedded in situ within the latter (Fig. S42).

Table S7: details of the quartz single aliquot and single grain equivalent dose measurements, and their resulting ages. Details of the dose rate calculations are presented in Appendix A; Table S12. *CAM D_e values and OD estimates following outlier removal. See Fig. S38 for values obtained using all grains.

Sample	Aliquots (grains) accepted/ measured	Central Age model D_e (Gy)	OD (%)	FMM 1 (Gy) (% grains)	FMM 2 (Gy) (% grains)	Total dose rate (Gy ka^{-1})	Age (ka)
Donga							
AF17/1/1	19/20	18 ± 1	14 ± 2	na	na	1.93 ± 0.08	**9 ± 1**
AF17/1/2	20/20	21 ± 1	18 ± 3	na	na	1.91 ± 0.08	**11 ± 1**
AF17/1/2 Single grain	95/687	19 ± 1*	37 ± 3*	na	na	1.91 ± 0.08	**10 ± 1**
Mauritzfontein							
AF17/5/3	25/28	3.4 ± 0.2	24 ± 4	na	na	1.31 ± 0.05	**2.6 ± 0.2**
AF17/5/2	13/14	5.8 ± 0.3	18 ± 4	na	na	1.35 ± 0.06	**4.3 ± 0.3**
AF17/5/2 Single grain	187/800	4.3 ± 0.2*	48 ± 5*	na	na	1.35 ± 0.06	**3.2 ± 0.2**
AF17/5/1	30/39	39 ± 2	30 ± 4	na	na	1.33 ± 0.06	**29 ± 2**
AF17/5/1 Single grain	181/916	21 ± 1*	66 ± 4	17 ± 1 (65%)	54 ± 4 (35%)	1.33 ± 0.06	**13 ± 1 (FMM1) 41 ± 3 (FMM2)**

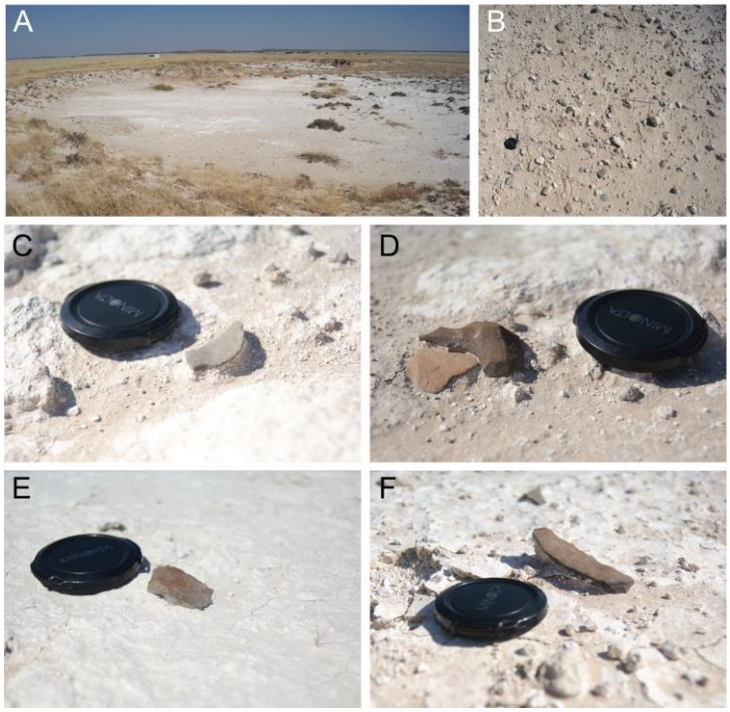


Fig. S39: A) A view southwest across the exposed MSA lithic-bearing MS2 calcrete at Mauritzfontein, onto whose pale surface aeolian erosion of overlying Holocene sands and terminal Pleistocene marls LSA artifacts are deflated. B) Here they mix with MSA artifacts eroding from the calcrete itself to form a high-density lag. A-D) Examples of in situ MSA artifacts actively emerging from the calcrete.

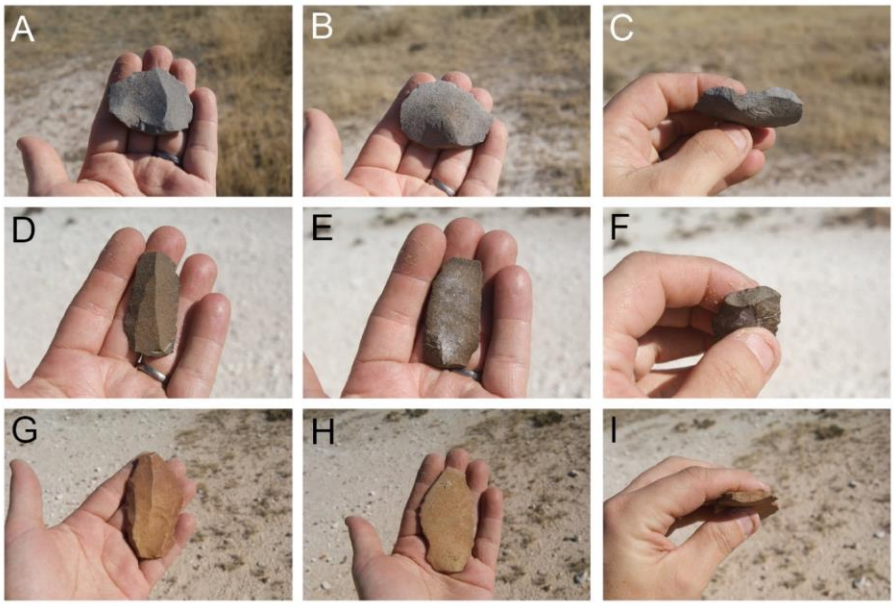


Fig. S40: Dorsal, ventral and proximal views of newly exposed Levallois flakes from the MS2 calcrete at Mauritzfontein. Note typical faceting on the platforms from preparation prior to striking (C, F, I). These raw materials are the site's most common – black hornfels patinated to light grey (A-C), black hornfels patinated to light brown (D-F) and reddish yellow hornfels (G-I).

This actively eroding, open-air MSA site of Mauritzfontein is extensive (~17,000 m^2) and in places extremely dense (~50 pieces/m^2). Spatial integrity is implied by the clear clustering of homogenous raw materials and technological products, some of which appear to have resulted from individual knapping events (a hammerstone was also encountered near to one). While a full understanding of its depositional context awaits further investigation, Mauritzfontein was a clear focus for human activities, potentially when the paleo-lake was between its +6 and +12 phases and certainly during the late Pleistocene. Butzer (48:41) likewise interpreted these MSA cultural materials as "coeval with … the MS-2 beach." We speculate that Mauritzfontein's high density relates to its proximity to both the lake and the aforementioned spring seep, which together rendered this locale more continuously attractive to MSA and LSA foragers than others ringing the Alexandersfontein pan.

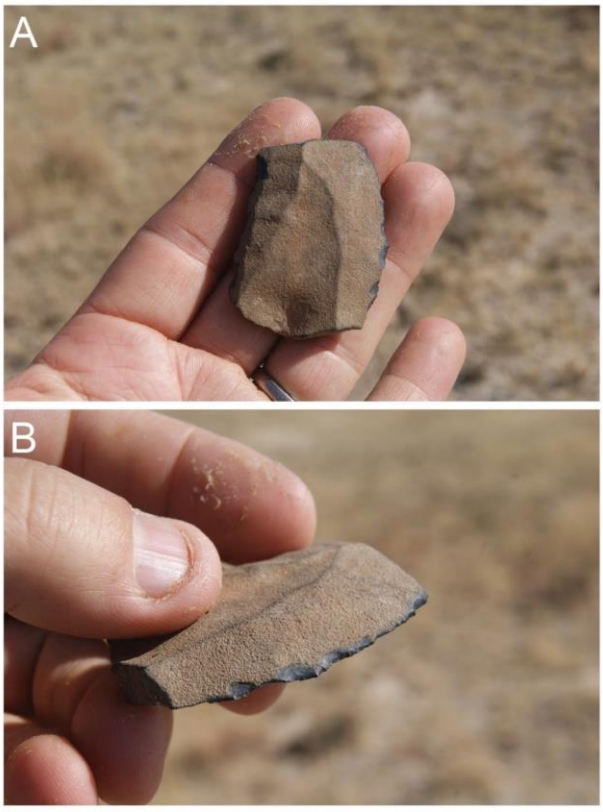


Fig. S41: Example of a large Levallois flake from Mauritzfontein that was subsequently recycled into a side/endscraper typical of the early Holocene Lockshoek (interior Oakhurst) technological tradition. The freshness of the black retouched edge contrasts sharply with the heavily patinated, edge-worn condition of the original MSA artefact.



Fig. S42: Examples of lower (A-D) and upper (E-F) grindstones from Mauritzfontein. A-B and E-F are whole or nearly so, while C-D are fragmentary; pecking damage is also visible on A. Note that D is embedded in-situ in the MSA artifact-bearing MS2 calcrete (ground surface facing away from camera).

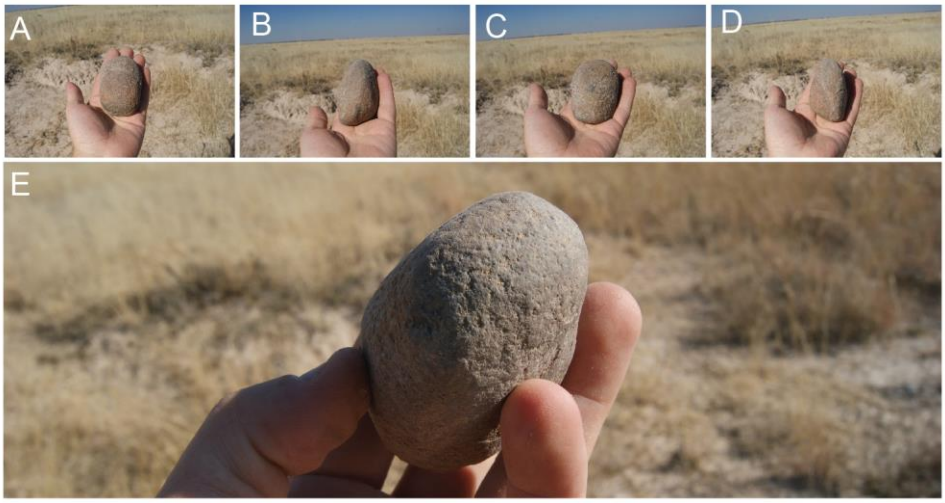


Fig. S43: Different views of a calcrete-encrusted hammerstone encountered near an in situ MSA knapping scatter at Mauritzfontein. Closeup (E) shows pitting damage from use at distal margin.

Vanwyksvlei: pan-associated archaeology

As part of a wider, albeit preliminary, survey of the relationship between palaeolakes and the regional archaeological record, we also conducted initial surveys of two pans to the northeast and east of Vanwyksvlei (Fig. S4). Today both pans are rarely (seasonally) inundated (36), but our hydrological modelling scenarios (see hydrological modelling section, Fig. S54) suggests these sites would have been inundated under the conditions required to inundate Alexandersfontein and Swatkolkvloer. Both exhibit extensive scatters of MSA artifacts. The geological context of this region was described by de Wit (50). Like the Brandvlei area, much of this region is underlain by Dwyka Group diamictites of the Karoo Supergroup. The most significant macroscale landscape feature is the Carnarvon Leegte River valley, which flows to the northeast (then via the Hartbees River to the Orange River), with its position constrained in places (particularly on its southern margin) by dolerite sills (50). Up to 18 m of largely Plio-Pleistocene fluvial sediment (Kleinvlei Formation), with a capping of aeolian sediment (50), lie within the Carnarvon Leegte valley, and unconformably overlie Dwyka Group diamictites (ibid).

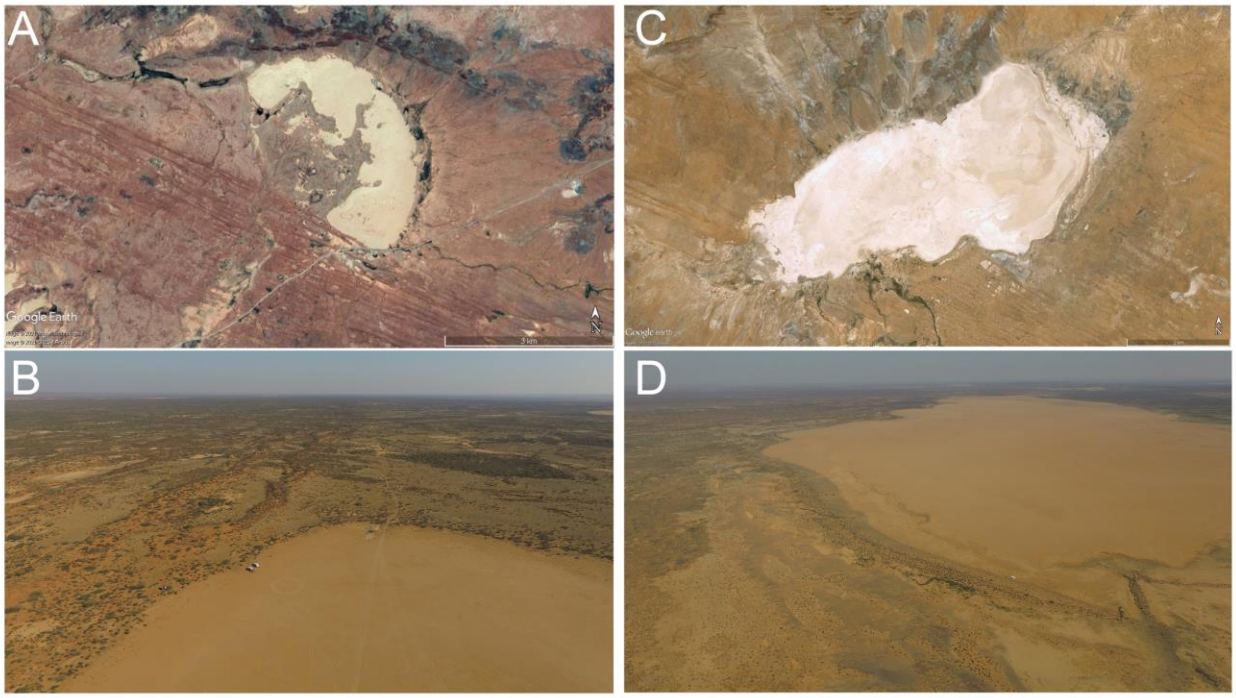


Fig. S44: Aerial images of the Nieuwdam Farm pan (A and B) and Verkeerdevlei (Fortuinskolk se Pan) (C and D). The view in B is towards the northwest at the southern margins of the pan. The encroaching linear dunes are visible to the left of the image. The view in D is to the west, viewed from the eastern margins of the pan. The dark sandy bar running along the pan margin was sampled for luminescence dating near the vehicles, barely visible in right foreground. Most of the archaeological materials described were observed on the surface behind (landward of) the sandy bar in the left foreground.

The first pan investigated lies east of Vanwyksvlei (-30.245° 21.945°). It is unnamed on some maps but lies on the farm Nieuwdam. Beaumont et al. (51; 241) referred to investigations of the pan "Vorentoe se Pan on the farm Nieudam", though it is not clear from the maps available to us whether this represents the pan investigated here, or another ~ 3 km northeast. The pan surface lies at 942m asl and is ~5 km^2. Its northern margin is clearly defined by rock geology, but its southern and eastern margins are obscured by aeolian sediment, some of which has migrated over the pan surface, as well as deposits (and associated

lag surfaces) of well-rounded gravels and pebbles (Fig. S44). The latter likely represent fluvial deposits from the Carnarvon Leegte River. Although partly blocked by dunes today, a connection to the main channel of the Carnarvon Leegte River is readily apparent on the southwestern margin of the pan. We infer the pan to have developed within the periodically inundated and unconsolidated sediments at the margins of the Carnarvon Leegte.

The pan is of particular interest because of it possesses an extensive, high-density scatter of heavily ventifacted MSA artifacts (Fig. S45). The scatter is composed of tens of thousands of artifacts in a crescent-shaped distribution that closely parallels the pan's southwestern corner. Densities drop steeply if one moves only ~10-15 meters towards the centre of the pan. MSA artifacts are dominant if not exclusive, with an emphasis on retouch into diverse tool types, including perforators, scrapers and retouched flakes (Fig. S34). No backed pieces were encountered, nor were any retouched points. Convergent and blade flakes are common, and medium sized blade cores are present. Levallois is the dominant reduction strategy. The latter clearly includes the recurrent form as several cores are exhausted and in one case a presumably final target flake accidentally plunged, taking with it a sizeable distal margin of the core (Fig. S46). Towards the easternmost extent of the scatter MSA artifacts persist, but there is a notable increase in less weathered pieces that include large D-shaped side scrapers and side/end scrapers, again signalling an early Holocene (Lockshoek) component. A flipped grindstone was also noted here.

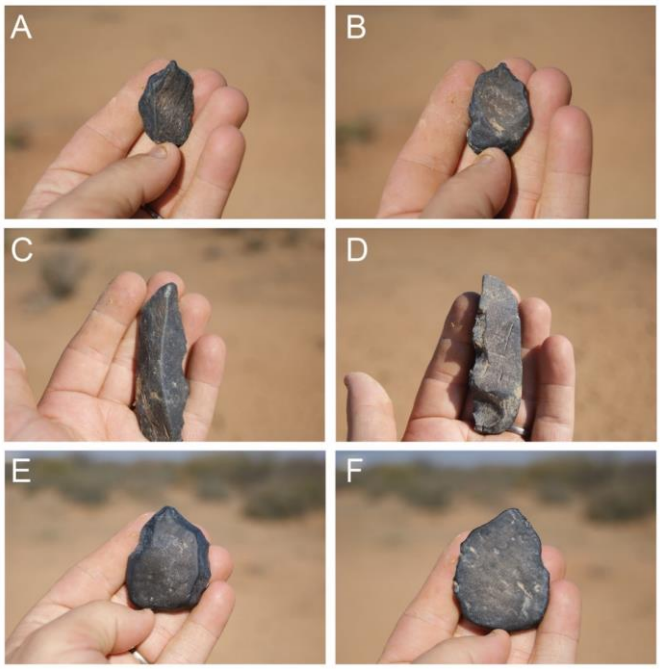


Fig. S45: Ventral and dorsal views of a selection of weathered tools common in the high-density MSA-dominated scatter at the southwestern corner of Nieuwdam farm pan. A-B: perforator; C-D retouched blade; E-F: side-scraper. All on fine-grained basalt.



Fig. S46: Different views of a weathered plunging flake that removed the entire striking surface of small Levallois core (fine-grained basalt).

Raw materials are dominated by a matt-black fine-grained basalt that wind polishes to a high sheen. The presence of a rounded, water-smoothed cortex suggests such nodules originated from the Carnarvon Leegte gravels, which outcrop beneath some of the recent linear dunes adjacent to this southwestern corner of the pan. Given that nearly every clast is an artifact, however, this scatter could not have derived from slopewash off the adjacent high gravels. Nor could it have accumulated here from slippage across pan-surface, as the pan dips west to east rather than vice versa. Instead, the artifacts likely deflated in situ from a previous pan-side occupation surface(s), perhaps composed of slackwater deposits. Together with the ready availability of high-quality siliceous rock and leeward setting relative to the prevailing westerlies, the wetland margin of this protected pan elbow may have been particularly attractive to MSA foragers.

The larger of the two Vanwyksvlei pans, Verkeerdevlei (also mapped as Fortuinskolk se Pan; -30.305° 22.078°), lies at 970 m asl and is ~20 km^2. The pan in endoreic, with a relatively small surrounding catchment (~ 80 km^2). It is separated from the Carnarvon Leegte valley by an area of higher topography (981 m). A prominent ~1.3 m high sandy bar can be traced from the northern corner of the pan around the eastern and southeastern margins. The bar is composed largely of homogenous silty sands. Three $pIRIR_{225}$ ages constrain the age of this bar to ~1.3 ka with the immediately underlying pan floor sediments dating to 2.2 ± 0.1 ka. In the eastern extremity of the pan, immediately landwards (east and southeast) of this bar, pan floor sediment is re-exposed for ~ 300 m before the topography rises away from the pan. This pan floor sediment is fine-grained slightly sandy silt-clay, with the quartz-rich gravel lag containing extensive MSA lithic artifacts.

As at Nieuwdam, this MSA lithic scatter at Verkeerdevlei is expansive, running ~400 m in a northeast-southwest orientation paralleling the pan margin and consisting of tens of thousands of heavily ventifacted artifacts. Matt-black to charcoal grey hornfels (or fine-grained basalt) predominate, most of which exhibit early stages of a light reddish orange patination forming on their surfaces. Also present is a light grey coarse-grained quartzite similar to, but notably more frequent than, that encountered at Nieuwdam, as well as a caramel-coloured chert, a bluish grey banded agate and rare banded ironstone. Diagnostic MSA elements include Levallois flakes, tools and cores (Fig. S48). Formal tools include marginally retouched Levallois scrapers and points with faceted platforms and numerous perforators. Long retouched blades, marginally retouched tabular pieces, and at least one heavily weathered, cracked dolerite hammerstone were also encountered. One elongated convergent flake was manufactured from a red chert-like material with the dorsal surface possessing a reflective, specuarlite-like inclusion. A

secondary, much less dense LSA component is also present, as evinced by extremely fresh large D-shaped scrapers, blades and blade cores. The site sits immediately adjacent to (south of) Verkeerdevlei's primary in wash area whose periodic flow the landowners capture using a small dam. Like the Nieuwdam scatter, this site is located at the margin of one of the highest points of this east to west sloping pan floor and would thus have been a nearshore setting only when this palaeo-lake was full. Again, this militates against the scatter having accumulated from pan floor slippage, and nor were artifacts noted on the adjacent slope forming pan's eastern boundary. We thus interpret this MSA scatter as having been deposited through subsequent aeolian removal of lakeside sediments, processes that also partly covered the resulting lag deposit in dune sands during the late Holocene. The luminescence ages, however, only attest to much later phases of pan filling in the late Holocene with a basal age of 2.2 ka and an overlying age of 1.3 ka.

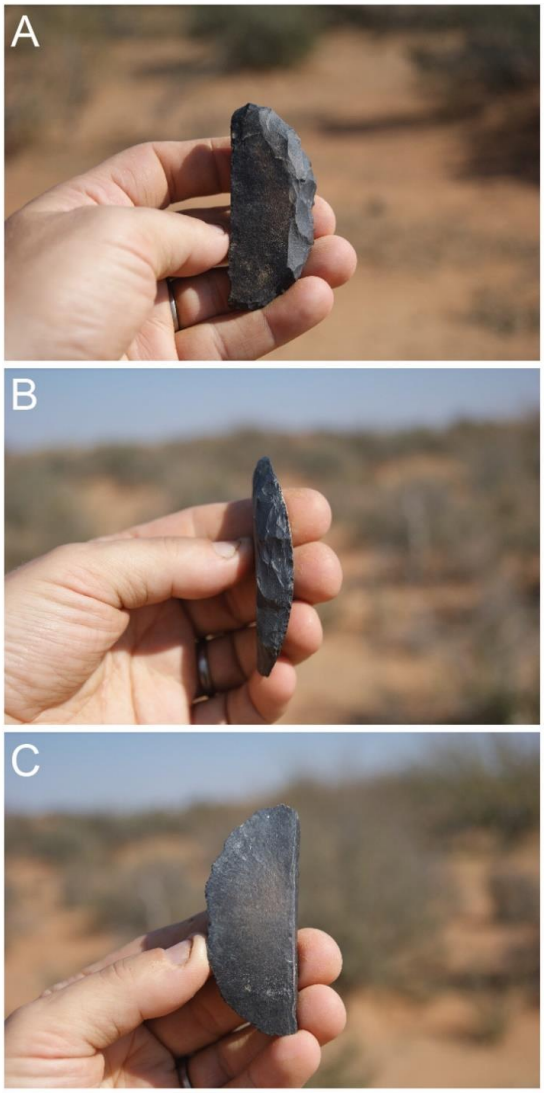


Fig. S47: Dorsal (A), marginal (B) and ventral (C) views of a fresh (non-weathered) Lockshoek (interior Oakhurst) D-shaped scraper encountered towards the eastern edge of the high-density Nieuwdam farm pan scatter (fine-grained basalt).

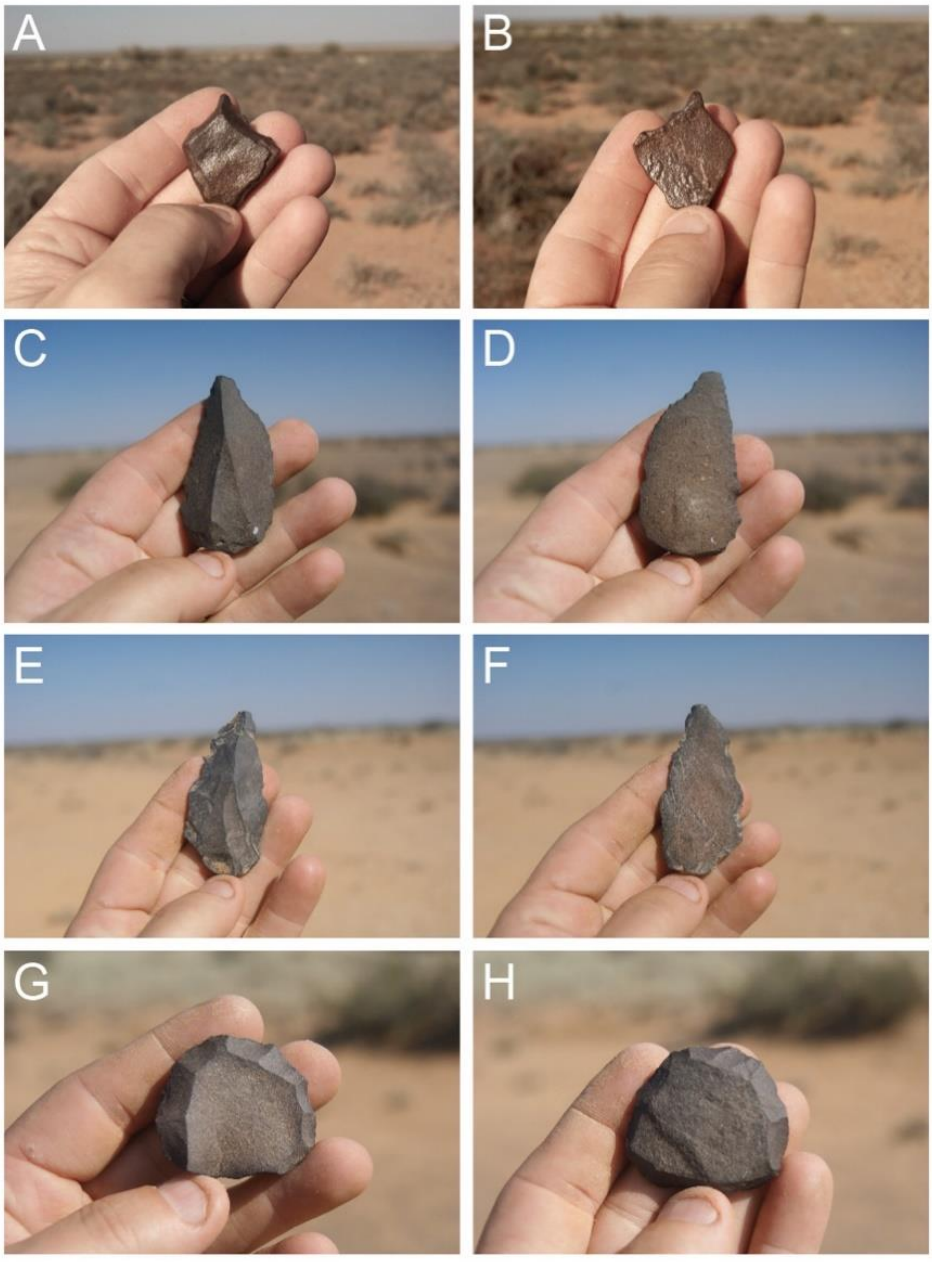


Fig. S48: Dorsal and ventral views of a selection of relatively common tools at the largely MSA scatter along the northeast margin of Verkeerdevlei pan. A-B: perforator; C-F: convergent scrapers (or marginally retouched points) on Levallois flakes; G-H: flat unilineal Levallois core. A-B is banded ironstone, C-D is fine-grained basalt, E-H are hornfels.

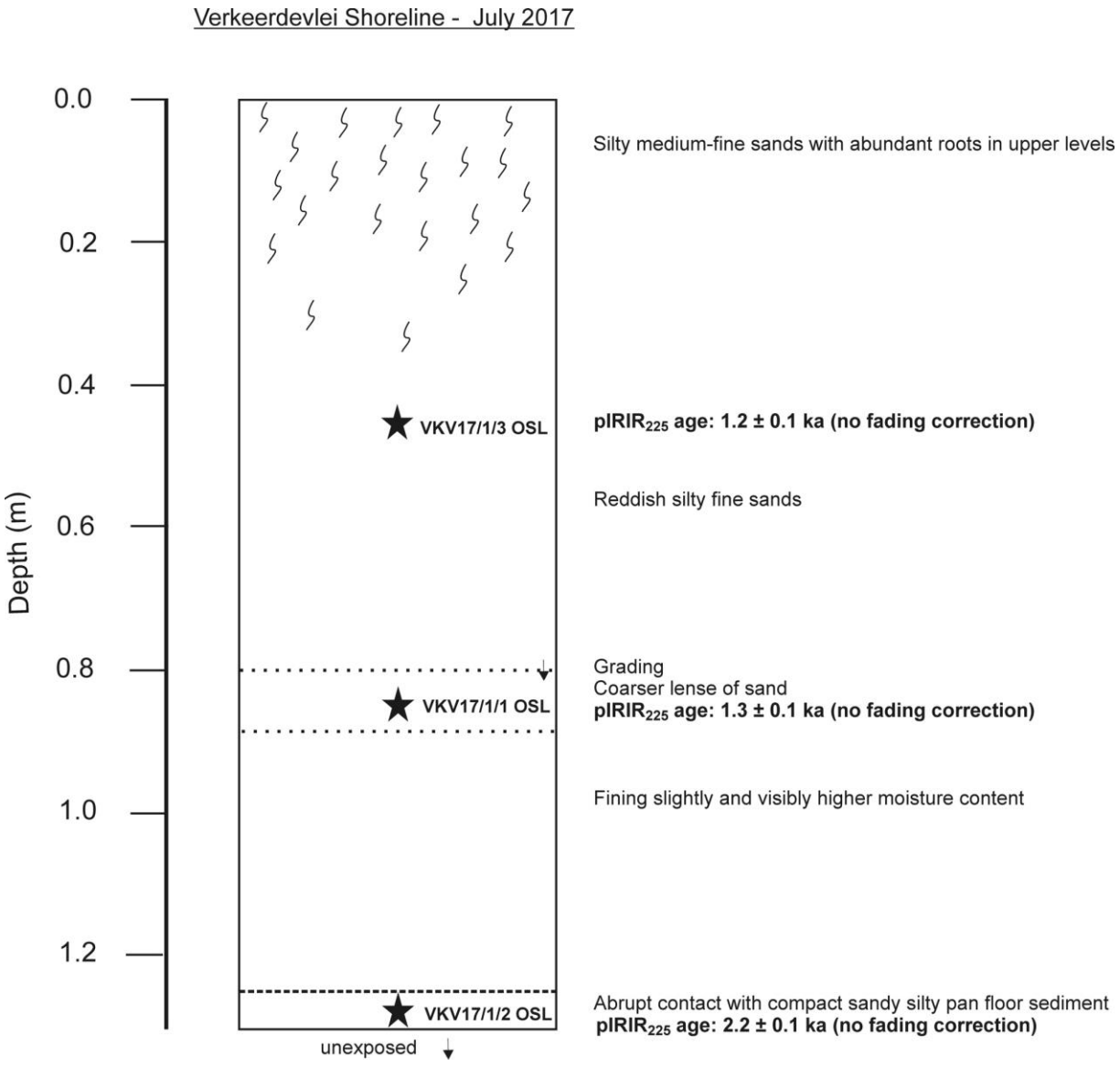


Fig. S49: Stratigraphy and dating of the shoreline bar at Verkeerdevlei.

Hydrological modelling

Hydrological modelling was employed to elucidate the palaeoclimatic conditions necessary to sustain the inundations and water depths implied by the geological observations. This was achieved by producing water surface depth or water table depths in the pans under current and a range of past climate scenarios.

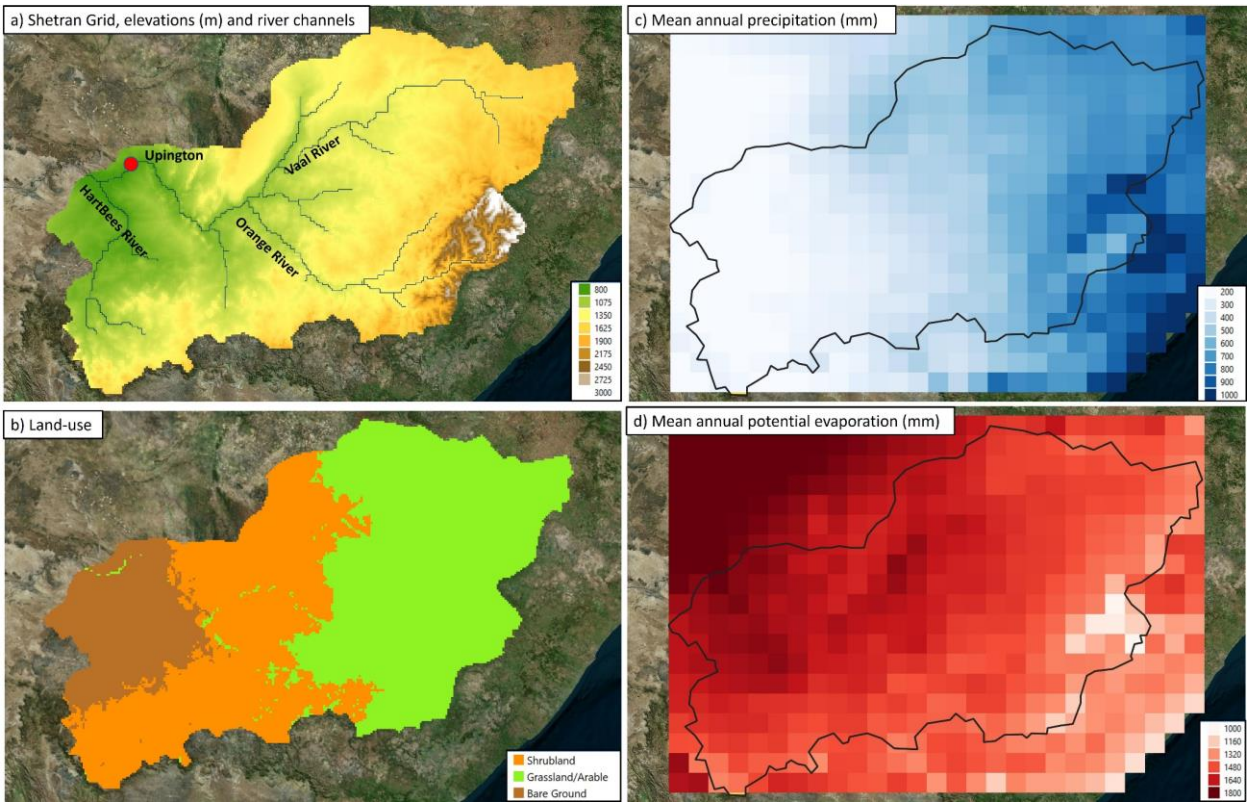


Fig. S50: Shetran setup for the simulation of the Orange/Vaal catchment 1970-2000.

The potential groundwater basin associated with each pan covers a considerable area, and as such the hydrological modelling considered the entire Orange and Vaal river catchments, downstream to the confluence of the Hartbees and Orange rivers (458,000 km^2) (Fig. S50). By considering hydrological basins, validation of the overall water balance within the models is possible as the modelled discharges can be compared to the measured ones. The Kalahari drainage basin, to the north of the Orange River, is not included as it is considered to be generally endorheic (52). The basin boundary on the west, south and east side was obtained from Hydrosheds (53). The northern boundary (excluding the Kalahari basin) was obtained by considering elevation and stream flow direction.

The pans are predominately groundwater fed with the Swartkolkvloer pan having a catchment of 1600 km^2 and Alexandersfontein a catchment of 430 km^2. In addition, there is direct rainfall onto the pans but this is less significant as the surface area of the pans is much smaller than the groundwater catchment. Within the model there is also the potential for surface water inflows and outflow from the pans into the

nearest river channel. However, in the hydrological modelling the water depths in the pans and rivers are insufficient for there to be a surface water connection between the rivers and the pans.

Table S8 shows the original source of spatial distributed data used by Shetran. Average spatially distributed monthly values of precipitation and potential evaporation were obtained from the WorldClim 2.1 dataset (6) and Global Aridity Index and Potential Evapotranspiration (ET0) Climate Database v2 (5). There is an increase in annual totals of precipitation going from west (<200 mm a^{-1}) to east (>1000 mm a^{-1}). Potential evaporation totals show a decrease from west (>1600 mm a^{-1}) to east (<1000 mm a^{-1}).

The aim of the hydrological simulation (outputs shown in Figs. S51, S53 and Table S9) is to produce the expected long-term water table and surface water depths, as observed and preserved at the lake basin study sites. To achieve this, the model was run for up to 100 years with the same annual precipitation and potential evaporation data until equilibrium conditions were achieved. Evapotranspiration is affected by the temporal distribution of the precipitation so the monthly precipitation data were converted to daily precipitation using the expected number of wet days per month (https://www.fao.org/aquastat/en/geospatial-information/climate-information). These are a number of dams and inter-basin transfers in the eastern part of Orange and Vaal basin (54), although their effect on the overall mass balance within the catchment is small.

For this study, considering basin size with long simulations needed in order to produce the long-term water table and surface water depths, a 24 hour timestep was selected as it produced good quality results with reasonable run times.

Table S8: Data sources for Shetran hydrological modelling. Data were aggregated up to a common 4 km resolution.

Data type	**Data source and original resolution**	**Link**
Elevations	SRTM (90m)	https://srtm.csi.cgiar.org/
Land cover	Copernicus Global Land Service (100m)	https://land.copernicus.eu/global/products/lc
Soil Type	HWSD (~1 km)	https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/harmonized-world-soil-database-v12/en/
Precipitation (monthly 1970-2000)	WorldClim 2.1	https://www.worldclim.org/data/worldclim21.html
Potential Evaporation (monthly 1970-2000)	Global Aridity Index and Potential Evapotranspiration (ET0) Climate Database v2	https://figshare.com/articles/Global_Aridity_Index_and_Potential_Evapotranspiration_ET0_Climate_Database_v2/7504448

The model was calibrated by manually changing the hydraulic conductivity of the aquifer, within physically realistic bounds, so that the modelled discharge corresponded with measured discharge at Upington. Long term simulations (up to 100 years) were run and equilibrium was classed as occurring when long-term changes in the water table had stopped occurring throughout the catchment. Decreases in the depth of the water table in response to the increased rainfall in the wet season were observed, but these values would subsequently decrease back to a constant baseline value.

Table S9: Climatic implications for hydrological modelling of Alexandersfontein and Swartkolkvloer palaeolakes. Time periods indicated are examples, and do not indicate that these are the only phases of comparable lake development. For Alexandersfontein, the 19-22 ka period was selected to provide a point of comparison with the palaeoclimatic estimates of Butzer et al. (3), who estimated similar paleo-precipitation levels (878 mm/yr) for this lake phase under what they deemed to be conditions slightly warmer than our estimates (-6°C as opposed to -6.2°C relative to present).

Lake basin	**Alexandersfontein**	**Swartkolkvloer**
Lake level (meters above pan floor)	19	18
Time period example	19-22 ka	40-48 ka
Mean annual temperature (°C, difference from modern)	-6.2	-4.7
Estimated potential evapotranspiration (mm/yr)	1249	1290
Precipitation:Potential evapotranspiration ratio	1:1.52	1:2.33
Precipitation required for target lake level (mm/yr)	821	554
Lake phase precipitation % of modern	189	317
Lake phase Aridity Index (P/PET)	0.66	0.43

To validate the model three different aspects were considered:

1. The modelled discharges at Upington on the Orange River (near the outlet of the modelled basin) were compared to the measured discharge averaged over the period from 1970-2000. The modelled mean annual discharge 236 $m^3 s^{-1}$ shows excellent correspondence to the measured discharge 224 $m^3 s^{-1}$. In both cases the highest flows are in March and the lowest are in September.
2. The Pekel et al. (2016) surface water occurrence data show (Figs. S5) that there is surface water recurrence at Swartkolkvloer 10% of the time, with recurrence most common in April. The very wet years that produce the surface water recurrence are not included in the modelled dataset, which has repeated average monthly values. Thus, if the model is working satisfactorily, it would be expected to simulate a water table just below the ground surface, or near the ground surface, in March/April. This is indeed the case, with the modelled water table depths at Swartkolkvloer between 0.6 m and 2 m below the ground surface.
3. There are a number of point measurements of groundwater levels measured in boreholes between 1949 and the present (as obtained from the National Groundwater Archives (NGA)). These correspond well with the modelled water levels (Fig. S52) given: a) spatial variability over the 4 km modelled grid square, and b) that the measured depths are only indicative of typical depths as there will be variations over time.

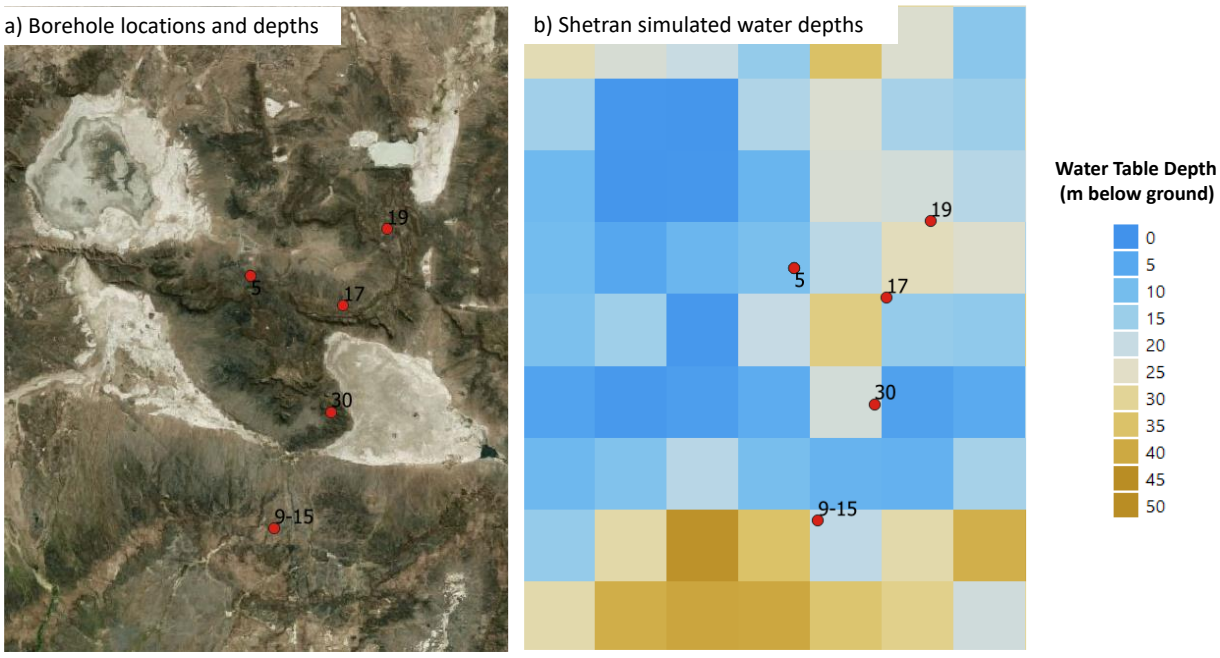


Fig. S51: Shetran hydrological model results for the impact of changes in potential evapotranspiration on lake levels in the Swarkolkvloer and Alexandersfontein basins. Negative values indicate groundwater depths below modern surface.

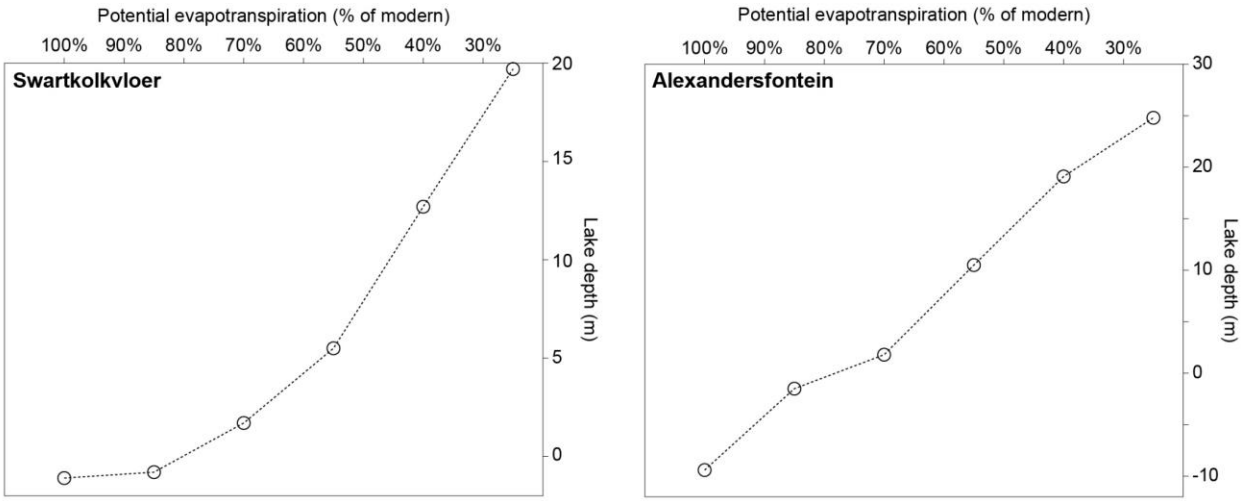


Fig. S52: a) water table depths and b) Shetran hydrological model results for the Swarkolkvloer basin. 9-15 refers to a range of depths with four measurements at this location

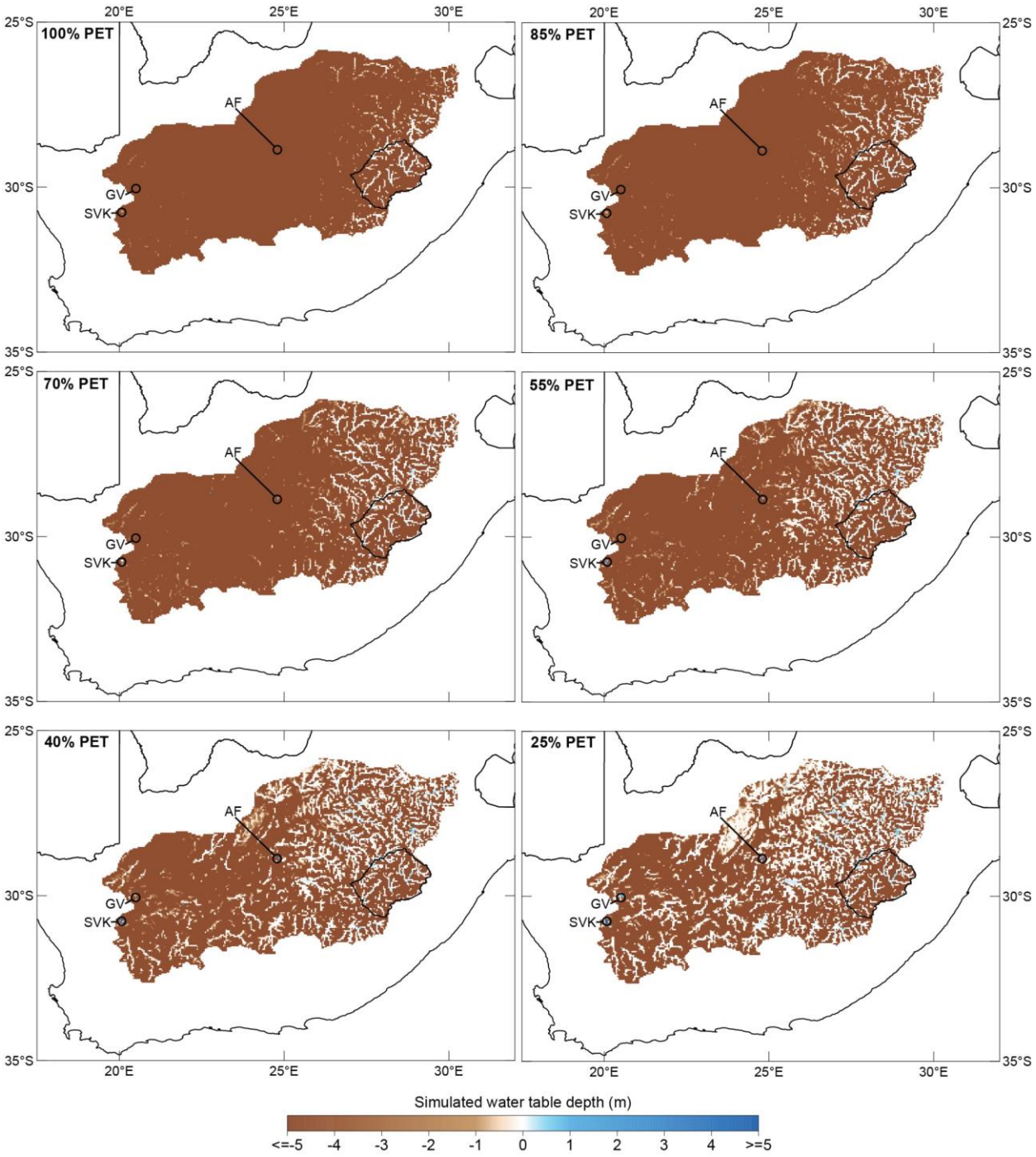


Fig. S53: Water table depths as simulated by the Shetran hydrological model under a range of different potential evapotranspiration scenarios (% of modern). Rainfall gradients for all simulations are as today.

Vegetation reconstruction

To consider potential vegetation changes associated with the inferred lacustrine periods, we have employed a modern analogue approach. The distribution of biomes in the interior is primarily determined by: 1) water availability (e.g. discriminating between Grassland and Nama-Karoo; Fig. S54) and 2) mean minimum temperature of the coldest month (TminColdM; a proxy for frost potential, e.g. discriminating between Grassland and Savanna; Fig. S54D) (55, 56). At the subcontinental scale, the seasonal distribution of rainfall is also significant, discriminating the winter-rainfall biomes of the Fynbos and Succulent Karoo of the southwestern Cape from the summer rainfall bioregions of the rest of southern Africa (Fig. S54C). Focusing on vegetation change in the South African interior, we have defined the modern relationships between bioregions and: 1) aridity, as defined by precipitation divided by potential evapotranspiration (PET), and 2) mean minimum temperature of the coldest month.

To estimate past changes in PET, we defined the relationship between PET and mean annual temperature (MAT) - a parameter that is reasonably well estimated in the palaeoclimatic record and a defining factor in establishing changes in PET (5, 57) - in the modern South African climate space. Simple linear regression indicates that in the South African interior the relationship between PET and MAT is reasonably defined as PET=59.22*MAT+548.8 (R^2=0.82, P= 0.0001) (Fig. S55).

To estimate palaeotemperature during the lacustrine phases, we considered the proxy-based palaeo-temperature data available from southern Africa. These are generally consistent in indicating a temperature depression of ~4-7°C during the Last Glacial Maximum (19-26.5 ka) (58-62), and an overall pattern of temperature change that broadly tracks global trends, as characterized by Antarctic ice core data (63). We thus use these patterns and data to create a temperature estimate time series for the region, calibrated using regional temperature estimates (for amplitude of change) and the EPICA Antarctic temperature reconstruction (64) (for the timing of change), as supported by the findings of Chevalier et al. (63) (Fig. S56). Applying a conservative estimate of a 6°C average LGM temperature depression, the resulting temperature estimates for our lacustrine phases are 16-17 ka = -4.7°C; 19-22 ka = -6.2°C; 31-35 ka = -5.6°C; 39-55 ka = -4.7°C.

Using these data, we established a temperature-based control on the potential influence of PET on the water balance during lacustrine phases, and then adjusted mean annual precipitation values to achieve the target P:PET ratios indicated by the hydrological model (Table S9). Climate data were obtained from the WorldClim2.1 dataset for TminColdM and MAP (6) and PET data (defined according to Hargreaves (65)) were obtained from the Global Potential Evapo-Transpiration (Global-PET) Geospatial Dataset (5). The MAP and PET data were used to calculate spatially continuous Aridity Index (AI) grids, the input values of which could be

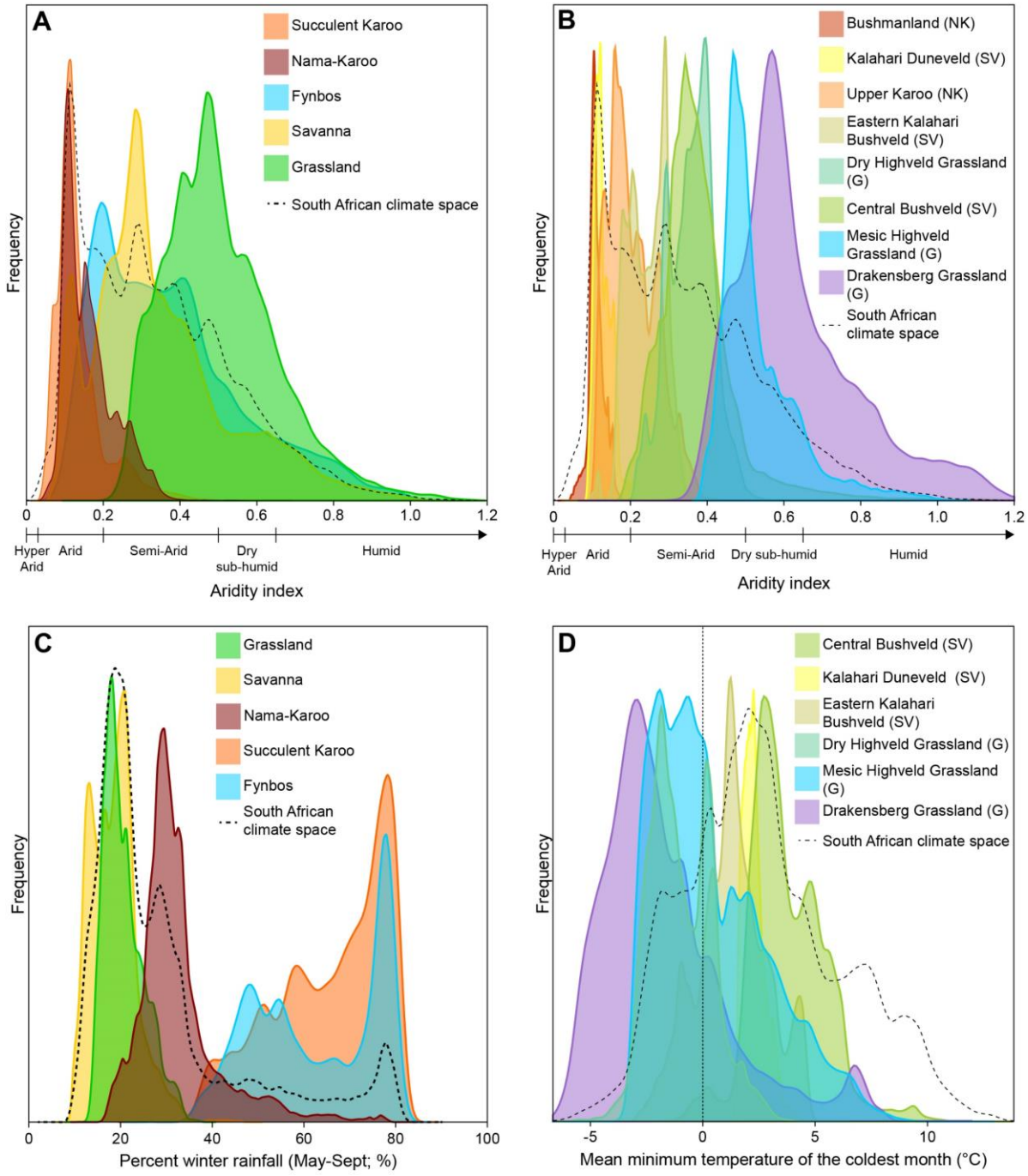


Fig. S54: Frequency distributions (scaleless gaussian kernel density) of: South Africa's major biomes (**A**) and the bioregions of the Nama-Karoo (NK), Savanna (SV) and Grassland (G) biomes of the South African interior (**B**) as a function of aridity, biome distributions as a function of rainfall seasonality (**C**) and the distribution of Grassland and Savanna bioregions according to mean temperature of the coldest month (**D**). As a generalized proxy for frost potential, the frost intolerance of savanna vegetation is evident in its distribution primarily in areas where the mean temperature of the coldest month is above 0°C. Biome and bioregion data from Mucina and Rutherford (4) and climate data from WorldClim 2.1 (6). The aridity index is calculated according to Trabucco and Zomer (5).

modified to reconstruct the water balance for the lacustrine phases in the South African interior.

Regarding the seasonal distribution of rainfall, in applying the defined relationships to reconstruct past vegetation distributions, we acknowledge that palaeo-precipitation gradients may have differed from those of today. It has been suggested that related changes may have been influenced by an increase in temperate moisture-bearing systems and a concurrent decrease in tropical rainfall during glacial periods (66, 67). The extent to which palaeo-precipitation regimes changed across the subcontinent is a matter of speculation (e.g. 68, 69), and suitable proxy records to adequately resolve this are yet to be recovered.

Despite their considerable potential to answer such questions, general circulation models (GCMs) cannot yet achieve the level of performance required to accurately simulate Pleistocene climate dynamics in southern Africa (70). This being the case, and until more robust scenarios can be defined, our reconstructions employ modern precipitation gradients. Retaining these gradients, however, leads to what we deem unrealistic extrapolations of environmental conditions when applying the climatic parameters required to sustain the lacustrine conditions observed at Swartkolkvloer (P:PET = 1:2.33, the equivalent of a reduction in PET to 25% of modern levels). Located on the arid western margin of the Orange River, extrapolation of these conditions across the modern precipitation gradient results in the attribution of levels of precipitation and humidity along the eastern interior margin (Fig. S57) that are inconsistent with the palaeoenvironmental record, which suggests more modest increases in humidity (57, 71). These results and apparent contradictions support the supposition that rainfall gradients during the last glacial period differed from today. While data from sites such as Equus Cave indicate that winter rainfall was unlikely to have dominated the region (Lee-Thorp and Beaumont, 1995), the increased incursion of frontal systems associated with the westerly storm track may have resulted in a flatter, if not inverted, precipitation gradient across the continental interior (cf. Fig. S57).

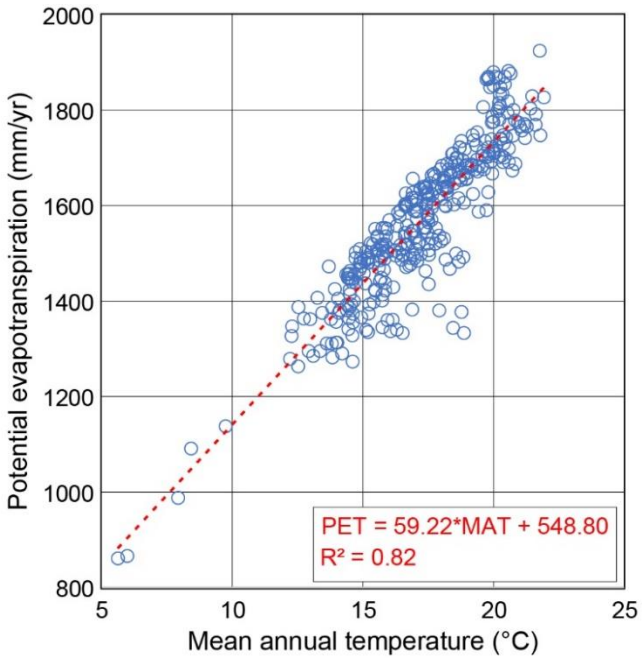
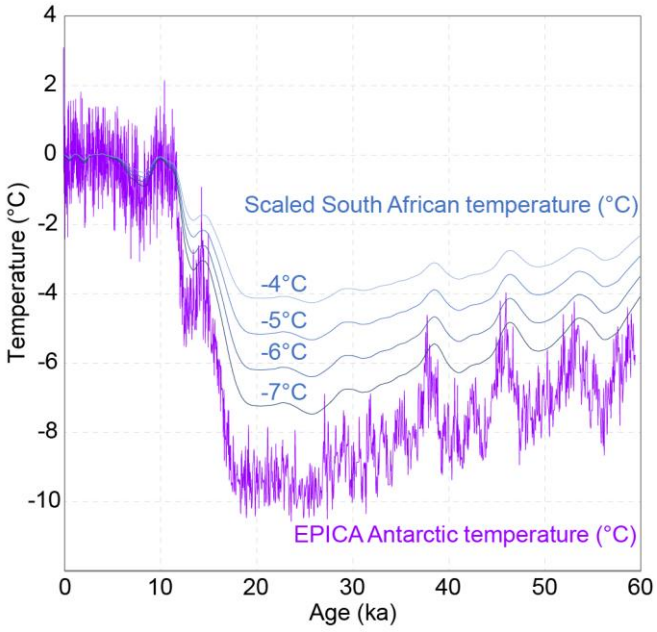


Fig. S55: Simple linear regression of potential evapotranspiration (Hargeraves) (PET; data from 5) and mean annual precipitation (MAT; data from 6, 39).

To mitigate the impact of potentially unrealistic rainfall gradients on vegetation reconstructions, we subdivided our vegetation reconstructions into eastern and western sub-regions (Fig. 3 main text). Conditions in the eastern sub-region are based on the 1:1.52 P:PET ratio necessary to sustain the lake level observed at Alexandersfontein. Conditions in the western sub-region are based on the 1:2.33 P:PET ratio required to sustain the +18 m Swartkolkvloer lake level.

Imposing these regional subdivisions, the resulting reconstructions of bioregion distributions represent plausible initial scenarios. In the western sub-region, it should be noted that our attribution of "grassland" vegetation may include bioregions whose closest analogue exist in the modern winter rainfall zone. If winter rainfall exceeded the 40-50% threshold for the establishment of Succulent Karoo and fynbos vegetation (Fig. 54C) the combination of relatively humid conditions and the interior's nutrient rich soils may have resulted in the development of high grass-cover renosterveld in the western portions of the interior (perhaps akin to the Langkloof or Humansdorp Shale Renosterveld (507 and 628 mm mean annual precipitation respectively); 72). Succulent Karoo vegetation would have likely been restricted under the more humid conditions indicated by our data, as denser vegetation may carry fire, which is detrimental to Succulent Karoo development (72).

Fig. S56: South African palaeotemperature estimate based on observed similarities between regional records (63) and the EPICA Antarctic temperature record (64). Regional proxy records indicate a ~4-7°C temperature depression during the Last Glacial Maximum (19-26.5 ka), and this has been used to scale the EPICA record to create continuous temperature estimates spanning the last 60 kyr. The estimations for 4-7°C scenarios have been smoothed using locally weighted smoothing. Lacustrine phases are derived from dating of unequivocal lacustrine features and deposits in the South African interior.



Lacustrine phase	Temperature depression
16-17 ka	-4.7°C
19-22 ka	-6.2°C
31-35 ka	-5.6°C
39-55 ka	-4.7°C

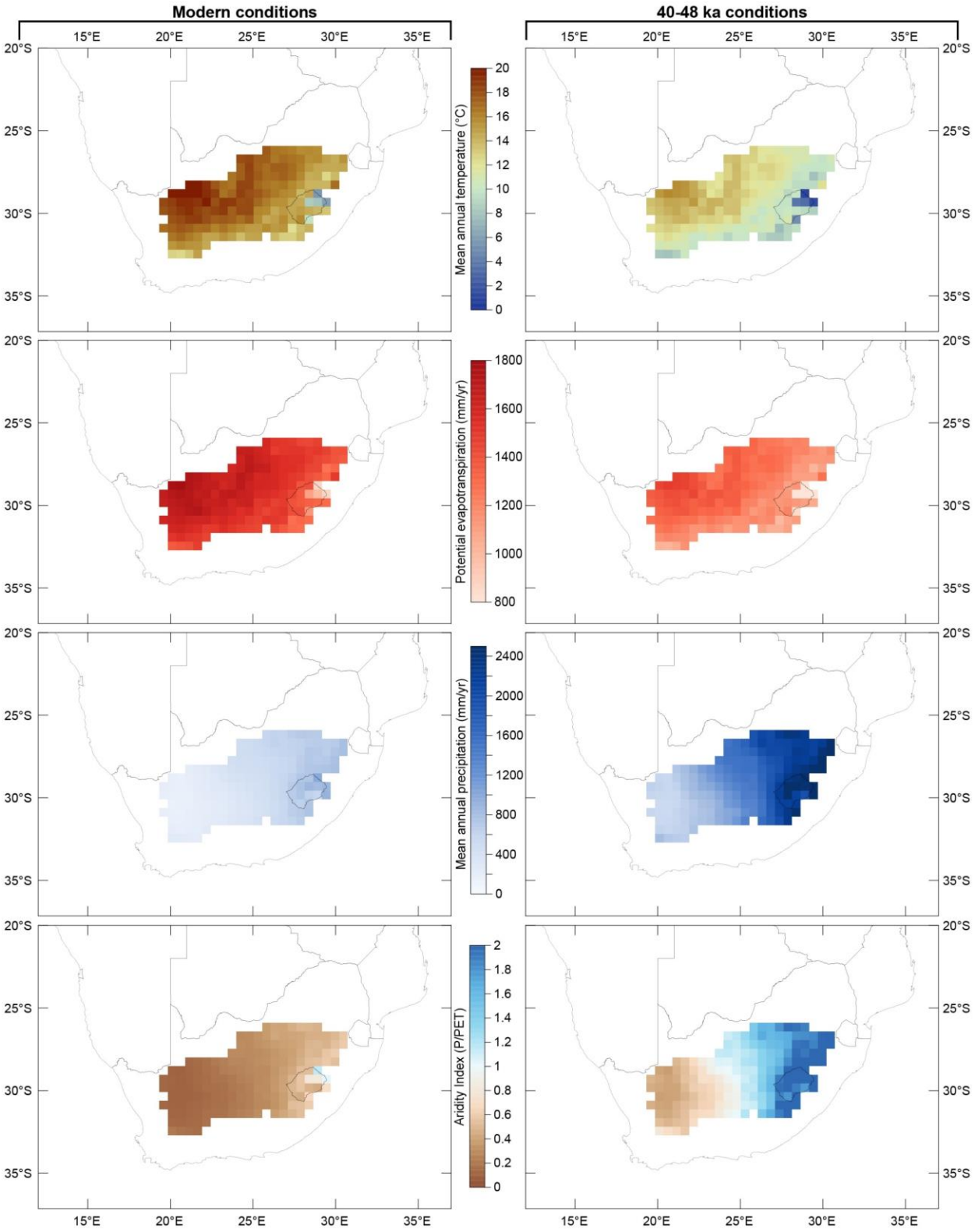


Fig. S57: Climate maps of South African interior study region showing modern values (right-hand column) and values for the 40-48 ka lake highstand period at Swartkolkvloer. Data from Trabucco and Zomer (5) and WorldClim 2.1 (6, 39).

Fig. S58: Assessment of vegetation reconstruction performance through the simulation of modern bioregion distributions (base map polygons; 4). Based on mean minimum temperature of the coldest month (6) and aridity (5), the reconstruction results (squares, same colour coding as observed bioregions) accord well with modern bioregion distributions, including in transitional regions of the Central Bushveld. Bioregions with strong edaphic constraints (Kalahari Duneveld) are simulated less accurately.

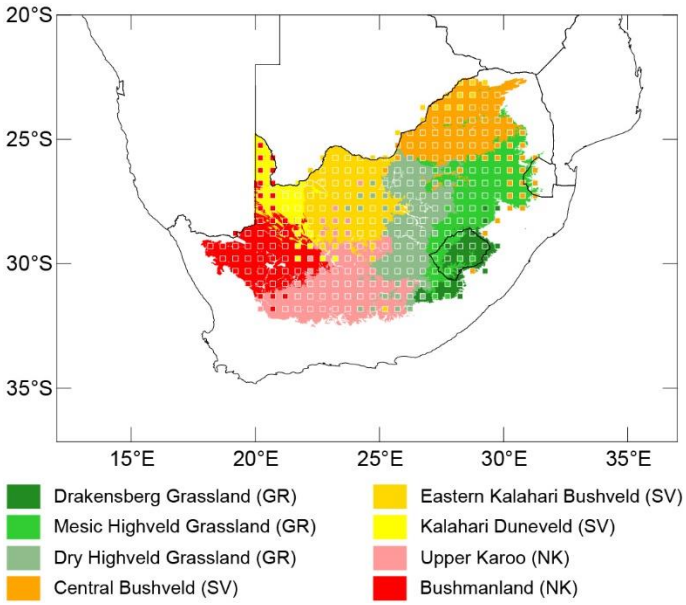
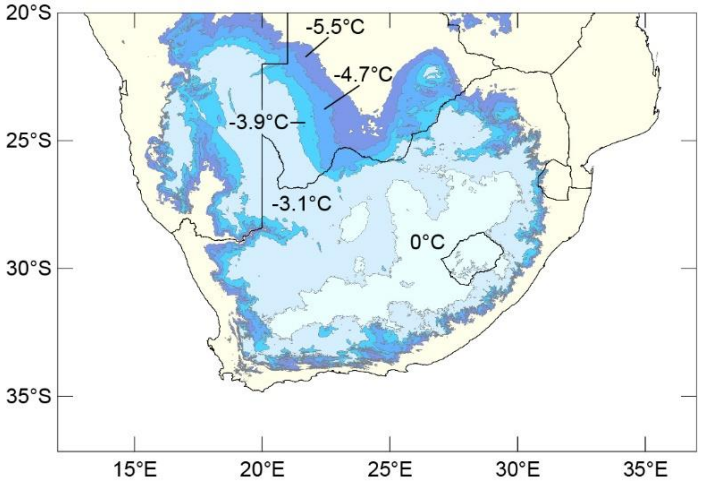


Fig. S59: Map of southern Africa showing zones where mean minimum temperature of the coldest month (TminColdM; data from 6) is equal to or below 0°C. Numbers indicate the degree of cooling predicted by the temperature reconstruction for the 39-55 ka lacustrine phase under each of the temperature depression scenarios shown in Fig. S56. Under all scenarios, it is unlikely that frost-intolerant savanna bioregions to developed in the South African interior.



From these data, and using the basic ruleset defined by modern climate-vegetation relationships, we estimated changes in bioregion distributions as a function of TminColdM and Aridity Index (Fig. S58). We have focused here only on the South African interior study region, acknowledging that the climatic and edaphic controls on other bioregion distributions (e.g. the Fynbos Biome) are more complex than in the continental interior. Considering the potential for the expansion of savanna bioregions during the more humid, but cooler, lacustrine phase of MIS 3-2, we find that under all temperature scenarios savanna vegetation was restricted by increased frost occurrence (Fig. S59) (73), although we do not rule out possibility that savanna-like vegetation structures may have occurred as a function of local hydrology under more mesic conditions (74). Indeed, the MIS 3-2 faunal assemblages from Equus Cave confirm the presence of both grazers and browsers, including giraffes, who favor open woodland vegetation (75). Considering the regional impact of cooler MIS 2-3 conditions, it is however most likely that it was the

Grassland bioregions that had the most potential for expansion into the interior under more humid conditions, with woodland likely to develop in riparian zones or in fire-protected areas.

In estimating the most likely Grassland bioregion equivalent, the model we employ is limited to modern bioregion distributions and their associated climatic controls. Based on temperature and aridity, the changes in water balance necessary to sustain the lakes described here predict that the Drakensberg Grassland bioregion would expand across much of the interior. This is because the high elevation regions of the Drakensberg are the coolest and most humid, receiving the most precipitation of the South African Grassland bioregions. However, the Drakensburg Grassland and adjacent Mesic Highveld Grassland bioregions are likely unrealistic analogues for the grass-rich environments of MIS 2-3 implied by the palaeoenvironmental data from Equus Cave (75, 76) and modelled here (Fig. 3 main text). While these bioregions occupy comparable climatic niches, they are characterized by leached soils and grasses of less palatable sourveld varieties (77, 78). In contrast, the soils of the arid to semi-arid interior are relatively unleached, and, with increased humidity, are likely to have supported more palatable, nutrient rich grasslands for which there are no direct analogues in southern Africa today (79). As such, we have maintained the climatic qualifier from the modern classifications of Mucina et al. (2006), but have renamed the simulated grassland bioregions as “Eutrophic Dry Grassland”, “Eutrophic Mesic Grassland” and “Eutrophic Cool/Cold Humid Grassland”.

Large ungulate resources

To consider changes in regional fauna, we re-analysed data from Equus Cave, the only faunal record from the South African interior that preserves remains clearly dated to MIS 1, 2 and 3. The site’s chronology does not permit detailed time series analyses, but is sufficient to assess late Pleistocene assemblages (75, 80). For our analysis, we compared the large ungulates reported by Klein et al. (75) on a presence/absence basis with comparable modern/historic data compiled by Hempson et al. (81) for Sub-Saharan Africa.

Considering the very different nature of MIS 2-3 environments and the extinction of taxa such as *Antidorcas bondi, Equus capensis*, *Megalotragus priscu*s and, regionally, *Phacochoerus aethiopicus*, it is predictable that a precise analogue cannot be identified in modern African environments. In the Equus Cave region, characterised today by Kalahari Xeric Savanna vegetation (4), and including a section of the Vaal River corridor, 11 of the 17 extant taxa (Simpson similarity score = 0.64) identified by Klein et al. (75) still occur (81). Absent species are *Hippopotamus amphibius* (hippopotamus), *Hippotragus* sp. (roan, (though present in the 1800s; 82) and/or sable antelope), *Tragelaphus strepsiceros* (greater kudu), *Pelea capreolus* (grey rhebok) and *Kobus leche* (lechwe). The best presence-based analogues for the MIS 2-3 faunal assemblages (Simpson similarity score = 0.76-0.88) are today found in the Central Bushveld and Highveld Grasslands to the east of Equus Cave (Fig. S60). Aggregating the half-degree grid cells used by Hempson et al. (81) according to the bioregions defined by Mucina and Rutherford (4) to include landscape scale topographic and hydrological diversity, the Central Bushveld and Highveld Grassland each support 94% of the taxa found in the MIS 2-3 deposits at Equus Cave, with the exception of *Kobus leche*, the closest occurrence of which is ≥900 km to the north, in the Okavango Delta and swamplands of southeastern Zambia (81, 83) (Fig. S60).

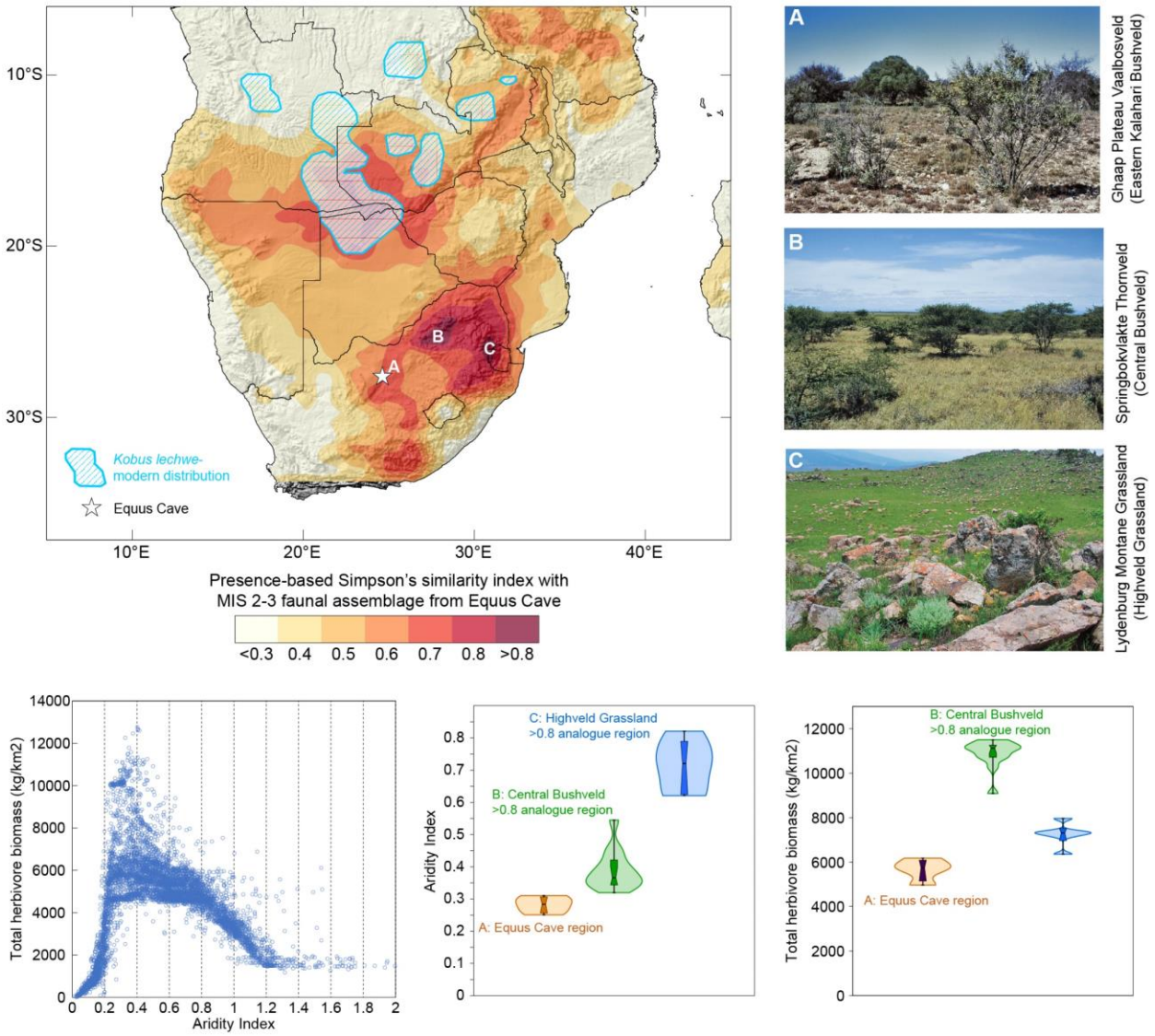


Fig. S60: Comparison of Equus Cave and potential analogue environments based on the site's MIS 2-3 large mammal herbivore fossil assemblage. A, B and C labels and accompanying photos (from 4; photo credit to M.C. Rutherford and M.C. Lötter) indicate the location and nature of the Equus Cave environment and those of the presence-based best-fit analogue regions in the Central Bushveld and Highveld Grasslands (4) as established using the African large mammal herbivore data from Hempson et al. (81). Lower panes show the relationship between large herbivore biomass and aridity, the aridity index values for the regions considered, and the associated change in faunal resources.

The best-fit regions (Fig. S60 B and C) are more humid and support significantly greater grass cover than the Equus Cave region today (4, 75). The relationship between large mammal herbivore biomass and aridity indicates a marked increase in biomass with even modest increases in humidity under arid and semi-arid conditions (aridity index values of 0.03-0.2 and 0.2-0.5 respectively) (81). This is exemplified in the approximate doubling of large herbivore biomass between the Equus Cave region and the >0.8 Simpson similarity analogue region in the Central Bushveld (Fig. S60). The relationship between biomass

and aridity is non-linear, however, and under wetter conditions biomass declines (84), as is the case with the higher elevations of the Highveld Grassland bioregion, which is only modestly more productive than the Equus Cave region today. However, soil nutrient levels must be considered, and as discussed above it is likely that the relatively unleached soils of the South African interior produced highly productive grasslands under the climatic conditions that sustained the high lake levels observed at Alexandersfontein and Swartkolkvloer. Considering these results in concert with the water-balance values simulated by the hydrological model employed and the derived vegetation reconstructions, we find independent corroboration for the model simulations and conclude that MIS 2-3 environments at Equus Cave were characterised by rich faunal resources, supported by more productive grassland (or grassier) vegetation and abundant perennial surface water.

Appendix A

Table S10: Details of the radiocarbon dating and calibrations. Samples were calibrated using CALIB 8.1.0 (40) and SHCAL20 (39). All Swartkolkvloer analyses were conducted on specimens of *Tomichia ventricosa.*

Sample	Material	Depth (m)	Code	^{14}C age (^{14}C yr)	2 sigma lower range (cal yr. BP)	2 sigma upper range (cal yr. BP)	Cal median (cal yr. BP)
Swartkolkvloer Quarry							
Quarry North (SKV18/2)	*T. ventricosa*	0.34	UBA-39022	41,995 ± 1560	42,602	47,254	**44,770**
Quarry North (SKV18/2)	*T. ventricosa*	0.52	UBA-39021	48,086 ± 3795	45,039	54,980	**50,310**
Quarry North (SKV18/2)	*T. ventricosa*	1.30	UBA-39020	44,644 ± 2270	43,050	52,334	**47,370**
Quarry North (SKV18/2)	*T. ventricosa*	1.80	UBA-39019	>47,922	-	-	-
Quarry South (SKV18/3)	*T. ventricosa*	1.00	UBA-39018	50,162 ± 5196	44,782	54,980	**50,650**
Swartkolkvloer dam (Breek Been Kolk)							
Dam 1 (SKV18/4/23)	*T. ventricosa*	0.23	Beta - 593951	38,100 ± 330	42,028	42,536	**42,290**
Dam 1 (SKV18/4/3)	*T. ventricosa*	0.85	UBA-39017	43,313 ± 1833	42,924	49,001	**45,930**
Dam 2 (SKV19/5/46)	*T. ventricosa*	0.46	Beta - 593952	30,040 ± 140	34,215	34,672	**34,450**
Dam 2 (SKV19/5/65)	*T. ventricosa*	0.65	Beta - 593953	38,140 ± 330	42,051	42,575	**42,320**
Dam 2 (SKV19/5/115)	*T. ventricosa*	1.15	Beta - 593954	43,220 ± 550	44,703	46,595	**45,580**
Locality A							
"Locality A" Lower shore	*T. ventricosa*	0.64	UBA-38533	27,443 ± 218	31,113	31,740	**31,400**
"Locality A" Lower shore	*T. ventricosa*	1.12	UBA-39023	29,979 ± 369	33,741	35,219	**34,420**
"Locality A" Upper shore	*T. ventricosa*	0.35	UBA-38532	35,921 ± 428	40,099	41,741	**40,960**
Swartkolkvloer pan floor							
SKV19/6/48	*T. ventricosa*	0.48	Beta - 593955	>43,500	-	-	-
SKV19/6/49	*T. ventricosa*	0.48	Beta - 593956	41,051 ± 450	43,140	44,610	**43,970**
Grootvloer							
GV18/4 Shell bed	*Unio caffer*	0.1	UBA-38535	17,186 ± 84	20,507	20,893	**20,700**
Alexandersfontein							
AF17/1/147	Bulk sediment	1.47	Beta - 593957	21,340 ± 70	25,533	25,865	**25,700**
AF17/1/220	Bulk sediment	2.20	UBA-38536	28,223 ± 260	31,634	33,111	**32,290**

Table S11: Sedimentological characteristics of the sampled materials, including the relative proportions of sand, silt and clay within each sample. These data were obtained using a Malvern Mastersizer 3000 laser particle sizer.

Sample	Mean (ϕ)	Median (ϕ)	Sorting (ϕ)	Skew	Kurtosis	Sand	Silt	Clay
Swartkolkvloer Quarry								
SKV18/3/1	2.6	2.3	2.0	-0.3	1.1	0.78	0.18	0.03
SKV18/2/9	2.5	2.0	2.6	-0.5	1.8	0.80	0.13	0.07
Swartkolkvloer Dam (Breek Been Kolk)								
SKV18/4/2	2.4	2.0	2.5	-0.5	2.1	0.82	0.08	0.09
SKV18/4/1	3.8	2.9	3.2	-0.5	1.3	0.66	0.19	0.15
Swartkolkvloer pan floor								
SKV19/6	8.0	8.2	3.3	0.1	0.8	0.15	0.33	0.52
Hoek van Spruit se Vloer								
HSV3-42	6.3	5.7	4.2	-0.2	0.6	0.42	0.19	0.39
HSV3-69	4.8	3.4	3.3	-0.6	1.3	0.65	0.18	0.17
HSV3-100	4.0	2.9	3.2	-0.5	1.9	0.73	0.12	0.15
HSV3-148	2.5	1.9	2.9	-0.5	2.0	0.81	0.08	0.10
Grootvloer – pan floor sediment								
GV18/5/3	9.9	10.9	4.4	0.3	0.7	0.14	0.25	0.61
GV17/3/1	9.9	10.6	4.1	0.2	0.7	0.10	0.29	0.61
Alexandersfontein (donga)								
AF17/1/1	5.9	4.1	4.4	-0.5	1.0	0.51	0.24	0.25
AF17/1/2	6.5	4.8	4.5	-0.5	0.7	0.47	0.19	0.33
AF17/1/3	10.4	11.2	4.1	0.3	0.6	0.06	0.36	0.58
Alexandersfontein (Mauritzfontein)								
AF17/5/2	7.7	7.3	4.5	-0.1	0.6	0.36	0.17	0.47
AF17/5/1	6.8	6.5	3.5	-0.1	0.6	0.32	0.33	0.36

Table S12: Dose rate calculations, sample depths, water contents, radionuclide concentrations and calculated dose rates for all of the luminescence dating samples reported.

Sample	Depth(m)	Water content (%)	Grain size fraction (µm)	U (ppm)	Th (ppm)	K (%)	Internal beta dose rate (Gy ka^{-1})	External beta dose rate (Gy ka^{-1})	External gamma dose rate (Gy ka^{-1})	Cosmic Dose rate (Gy ka^{-1})	Total dose rate (Gy ka^{-1})
SKV Quarry											
SKV18/2/9	0.2	0.6[a]	180-212	1.9	3.9	0.9	0.85 ± 0.06	0.86 ± 0.06	0.58 ± 0.03[#]	0.25 ± 0.03	**2.54 ± 0.09**
SKV19/2/10	1.1	0.6 [a]	180-212	2.3	7.7	1.5	0.85 ± 0.06	1.38 ± 0.10	0.97 ± 0.06	0.21 ± 0.03	**3.41 ± 0.13**
SKV19/2/11	1.5	5.0 [a]	180-212	3.1	9.2	2.0	0.85 ± 0.06	1.78 ± 0.12	1.17 ± 0.06[b]	0.19 ± 0.02	**3.99 ± 0.15**
SKV18/3/1	1.5	0.6 [a]	180-212	2.0	6.4	1.4	0.85 ± 0.06	1.21 ± 0.09	0.84 ± 0.05	0.19 ± 0.02	**3.09 ± 0.12**
SKV Dam (Breek Been Kolk)											
SKV18/4/2	1.1	2.2 [a]	180-212	1.9	6.2	1.5	0.85 ± 0.06	1.30 ± 0.09	0.86 ± 0.05	0.20 ± 0.02	**3.21 ± 0.12**
SKV18/4/1	1.4	1.7 [a]	180-212	1.7	5.7	1.5	0.85 ± 0.06	1.24 ± 0.09	0.80 ± 0.04	0.20 ± 0.02	**3.09 ± 0.11**
SKV19/5/1	0.7	1.8 [a]	180-212	2.0	5.7	1.2	0.85 ± 0.06	1.11 ± 0.07	0.77 ± 0.04	0.22 ± 0.02	**2.95 ± 0.11**
SKV Locality A											
SKV17/1/2[c]	0.35	5.4	212-250	2.7	10.1	1.7	0.99 ± 0.07	1.67 ± 0.11	1.14 ± 0.05	0.23 ± 0.02	**4.25 ± 0.15**
SKV pan floor											
SKV19/6/1	0.52	15.5	90-250*	4.2	16.4	4.7	0.69 ± 0.28	3.36 ± 0.23	2.06 ± 0.11	0.22 ± 0.02	**6.33 ± 0.38**
Hoek van spruit se vloer											
HSV19/3/1	0.40	3.9	180-212	2.5	7.60	1.6	0.85 ± 0.06	1.42 ± 0.09	0.99 ± 0.06	0.23 ± 0.02	**3.49 ± 0.13**
HSV19/3/3	1.32	2.6	180-212	2.7	9.4	2.0	0.85 ± 0.06	1.76 ± 0.12	1.21 ± 0.06	0.20 ± 0.02	**4.02 ± 0.15**
HSV19/3/4	1.85	3.0	180-212	3.2	11.7	2.3	0.85 ± 0.06	2.07 ± 0.14	1.45 ± 0.08	0.19 ± 0.02	**4.56 ± 0.17**

Sample	Depth(m)	Water content (%)	Grain size fraction (µm)	U (ppm)	Th (ppm)	K (%)	Internal beta dose rate (Gy ka^{-1})	External beta dose rate (Gy ka^{-1})	External gamma dose rate (Gy ka^{-1})	Cosmic Dose rate (Gy ka^{-1})	Total dose rate (Gy ka^{-1})
Grootvloer – sand ridge (Hoezar Oos)											
GV17/1/1	0.94	1.3	180-212	1.25	4.98	1.58	-	1.27 ± 0.08	0.76 ± 0.04	0.21 ± 0.02	**2.24 ± 0.09**
Grootvloer – pan floor sediment fill (Hoezar Wes)											
GV17/3/1[d]	0.48	8.7	90-180*	2.6	12.9	1.9	0.56 ± 0.05	2.11 ± 0.12	1.25 ± 0.06	0.22 ± 0.02	**4.14 ± 0.21**
GV18/4/2	0.65	3.9	180-212	2.0	7.7	1.9	0.85 ± 0.06	1.55 ± 0.11	1.01 ± 0.06	0.21 ± 0.02	**3.62 ± 0.14**

Grootvloer – aeolian sand ridge and subjacent pan floor sediment fill (Hoezar Wes)											
GV18/5/1	0.97	4.5	180-212	2.6	6.4	1.6	0.85 ± 0.06	1.41 ± 0.10	0.95 ± 0.06	0.21 ± 0.02	**3.42 ± 0.13**
GV18/5/2	1.70	8.2	180-212	0.8	3.9	0.9	0.85 ± 0.06	0.68 ± 0.05	0.45 ± 0.03	0.19 ± 0.02	**2.17 ± 0.08**
GV18/5/2	2.5	11.5	180-212	3.7	10.8	2.4	0.85 ± 0.06	1.93 ± 0.13	1.34 ± 0.08	0.17 ± 0.02	**4.29 ± 0.17**
Alexandersfontein (Donga)											
AF17/1/1	0.62	3.8	180-250	2.0	7.6	0.9	-	0.92 ± 0.07	0.78 ± 0.04	0.23 ± 0.02	**1.93 ± 0.08**
AF17/1/2	1.22	2.8	180-212	2.4	8.6	0.7	-	0.88 ± 0.05	0.82 ± 0.06	0.21 ± 0.02	**1.91 ± 0.08**
Alexandersfontein (Mauritzfontein)											
AF17/5/3	0.35	5.3	180-212	1.4	4.0	0.6	-	0.59 ± 0.05	0.47 ± 0.02	0.25 ± 0.03	**1.31 ± 0.05**
AF17/5/2	0.60	8.3	180-212	2.0	4.0	0.6	-	0.61 ± 0.05	0.50 ± 0.03	0.23 ± 0.02	**1.35 ± 0.06**
AF17/5/1	1.10	8	180-212	1.4	4.0	0.7	-	0.63 ± 0.04	0.48 ± 0.03	0.22 ± 0.02	**1.33 ± 0.06**
Verkeerdevlei (Vanwyksvlei)											
VKV17/1/3	0.45	1.7	180-212	1.1	3.7	1.5	0.85 ± 0.06	1.15 ± 0.08	0.65 ± 0.03	0.23 ± 0.03	**2.88 ± 0.11**
VKV17/1/1	0.86	3.9	180-212	2.8	6.8	1.5	0.85 ± 0.06	1.37 ± 0.09	0.96 ± 0.04	0.22 ± 0.03	**3.40 ± 0.12**
VKV17/1/2	1.20	2.8	180-212	4.6	11.4	2.0	0.85 ± 0.06	2.00 ± 0.13	1.36 ± 0.08[d]	0.21 ± 0.02	**4.42 ± 0.16**

*Grain size attenuation using mean grain size and error calculated following Durcan et al (85). DRAC; [#]gamma dose adjusted for shallow burial following Aitken (15); [a]Due to possible drying of these sediments following 20th century excavation, a water content of 3 ± 3% was applied to these samples; [b]gamma dose adjusted for overlying (from 7 cm above sample) shelly layer composition as gamma spectrometry was not available [c]sample not etched and beta dose adjusted accordingly, with additional 0.22 ± 0.05 Gy ka^{-1} eternal alpha dose rate included in total dose rate. [d]Sample gamma dose rate adjusted to account for underlying pan floor sediment, following Aitken (15). The dose rates are for K-feldspars for all samples except GV17/1/1, AF17/1/1, AF17/1/2 and AF17/5/1, 5/2 and 5/3 for which the total dose rate to quartz is reported.

Table S13: Details of the equivalent dose measurements and fading rate (g-values) obtained for all of the K-feldspar and quartz luminescence analyses at Swartkolkvloer and Grootvloer. n/N refers to accepted/measured aliquots (as per the criteria outlined the text above). OD – overdispersion parameter, as derived from the calculation of the Central Age Model weighted average D_e (24).

Sample	Quartz D_e (Gy) (n/N)	Quartz OD (%)	K Feldspar IR50 D_e (Gy) (n/N)	K Feldspar IR50 OD (%)	K Feldspar IR50 g-value$_{2days}$ (%)	K Feldspar pIRIR$_{225}$ D_e (Gy) (n/N)	K Feldspar pIRIR$_{225}$ OD (%)	K Feldspar pIRIR$_{225}$ g-value$_{2days}$ (%)	K Feldspar pIRIR$_{290}$ D_e (Gy) (n/N)	K Feldspar pIRIR$_{290}$ OD (%)	K Feldspar pIRIR$_{290}$ g-value$_{2days}$ (%)
Swartkolkvloer Quarry											
SKV18/2/9	nd	nd	93 ± 4 (14/14)	8 ± 2	3.4 ± 0.7	119 ± 4 (14/14)	7 ± 2	1.2 ± 0.7	170 ± 11 (11/14)	18 ± 4	nd.
SKV19/2/10	nd	nd	112 ± 5 (13/13)	11 ± 2	3.9 ± 0.4	159 ± 6 (13/13)	8 ± 2	1.6 ± 0.4	182 ± 7 (15/17)	9 ± 1	1.3 ± 0.6
SKV19/2/11	nd	nd	106 ± 9 (14/15)	28 ± 5	6.8 ± 0.7	154 ± 11 (13/15)	22 ± 4	2.7 ± 0.8	177 ± 8 (6/7)	8 ± 3	nd.
SKV18/3/1	nd	nd	74 ± 4 (17/17)	19 ± 3	5.2 ± 0.5	126 ± 5 (17/17)	8 ± 2	1.7 ± 0.6	163 ± 8 (9/11)	12 ± 3	nd.
Swartkolkvloer Dam (Breek Been Kolk)											
SKV18/4/2	nd.	nd.	37 ± 5 (25/25)	63 ± 9	9 ± 1	76 ± 8 (23/25)	48 ± 7	1.6 ± 0.9	nd.	nd.	nd.
SKV18/4/1	70 ± 5 (9/17)	15 ± 6	83 ± 3 (12/13)	9 ± 2	3.8 ± 0.8	117 ± 5 (12/13)	8 ± 2	1.3 ± 0.7	147 ± 6 (15/16)	9 ± 2	nd.
SKV19/5/1	nd	nd	80 ± 4 (17/17)	16 ± 2	4.0 ± 0.4	110 ± 6 (17/17)	18 ± 3	2.1 ± 0.4	140 ± 7 (13/13)	13 ± 3	2.0 ± 0.8
Swartkolkvloer Locality A											
SKV17/1/2			119 ± 6 (22/24)	21 ± 3	nd	137 ± 8 (23/23)	24 ± 4	nd.	nd.	nd.	nd.
Swartkolkvloer pan floor											
SKV19/6/1			61 ± 6 (16/16)	34 ± 6	4.6 ± 0.5	94 ± 7 (15/16)	27 ± 5	2.0 ± 0.6	nd	nd	nd
Hoek van spruit se vloer											
HSV19/3/1			137 ± 6 (15/16)	13 ± 2	4.6 ± 0.7	183 ± 12 (14/16)	21 ± 4	1.8 ± 0.7			

HSV19/3/3			145 ± 8 (13/13)	16 ± 2	nd	201 ± 11 (13/13)	15 ± 3	nd			
HSV19/3/4			169 ± 9 (15/15)	16 ± 3	4.0 ± 1.7	223 ± 12 (13/15)	14 ± 3	1.6 ± 0.7	305 ± 14 (14/15)	14 ± 3	
Grootvloer – sand ridge (Hoezar Oos)											
GV17/1/1	12 ± 1 (12/24)	17 ± 4									
Grootvloer – pan floor sediment fill (Hoezar Wes)											
GV17/3/1			59 ± 3 (13/15)	11 ± 2	5.0 ± 0.6	85 ± 4 (13/15)	13 ± 3	1.7 ± 0.6	nd.	nd.	nd.
GV18/4/2	57 ± 3 (15/32)	17 ± 4	42 ± 2 (15/15)	13 ± 3	3.9 ± 0.6	55 ± 2 (14/15)	8 ± 1	2.2 ± 0.6	72 ± 3 (11/14)	12 ± 3	2.5 ± 0.6
Grootvloer – aeolian sand ridge and subjacent pan floor sediment fill (Hoezar Wes)											
GV18/5/1			2.8 ± 0.1 (13/13)	9 ± 2	nd.	3.7 ± 0.1 (12/13)	8 ± 2	nd.	nd.	nd.	nd.
GV18/5/2			24 ± 1 (16/17)	18 ± 3	7.6 ± 0.6	37 ± 2 (17/17)	17 ± 3	1.9 ± 0.7	nd.	nd.	nd.
GV18/5/3			55 ± 2 (12/14)	10 ± 2	5.8 ± 0.6	86 ± 3 (14/14)	10 ± 3	2.0 ± 0.7	90 ± 5 (7/8)	10 ± 2	nd
Verkeerdevlei (Vanwyksvlei)											
VKV17/1/3	3.4 ± 0.2 (12/38)	0 ± 0	2.9 ± 0.1 (12/12)	4 ± 1	nd.	3.5 ± 0.1 (12/12)	1 ± 2	nd.			
VKV17/1/1	nd.	nd.	3.7 ± 0.1 (9/9)	5 ± 1	nd.	4.5 ± 0.2 (9/9)	5 ± 1	nd.			
VKV17/1/2	nd.	nd.	7.7 ± 0.3 (17/18)	8 ± 2	6.3 ± 0.5	9.9 ± 0.4 (17/18)	7 ± 1	2.1 ± 0.6			

Sites included in Figure 1 of main text

Site	Indicative references
Makgadikgadi Basin	(86)
Lobatse Cave	(87)
Apollo 11 Rockshelter	(88, 89)
Blombos Cave	(123)
Border Cave	(91, 92)
Cango Cave	(59, 93)
Die Kelders Cave	(94, 95)
Diepkloof Rockshelter	(96, 97)
Elands Bay Cave	(98, 124)
Efflux Cave	(99)
Equus Cave	(76)
Ga-Mohana Rockshelter	(121, 122)
Hollow Rock Shelter	(100)
Klipdrift Cave	(101)
Klipfonteinrand	(102)
Kathu Pan	(103, 104)
Klasies River Mouth	(94, 105)
Melikane Rockshelter	(106)
Pinnacle Point sites	(125)
Putslaagte 8	(108)
Rose Cottage Cave	(109, 110)
Sibudu Cave	(126)
Sehonghong Rockshelter	(112, 127, 128)
Tswaing Crater	(114)
Varsche River 3	(116, 129)
Wonderwerk Cave	(117, 118)
Bundu Farm	(119)
Florisbad	(120)

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